

Radial velocities

The motion of a single planet in orbit around a star causes the star to undergo a reflex motion about the star–planet barycentre (centre of mass). This results in the periodic perturbation of three observable properties of the star, all of which have been detected (albeit typically in different systems): in radial velocity, in angular (or astrometric) position on the sky, and in the time of arrival of some periodic reference signal.

2.1 Description of orbits

As in all orbiting systems, both star and planet orbit the common system barycentre. Under the inverse square law of gravity¹, each moves in a closed elliptical orbit in inertial space, with the centre of mass at one focus (Figure 2.1). Such an ellipse is described in polar coordinates (with respect to a focus) by

$$r = \frac{a(1-e^2)}{1+e\cos v}, \quad (2.1)$$

or in Cartesian coordinates (with respect to the centre)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad (2.2)$$

with the semi-major axis a and the semi-minor axis b related to the *eccentricity* e by

$$b^2 = a^2(1-e^2). \quad (2.3)$$

[The ellipticity, $\eta = (a-b)/a = 1 - \sqrt{1-e^2}$, is an alternative measure of non-circularity, not used further.]

The pericentre² distance q and apocentre distance Q are given by

$$q = a(1-e),$$

¹The relations given in this section, their derivations, and more extensive dynamical considerations, can be found in various recent texts (e.g., Murray & Dermott 2000, Chapter 2; Hilditch 2001; Cole & Woolfson 2002, Section M) as well as in many earlier treatments of orbits (e.g. Heintz, 1978a; Roy, 1978)

²An *apsis* (plural *apsides*) is the point on the orbit of minimum distance (*pericentre*, or *periapsis*) or maximum distance (*apocentre*, or *apoapsis*) from the barycentre. The line connecting the two is the *line of apsides*, which defines the orbit's

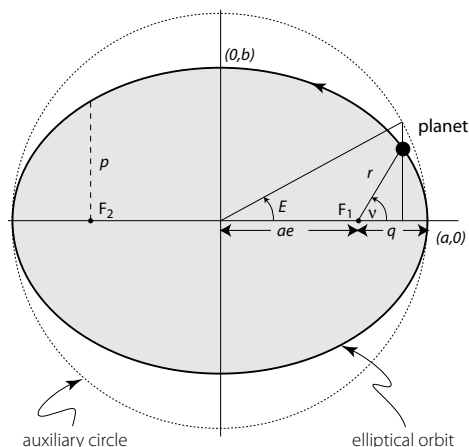


Figure 2.1: Characteristics of an elliptical orbit. The auxiliary circle is of radius equal to the semi-major axis a . Points on the orbit can be described either in terms of the true anomaly v (with respect to the ellipse) or the eccentric anomaly E (with respect to the auxiliary circle). The focus F_1 is the system barycentre. F_2 is the 'empty' focus.

$$Q = a(1+e). \quad (2.4)$$

A line through a focus and parallel to the minor axis defines the latus rectum, with the *semi-latus rectum* (related to the planet's angular momentum) of length

$$p = a(1-e^2). \quad (2.5)$$

True, eccentric, and mean anomaly Various angles in the orbit plane, referred to as 'anomalies' (the word dating from the time that planetary motions were considered anomalous), are used to describe the position of a planet along its orbit at a particular time (Figure 2.1).

Derivative terms refer to the body being orbited: perigee/apogee around the Earth, perihelion/aphelion around the Sun, periastron/apastron around a star, peribac/apobac around a barycentre, and perigalacticon/apogalacticon around a galaxy. The general terms pericentre/apocentre are used preferentially here.

The *true anomaly*, $v(t)$, also frequently denoted $f(t)$, is the angle between the direction of pericentre and the current position of the body measured from the barycentric focus of the ellipse. It is the angle normally used to characterise an observational orbit.

The *eccentric anomaly*, $E(t)$, is a corresponding angle which is referred to the *auxiliary circle* of the ellipse. The true and eccentric anomalies are geometrically related by

$$\cos v(t) = \frac{\cos E(t) - e}{1 - e \cos E(t)}, \quad (2.6)$$

or, equivalently,

$$\tan \frac{v(t)}{2} = \left(\frac{1+e}{1-e} \right)^{1/2} \tan \frac{E(t)}{2}. \quad (2.7)$$

The *mean anomaly*, $M(t)$, is an angle related to a fictitious mean motion around the orbit, used in calculating the true anomaly. Over a complete orbit, during which the real planet (or the real star) does not move at a constant angular rate, an average angular rate can nevertheless be specified in terms of the *mean motion*

$$n \equiv 2\pi / P, \quad (2.8)$$

where P is the orbital period. The mean anomaly at time $t - t_p$ after pericentre passage is then defined as

$$M(t) = \frac{2\pi}{P}(t - t_p) \equiv n(t - t_p). \quad (2.9)$$

The relation between the mean anomaly, $M(t)$, and the eccentric anomaly, $E(t)$, can be derived from orbital dynamics. This relation, Kepler's equation, is given by

$$M(t) = E(t) - e \sin E(t). \quad (2.10)$$

The position of an object along its orbit at any chosen time can then be obtained by calculating the mean anomaly M at that time from Equation 2.9, (iteratively) solving the transcendental Equation 2.10 for E , and then using the geometrical identity Equation 2.6 to obtain v .

Orbit specification A Keplerian orbit in three dimensions (Figure 2.2) is described by seven parameters: $a, e, P, t_p, i, \Omega, \omega$. The first two, a and e , specify the size and shape of the elliptical orbit. P is related to a and the component masses through Kepler's third law (see below), while t_p corresponds to the position of the object along its orbit at a particular reference time, generally with respect to a specified pericentre passage.³

The three angles (i, Ω, ω) represent the projection of the true orbit into the observed (apparent) orbit; they

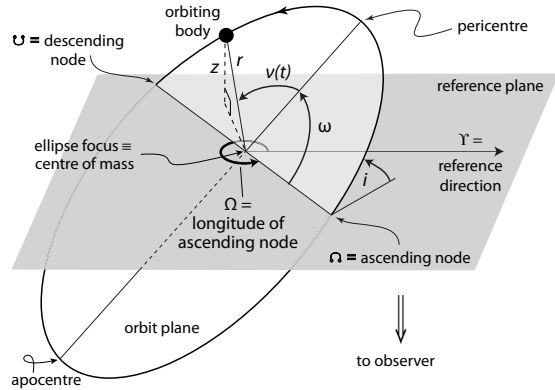


Figure 2.2: An elliptical orbit in three dimensions. The reference plane is tangent to the celestial sphere. i is the inclination of the orbit plane. The nodes are the points where the orbit and reference planes intersect. Ω defines the longitude of the ascending node, measured in the reference plane. ω is the fixed angle defining the object's argument of pericentre relative to the ascending node. The true anomaly, $v(t)$, is the time-dependent angle characterising the object's position along the orbit.

depend solely on the orientation of the observer with respect to the orbit. In general, the semi-major axis of the true orbit does not project into the semi-major axis of the apparent orbit.

i specifies the *orbit inclination* with respect to the reference plane, in the range $0 \leq i < 180^\circ$. $i = 0^\circ$ corresponds to a face-on orbit. In the discussion of binary orbits, motion is referred to as prograde (in the direction of increasing position angle on the sky, irrespective of the relation between the rotation and orbit vectors) if $i < 90^\circ$, retrograde if $i > 90^\circ$, and projected onto the line of nodes if $i = 90^\circ$.

Ω specifies the *longitude of the ascending node*, measured in the reference plane. It is the node where the measured object moves away from the observer through the plane of reference. [For solar system objects, it is the node where an orbiting object moves north through the plane of reference.]

ω specifies the *argument of pericentre*, being the angular coordinate of the object's pericentre relative to its ascending node, measured in the orbital plane and in the direction of motion. [For $e = 0$, where pericentre is undefined, $\omega = 0$ can be chosen such that t_p gives the time of nodal passage.]

to be in linear measure unless otherwise noted. If a is determined in angular measure, as in the relative astrometry of binary stars, the system distance d (equivalently the parallax ϖ) is required to establish the linear scale; (iii) the parameters of the two co-orbiting bodies (e.g. a star and planet) with respect to the barycentre are identical, with the exception of their values of a which differ by a factor $M_p/M_\star$, and their values of ω which differ by 180° .

³A few remarks are in order: (i) some texts state that just six parameters are required, and omit P , implicitly invoking the relation between a and P (and the component masses) as given by Kepler's third law; (ii) a is the semi-major axis of the orbiting body with respect to the system barycentre, assumed here

Three other angles with respect to the adopted reference direction are used in the specification of practical orbits:

$$\tilde{\omega} = \Omega + \omega \quad \text{the longitude of pericentre} \quad (2.11)$$

$$\theta = \tilde{\omega} + \nu \quad \text{the true longitude} \quad (2.12)$$

$$\lambda = \tilde{\omega} + M \quad \text{the mean longitude} \quad (2.13)$$

Since Ω and ω are measured in different planes, the longitude of pericentre, $\tilde{\omega}$, is a ‘dog-leg’ angle. The true longitude and mean longitude are correspondingly offset with respect to the true anomaly and mean anomaly, respectively. Despite its name, the mean longitude is again a linear function of time and, as for the mean anomaly, has only an auxiliary geometrical interpretation.

Kepler’s laws Kepler’s three laws of planetary motion are: (1) the orbit of a planet is an ellipse with the Sun at one focus; (2) the line joining a planet and the Sun sweeps out equal areas in equal intervals of time; (3) the squares of the orbital periods of the planets are proportional to the cubes of their semi-major axes.

The first and third laws are consequences of the inverse square law of gravity, while the second follows from conservation of angular momentum (and is true for any radial law of attraction). Kepler’s laws originally referred to relative orbits with respect to the Sun, but corresponding formulations apply also to ‘absolute orbits’ defined with respect to the barycentre.

For the general two-body problem where the mass of the secondary is not neglected, both orbits are ellipses with their foci at their common barycentre (Figure 2.3). Kepler’s third law takes the general form

$$P^2 = \frac{4\pi^2}{GM} a^3, \quad (2.14)$$

with M and a taking different values according to the type of orbit being measured:

(a) relative orbits: the motion of the planet, now relative to the star rather than the barycentre, can be found by applying an acceleration to the system which cancels that of the star, viz. GM_p/r^2 where r is their instantaneous separation (Figure 2.3). Then

$$P^2 = \frac{4\pi^2}{G(M_\star + M_p)} a_{\text{rel}}^3, \quad (2.15)$$

where the coordinate origin is now the star, not the barycentre, and a_{rel} is the semi-major axis of the relative orbit, i.e. the planet around the star.⁴

⁴This measurement of *relative* separation does not arise for exoplanet orbits when the planet is unseen. It is, in contrast, a situation relevant for the relative astrometry of binary stars, where an orbit is measured as a separation and position angle of one star with respect to another; then, the combined system mass can be determined if P and a_{rel} are measurable, while individual masses can only be determined if the mass ratio can be established, either from the ratio of the distances from the barycentre, or the ratio of their speeds around it.

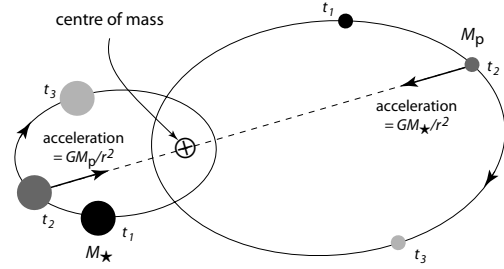


Figure 2.3: Two orbiting bodies, shown at times t_1, t_2, t_3 , move about their common barycentre. Both bodies follow orbits having the same shape and period, but of different sizes and with ω differing by 180° .

For $M_p \ll M_\star$ and in units of Earth’s orbit of 1 AU

$$P \approx 1 \text{ yr} \left(\frac{a_{\text{rel}}}{\text{AU}} \right)^{3/2} \left(\frac{M_\star}{M_\odot} \right)^{-1/2}. \quad (2.16)$$

(b) absolute orbits: the orbit of the star around the system (star–planet) barycentre is given by

$$P^2 = \frac{4\pi^2}{GM'} a_\star^3, \quad (2.17)$$

where

$$M' \equiv \frac{M_p^3}{(M_\star + M_p)^2}, \quad (2.18)$$

and $a_\star$ is the semi-major axis of the stellar orbit around the system barycentre. An equivalent expression gives the orbit of the planet around the system barycentre in terms of the semi-major axis of the planet around the system barycentre a_p .

It follows that the sizes of the three orbits are in proportion $a_\star : a_p : a_{\text{rel}} = M_p : M_\star : (M_\star + M_p)$, with $a_{\text{rel}} = a_\star + a_p$. Furthermore, $e_{\text{rel}} = e_\star = e_p$, $P_{\text{rel}} = P_\star = P_p$, the three orbits are coplanar, and the orientations of the two barycentric orbits (ω) differ by 180° .

Since the planet is assumed to be invisible (which is the case for essentially all observations at the present time), the orbital motion of the star around the system barycentre is only correctly determined by astrometry if its position is measured with respect to an ‘absolute’ (quasi-inertial) reference frame.

That all seven orbital elements are accessible to astrometric measurements for an arbitrary projection geometry essentially follows from two principal considerations: the star position versus time allows the maximum and minimum angular rates to be determined, and hence the position of the line of apsides. With the orientation of the major axis so established, appeal to Kepler’s second law fixes the orbit inclination.

Radial velocity measurements of the host star also give information on its barycentric orbital motion, although not all seven Keplerian orbital elements are accessible from the line-of-sight velocity variations alone.

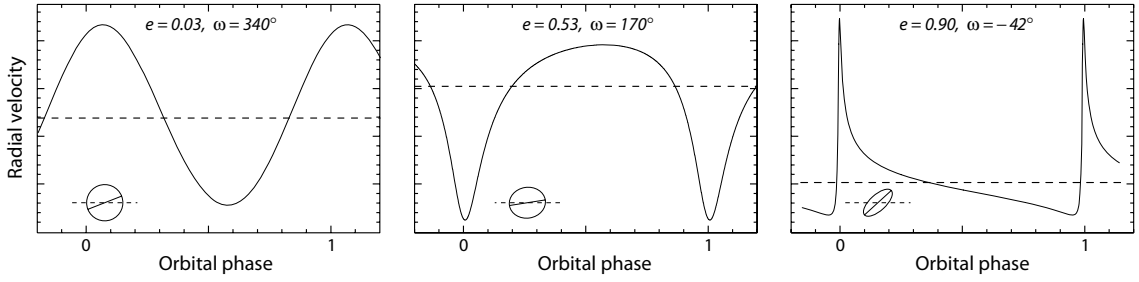


Figure 2.4: Example stellar radial velocity curves, illustrating their dependency on e and ω , for HD 73256 (Udry et al., 2003a, Figure 2), HD 142022 (Eggenberger et al., 2006, Figure 4), and HD 4113 (Tamuz et al., 2008, Figure 1) respectively. Horizontal dashed lines show the systemic velocity (radial velocity of the barycentre). The ellipses at lower left show the viewing geometries.

2.1.1 Orbits from radial velocity measurements

Radial velocity (Doppler) measurements describe the projected motion, along the line-of-sight, of the primary star as it orbits the system barycentre.

Radial velocity semi-amplitude With Figure 2.2 now considered as representing the orbit of the star around the barycentre, the star's z -coordinate along the line-of-sight can be derived from trigonometry

$$z = r(t) \sin i \sin(\omega + \nu), \quad (2.19)$$

where $r(t)$ is the distance from the barycentre. Then

$$v_r \equiv \dot{z} = \sin i [\dot{r} \sin(\omega + \nu) + r \dot{\nu} \cos(\omega + \nu)]. \quad (2.20)$$

Some algebraic substitutions for r and $\dot{r}$ lead to

$$v_r = K [\cos(\omega + \nu) + e \cos \omega], \quad (2.21)$$

where the *radial velocity semi-amplitude*⁵ is given by

$$K \equiv \frac{2\pi}{P} \frac{a_\star \sin i}{(1 - e^2)^{1/2}}. \quad (2.22)$$

From Equation 2.21, it can be seen that v_r varies around the orbit between limits of $K(1 + e \cos \omega)$ and $K(-1 + e \cos \omega)$. This expression for v_r as a function of the true anomaly $\nu(t)$ can be transformed into an expression for v_r as a function of time through Equations 2.6–2.10.

The shape of the radial velocity curve is determined by e and ω (Figure 2.4). Together with P , their combination constrains the value of $a_\star \sin i$ (Equation 2.22). But neither $a_\star$ nor $\sin i$ can be determined separately.

Two alternative expressions for K are instructive. Substituting Equations 2.17 and 2.18 into 2.22 gives

$$K^2 = \frac{G}{(1 - e^2)} \frac{1}{a_\star \sin i} \frac{M_p^3 \sin^3 i}{(M_\star + M_p)^2}. \quad (2.23)$$

⁵Often denoted as K_1 in binary star work to emphasise that it describes the motion of the primary, this amplitude of radial motion is analogous to the projected semi-major axis measured astrometrically.

With the product $a_\star \sin i$ determined as above, it follows that radial velocity measurements provide a value for the *mass function* (i.e. Equation 2.18 modified by a dependency on the orbit inclination)

$$\mathcal{M} \equiv \frac{M_p^3 \sin^3 i}{(M_\star + M_p)^2}. \quad (2.24)$$

If $M_p \ll M_\star$, then Equation 2.24 reduces to

$$\mathcal{M} \approx \frac{M_p^3 \sin^3 i}{M_\star^2}. \quad (2.25)$$

Further, if $M_\star$ can be estimated from its spectral type and luminosity class (or otherwise), then $M_p \sin i$ can be determined. The mass of the planet nevertheless remains uncertain by the unknown factor $\sin i$.

Equations 2.17, 2.18 and 2.22 can be combined to give an alternative expression for K without the explicit appearance of $a_\star$ (e.g. Cumming et al., 1999, eqn 1)

$$K = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{M_p \sin i}{(M_\star + M_p)^{2/3}} \frac{1}{(1 - e^2)^{1/2}}, \quad (2.26)$$

which can also be written (Torres et al., 2008, eqn 1)

$$\frac{M_p \sin i}{M_J} = 4.919 \times 10^{-3} \left(\frac{K}{\text{km s}^{-1}} \right) (1 - e^2)^{1/2} \left(\frac{P}{\text{days}} \right)^{1/3} \left(\frac{M_\star + M_p}{M_\odot} \right)^{2/3} \quad (2.27)$$

For a circular orbit with $M_p \ll M_\star$, for example, the stellar velocity variations are sinusoidal with amplitude

$$K = 28.4 \text{ m s}^{-1} \left(\frac{P}{1 \text{ yr}} \right)^{-1/3} \left(\frac{M_p \sin i}{M_J} \right) \left(\frac{M_\star}{M_\odot} \right)^{-2/3}. \quad (2.28)$$

For Jupiter around the Sun ($a = 5.2 \text{ AU}$, $P = 11.9 \text{ yr}$) $K_J = 12.5 \text{ m s}^{-1}$. For Earth, $K_\oplus = 0.09 \text{ m s}^{-1}$.

Keplerian observables Of the seven elements describing an orbit in three-dimensions ($a, e, P, t_p, i, \Omega, \omega$), Ω cannot be determined from radial velocity measurements. Furthermore, only the combination $a_\star \sin i$ is determined, with neither $a_\star$ nor $\sin i$ individually.

Radial velocity measurements alone do not provide individual planetary masses, their sum, or their ratio.⁶ Taken alone, small periodic radial velocity modulation is insufficient to infer, unambiguously, the presence of a companion of planetary mass. Without additional information, such as an astrometric orbit, a constraint on the orbit inclination i from photometric transits, spectroscopic line profiles, or statistical deconvolution (e.g. Jorissen et al., 2001), small values of K could indicate either a low-mass planet, or an object of significantly higher mass with small orbital inclination, i.e. with the plane of the orbit almost face-on to the line of sight.

Fitting a single planet There are, in consequence, five observables related to the star's Keplerian orbit which can be fit for each orbiting planet on the basis of radial velocity measurements alone: e , P , t_p , and ω , and the combination $K = f(a, e, P, i)$.

Two further terms are usually taken into account: a *systemic velocity* γ describing the constant component of the radial velocity of the system's centre of mass relative to the solar system barycentre, which may also include an instrument-dependent radial velocity offset; and a linear trend parameter, d , which may accommodate instrumental drifts as well as unidentified contributions from massive, long-period companions.

From Equation 2.21, the radial velocity signal of a star with an orbiting planet can then be expressed as a function of the true anomaly as (e.g. Wright & Howard, 2009, eqn 1)

$$v_r(t) = K [\cos(\omega + \nu(t)) + e \cos \omega] + \gamma + d(t - t_0). \quad (2.29)$$

Establishing the astrometric or radial velocity parameters for a single orbiting planet is typically based on the search of parameter space using χ^2 minimisation.

Good period estimates are invariably used to simplify the search, for example using the well-established Lomb–Scargle algorithm (Figure 2.5). This is a modified periodogram analysis, equivalent to least-squares fitting of sine waves, which targets the efficient and reliable detection of a periodic signal in the case of unevenly-spaced observation times and in the presence of noise (Scargle, 1982; Horne & Baliunas, 1986; Cumming, 2004; Baluev, 2008a). Zechmeister & Kürster (2009) give an analytic solution for the generalisation to a full sine wave fit, taking account of measurement errors and an offset. They include a specific algorithm to search for the period of the best-fit Keplerian orbit to radial velocity data.

⁶For double-lined spectroscopic binaries in which two distinct spectra can be measured (not the case for exoplanet orbits), the mass function for each component can be established separately. Then the component mass ratio can also be estimated, but still neither the separate masses, nor the orbit inclination. Further inferences can be made if the system is eclipsing (such that $i \approx 90^\circ$), or if the system can also be resolved astrometrically.

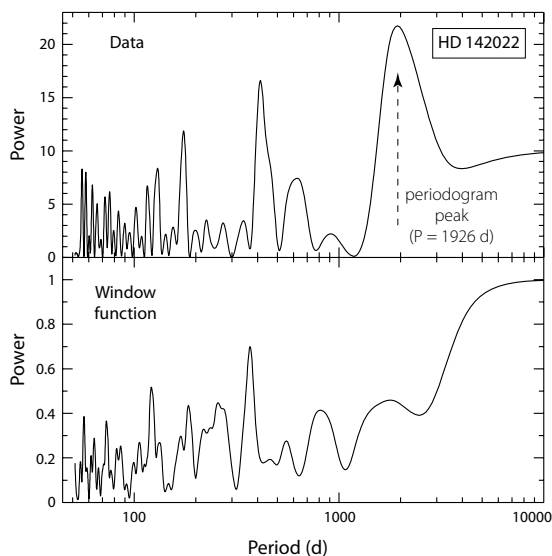


Figure 2.5: The Lomb–Scargle periodogram analysis applied to the radial velocity observations for HD 142022 obtained with the CORALIE échelle spectrograph, La Silla. The power is a measure of the statistical significance of the signal, not of its true amplitude. The window function, determined by the observation epochs, is shown below. The highest peak corresponds to $P = 1926$ d, while the final best-fit solution gives $P = 1928$ d. From Eggenberger et al. (2006, Figure 2), reproduced with permission © ESO.

Multiple planets: kinematic fitting For a system of n_p planets, the total radial velocity signal can be approximated as a linear sum over n_p terms of the form contained in Equation 2.29, giving a total of $5n_p + 1$ Keplerian parameters to be fit, including γ (and optionally d).

This first-order approach to fitting multiple systems considers that the reflex motions of the star caused by each planet are independent, i.e. ignoring the effects of planet–planet gravitational interactions. In this kinematic (or Keplerian) fitting, the dominant planet signal is identified, its Keplerian contribution subtracted from the observational data, and the process repeated until all significant planet signals have been accounted for. A more rigorous multi-planet χ^2 fit to the original data can then be made using these results as starting values. Such an approach was adopted for the five-planet fit to the 18 years of Doppler data for 55 Cnc, with weights assigned to account for signal-to-noise, photospheric motions, and instrument errors (Fischer et al., 2008).

Algorithmic implementation Because the equations describing an astrometric or radial velocity orbit are nonlinear, an unstructured search of parameter space for multiple planets may be computationally prohibitive, and with many false local χ^2 minima. More sophisticated search schemes are therefore desirable.

The Levenberg–Marquardt method is an efficient algorithm for finding a local χ^2 minimum for non-linear

models. It varies smoothly between the steepest descent method far from the minimum, and the inverse-Hessian method as the minimum is approached. It has been widely adopted for both radial velocity and astrometric orbit fitting. Implementations are given by Press et al. (2007, *mrqmin*), while an IDL implementation *MPFIT* (Markwardt, 2009) applied to radial velocity orbit fitting has been described by Wright & Howard (2009).

Due to sparse sampling, measurement errors, parameter degeneracy and model limitations, there are frequently no unique values of the basic model parameters such as period and eccentricity, and a Bayesian approach can provide more robust estimates of parameter uncertainties (e.g. Brown, 2004; Cumming & Dragomir, 2010). The Markov Chain Monte Carlo method (MCMC) has been used for orbit fitting (e.g. Ford, 2004b; Gregory, 2005; Ford, 2006b; Gregory, 2007a,b), and an implementation specifically tailored to exoplanet radial velocity fitting for a one- or two-planet system, *EXO-FIT*, has been made available by Balan & Lahav (2009).

Various searches have employed genetic algorithms (e.g. Goldberg, 1989; Charbonneau, 1995) to explore the global parameter space, for example in the case of ν And (Butler et al., 1999; Stepinski et al., 2000), GJ 876 (Laughlin & Chambers, 2001), 55 Cnc (Marcy et al., 2002), HD 12661 (Goździewski & Maciejewski, 2003, *PIKAIA*), and μ Ara (Pepe et al., 2007, *Stakanof*). According to Stepinski et al. (2000) and Goździewski & Maciejewski (2003), the method is inefficient in pin-pointing very accurate best-fit solutions, but provides good starting points for more precise gradient methods such as Levenberg–Marquardt.

Linearisation An alternative to least-squares fitting of Keplerian motions to the radial velocity measurements is to seek linear parameters which can be constructed from certain of the orbital elements.

Konacki & Maciejewski (1999) described a harmonic component analysis which they applied to 16 Cyg. This is based on a Fourier expansion of the Keplerian motion in which the harmonic coefficients are functions of all orbital elements. The coefficients, obtained by linear least-squares, are then used as starting estimates for a local minimisation of the non-linear problem.

Wright & Howard (2009) formulated a linearisation which reduces the search space for n_p planets from the $5n_p + 1$ Keplerian parameters to $3n_p$ non-linear variables corresponding to (e, P, t_p) for each planet. In this approach Equation 2.29, generalised to the case of n_p planets, is re-cast as (Wright & Howard, 2009, eqn 4–7)

$$v_r(t) = \sum_{j=1}^{n_p} \left[h_j \cos v_j(t) + c_j \sin v_j(t) \right] + v_0 + d(t - t_0), \quad (2.30)$$

where

$$h_j = K_j \cos \omega_j,$$

$$c_j = -K_j \sin \omega_j,$$

$$v_0 = \gamma + \sum_{j=1}^n K_j e_j \cos \omega_j, \quad (2.31)$$

with K_j denoting the semi-amplitudes for each planet.

The non-linear terms (e, P, t_p) are then searched for algorithmically (e.g. using the Levenberg–Marquardt or Markov Chain Monte Carlo methods), with the linear parameters (h, c, v_0) , which are transformable back to (ω, K, γ) , solved for analytically at each search step.

Multiple planets: dynamical fitting In numerous multiplanet systems, gravitational interaction between two or more planets further modifies the total radial velocity or astrometric signature as a function of time. Such interaction can result in detectable variations in the planetary orbits even over short intervals measured in years. A more complete dynamical (or Newtonian) fit involving N-body integrations must then be used.

Various self-consistent algorithms, which incorporate mutual perturbations in fitting the radial velocity data, were developed for the first resonantly interacting system GJ 876, resulting in substantially improved fits to the radial velocity data (Laughlin & Chambers, 2001; Rivera & Lissauer, 2001; Nauenberg, 2002a; Lee & Peale, 2002). The latter was verified against full hydrodynamic evolution of embedded planets by Kley et al. (2004).

Even good dynamical fits can result in orbital parameter solutions which might be stable over years or decades, but unstable on time scales comparable to the age of the planetary system. Since such formal solutions may be considered as implausible, short-term and long-term dynamical stability is a further constraint that must be satisfied by multiple planet orbit fitting. The issue is considered further in §2.6.

Detectability and selection effects Various studies have been made of planet detectability from radial velocity data as a function of period and signal amplitude (e.g. Nelson & Angel, 1998; Eisner & Kulkarni, 2001b; Cumming, 2004; Narayan et al., 2005).

Cumming (2004) provides analytic expressions for planet detectability as a function of period and eccentricity, and some insight into the detection limits. For short orbital periods, $P \leq T$ where T is the duration of the observations, the radial velocity semi-amplitude threshold alone characterises detectability. At long periods, $P \geq T$, the observations cover only part of the orbit by definition, and detectability depends on which part of the orbit is being sampled. If the orbit is close to a velocity maximum/minimum, or to a zero crossing, velocity variations are ‘sine-like’ or ‘cosine-like’ respectively, and the velocity variations are

$$\Delta v = K \sin \left(\frac{2\pi T}{P} \right) \approx K \left(\frac{2\pi T}{P} \right), \quad \text{or} \quad (2.32)$$

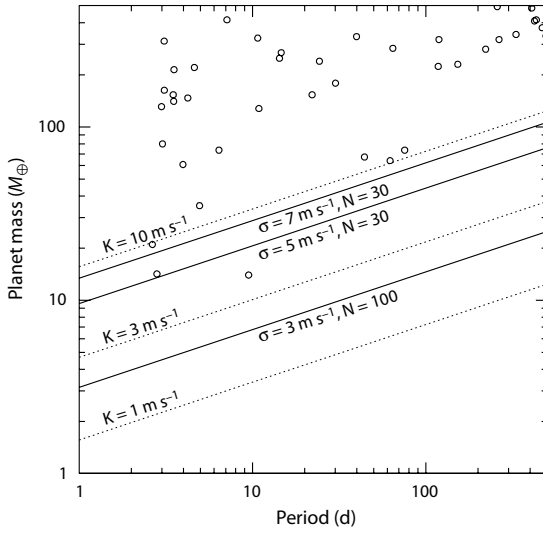


Figure 2.6: Minimum mass limits for a 50% detection threshold as a function of the number of observations, N , and the combined error (measurement and stellar jitter) σ , for $M_\star = 1M_\odot$ (Equation 2.34). The observation duration is assumed longer than the orbital period. Circles show those planets known in 2005. Dotted lines show various radial velocity semi-amplitudes, K . From Narayan et al. (2005, Figure 7), reproduced by permission of the AAS.

$$\Delta v = K \cos\left(\frac{2\pi T}{P}\right) \approx \frac{K}{2} \left(\frac{2\pi T}{P}\right)^2 \quad (2.23)$$

respectively. Averaging over phase introduces a dependency on the adopted detection efficiency, ϵ_D : for $\epsilon_D = 0.5$, the amplitude must be large enough that sine-like phases are detected, but cosine-like phases do not have to be, and the velocity threshold scales as $K \propto P$. For $\epsilon_D = 0.99$, almost all phases must be detected, requiring a large amplitude, which then scales as $K \propto P^2$.

While the dependence of K on e given by Equation 2.26 results in large K for highly eccentric orbits, in practice eccentricity acts to make detection more difficult at short periods, where an uneven sampling often results in poor phase coverage during rapid pericentre passages. At longer periods, the increased velocity amplitude and acceleration near the pericentre increase detectability. The transition to the long-period regime occurs for orbital periods $P \approx T/(1-e)^2$. The analysis also allows the completeness of existing surveys to be assessed (Cumming et al., 2008). The results emphasise that there remains a significant selection effect against detecting eccentric orbits for $e \geq 0.6$.

Shen & Turner (2008) found that, once a planet is detected, the eccentricities derived from Keplerian fitting are biased upwards for low signal-to-noise and moderate numbers of observations. They suggest that the numbers of exoplanets with low-eccentricity may be underestimated in current samples. The effect may be ev-

ident in results from Keck, for example, where for $P > 10$ d the observed eccentricity distribution is nearly flat for large-amplitude systems, $K > 80 \text{ m s}^{-1}$, but rises linearly towards low eccentricity at lower amplitudes, $K > 20 \text{ m s}^{-1}$ (Valenti et al., 2009).

Rodigas & Hinz (2009) showed that there is an additional bias due to the presence of an undetected outer companion. For moderate eccentricity $0.1 < e < 0.3$, for example, there is a 13% probability that a modeled eccentric orbit is in fact circular, with the model fit confused by the undetected outer companion.

Narayan et al. (2005) evaluated the detection probability for low mass planets, either in isolation or near mean motion resonance with a hot Jupiter. For a 50% detection rate their expression for the minimum detectable mass is

$$M_{p,\min} \sim 4M_\oplus \left(\frac{N}{20}\right)^{-\frac{1}{2}} \left(\frac{\sigma}{1 \text{ m s}^{-1}}\right) \left(\frac{P}{1 \text{ d}}\right)^{\frac{1}{3}} \left(\frac{M_\star}{M_\odot}\right)^{\frac{2}{3}}, \quad (2.34)$$

for $N \geq 20$, where N is the number of observations, and σ is the quadratic sum of the Doppler velocity measurement error and stellar jitter (Figure 2.6).

Scheduling In a study for the Space Interferometric Mission SIM, Ford (2004a) considered several non-adaptive observing schedules (i.e. schedules fully defined *a priori*) for a targeted astrometric planet search. These included time intervals which are regular periodic (at constant spacing); Golomb ruler⁷; regular power law; regular logarithmic; regular geometric; and random uniform. The efficiency for planet searches was found to be relatively insensitive to the actual observing schedule.

Adaptive scheduling, in which information from previous observations is used to plan future ones most efficiently, can optimise their information content, and can perform significantly better. The objective is to predict future epochs which yield, for example, maximum improvement in orbit parameters and planet masses, the most favourable epochs for distinguishing between alternative models, or an optimum series of future observations. For multiple planet systems, and in particular for resonant orbits, such optimal timings may be concentrated in rather narrow time intervals.

Loredo (2004) outlined a maximum entropy sampling strategy for determining an exoplanet orbit based on Bayesian inference, and Ford (2008) developed a more rigorous adaptive scheduling algorithm exploiting information theory. Based on a small number of initial observations, this proceeds by assuming a prior

⁷A Golomb ruler has unique integer intervals between each pair of marks, thus providing the maximum number of unique baselines. The number of marks defines its order; it is 'perfect' if it includes all distances up to its length, and 'optimal' if no shorter ruler of that exists. Thus the (optimal and perfect) Golomb ruler of order 4 and length 6 has marks at 0, 1, 4, 6.

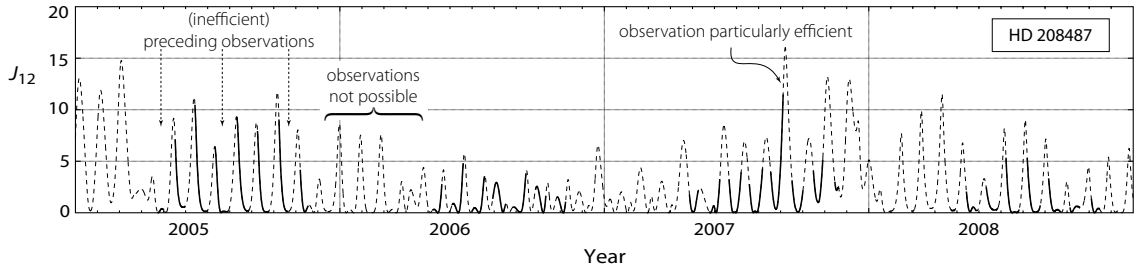


Figure 2.7: Scheduling future radial velocity observations to discriminate between two orbit models for HD 208487. The parts of the curve shown dashed correspond to times when observations are impossible. Observations are optimal when scheduled to coincide with the information peaks J_{12} . Arrows mark the three preceding radial velocity measurements, which in retrospect are seen to have been of little additional information value. From Baluev (2008b, Figure 2), © John Wiley & Sons, Inc.

for the distribution of orbital periods and masses, using Bayesian inference to calculate the posterior probability distribution for all model parameters, using this to calculate the predictive distribution for the radial velocities at some future time, and finally choosing observing times at which additional observations would be most valuable. Monte Carlo simulations demonstrated that such searches are more efficient in terms of detection, and can measure orbital parameters more accurately, than non-adaptive algorithms.

Baluev (2008b, 2010) used criteria from optimal design theory (based on Fisher information, Shannon information, and Kullback–Leibler divergence) to design similar general scheduling formalisms. As an example, Figure 2.7 illustrates the problem of deciding between two orbital solutions based on the 35 measurements of HD 208487 from Butler et al. (2006). Wright et al. (2007) identified periodicity attributable to an additional planet at either 28.6 or 900 d, but their data was inadequate to rule between them due to aliasing.

If the probability densities $p_{1,2}(v)$ describe the distribution of the radial velocity predictions for two possible orbit fits, the expectation of the likelihood ratio statistic considering the first or the second model as true are given by (Baluev, 2008b, eqn 3)

$$I_{2|1} = \int p_1(v) \ln \frac{p_1(v)}{p_2(v)} dv, \quad I_{1|2} = \int p_2(v) \ln \frac{p_2(v)}{p_1(v)} dv \quad (2.35)$$

The maximum information is given by (Baluev, 2008b, eqn 11)

$$J_{12} = I_{2|1} + I_{1|2} \\ = -1 + \frac{1}{2} \left(\frac{\sigma_1^2}{\sigma_2^2} + \frac{\sigma_2^2}{\sigma_1^2} \right) + \left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} \right) \frac{(v_1 - v_2)^2}{2}, \quad (2.36)$$

under the assumption that the distributions of v_i are close to Gaussian. The largest values of J_{12} then correspond to the most promising times for future observations to rule out one of the alternative models. Although Equation 2.36 is derived by Baluev (2008b) from a formal consideration of information content, its interpretation

is straightforward: observations should be made when the two models imply the largest predicted differences in the measured radial velocity, while the uncertainties of these predictions, σ , should be small enough to avoid statistically insignificant differences.

In the case of HD 208487 (Figure 2.7) the radial velocity predictions for the two different orbit models differ by up to 20 m s^{-1} . The function J_{12} identifies the epochs which maximise the discrimination between them. It turns out that a single observation at one of the peaks during the 2005 observing season could have decided between the competing models at the $\geq 2\sigma$ level, while the actual observations (indicated by arrows) fell at epochs of low information content, and added little new knowledge to the contested two-planet solution. The subsequent season (2006) offered uniformly limited opportunity for further discrimination.

2.2 Measurement principles and accuracies

2.2.1 Introduction

An instantaneous measurement of the stellar radial velocity about the star–planet barycentre is given by the small, systematic Doppler shift in wavelength of the many absorption lines that make up the star's spectrum.

If, in the observer's reference frame, the source is receding with velocity v at an angle θ relative to the direction from observer to source, the change in wavelength

$$\Delta\lambda = \lambda_{\text{obs}} - \lambda_{\text{em}}, \quad (2.37)$$

is related to the velocity by the expression for the relativistic Doppler shift (e.g. Lang, 1980, eqn 2–226)

$$\lambda_{\text{obs}} = \lambda_{\text{em}} \frac{(1 + \beta \cos \theta)}{(1 - \beta^2)^{1/2}}, \quad (2.38)$$

where λ_{obs} , λ_{em} are observed and emitted wavelengths, and $\beta = (v/c)$. For $v \ll c$ and $\theta \ll \pi/2$, the expression reduces to the classical form

$$v_r = v \cos \theta \approx \left(\frac{\Delta\lambda}{\lambda_{\text{em}}} \right) c, \quad (2.39)$$

where, conventionally, positive values indicate recession. Special relativistic terms correspond to changes in v_r of several m s^{-1} , and are therefore significant. Equation 2.38 omits the effect of the refractive index of air at the spectrograph, n_{air} (1.000277 at standard temperature and pressure, STP), which introduces errors of $\leq 1 \text{ m s}^{-1}$ (Marcy & Butler, 1992, eqn 3).

With the radial velocity semi-amplitude values of $K_j \approx 12.5 \text{ m s}^{-1}$ and $K_{\oplus} \approx 0.09 \text{ m s}^{-1}$ as indicative goals, the detection of planets around solar-type stars has demanded long-term radial velocity accuracies of some 15 m s^{-1} or preferably significantly better, corresponding to an accuracy of a few parts in 10^8 in wavelength. This must be maintained over months or years.

High-accuracy radial velocities for exoplanet detection are typically acquired using échelle spectrographs with high spectral resolving power (typically $R \equiv \lambda/\Delta\lambda \sim 50\,000 - 100\,000$), and operated in the optical region (450–700 nm). Many diffraction orders are cross-dispersed, and recorded simultaneously on rectangular format CCDs providing large numbers of resolved absorption lines (Figure 2.8). The principles of an échelle spectrograph are described by, e.g., Vogt (1987) in the case of the Lick Observatory Hamilton spectrometer, by Vogt et al. (1994) in the case of Keck-HIRES, and by Baranne et al. (1996) in the case of ELODIE.

High instrumental stability and accurate wavelength calibration is demanded to minimise effects of gravitational and thermal telescope flexure, and other instrument drifts. Large telescopes and long integration times are still required to achieve the necessary high signal-to-noise, and corresponding sub-pixel accuracies.

Ultra-high spectral resolution Throughput is an important requirement, and ultra-high resolution spectrographs (e.g. the $R = 10^6$ AAT-UHRE, Diego et al. 1995; or the $R = 600\,000$ échelle at Steward Observatory, Ge et al. 2002a) do not provide the efficiencies of $\geq 10\%$ needed for competitive planet searches.

2.2.2 Cross-correlation spectroscopy

Information about the instantaneous Doppler shift is contained in the many thousands of absorption lines present in the high-resolution optical spectrum of solar-type stars. This information can be concentrated into a few parameters by cross-correlation, even at low signal-to-noise levels (Figure 2.9). The technique, using a physical mask, was proposed by Fellgett (1955), first demonstrated by Griffin (1967), and extended to échelle spectroscopy using CORAVEL by Baranne et al. (1979).⁸

⁸As Fellgett phrased it, 'It is uneconomical, both in telescope time and in labour of reduction, to observe the details of a stellar spectrum if the sole object of the observation is to measure a radial velocity.'

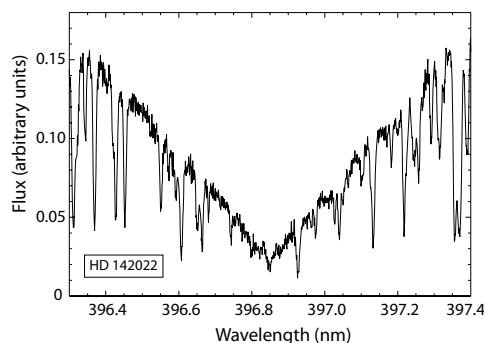


Figure 2.8: The Ca II H absorption line region of the summed HARPS spectra for HD 142022. The figure illustrates the density of absorption lines in 1 nm of the entire 378–691 nm échelle spectrum. From Eggenberger et al. (2006, Figure 1), reproduced with permission © ESO.

Limitations of a fixed physical template, in accuracy and adaptability to spectral type, led to the cross-correlation subsequently implemented numerically as a box-shaped template (Queloz, 1995; Baranne et al., 1996). Finer details of the method have been developed progressively (e.g. Simkin, 1974; Sargent et al., 1977; Tonry & Davis, 1979; Bender, 1990). The essentials are to determine the value of ϵ minimising (Queloz, 1995)

$$C(\epsilon) \propto \int_{-\infty}^{+\infty} S(v) M(v - \epsilon) dv, \quad (2.40)$$

where S is the spectrum and M is the mask, both expressed in velocity space v . Associated errors are established from Monte Carlo modeling. Weighting according to the relative line depths further optimises the signal-to-noise, and can also reduce the perturbing effects of telluric lines (Pepe et al., 2002).

The precise shape of the resulting *cross-correlation function* depends on the intrinsic spectral line shapes and on the template line widths, and overall represents a mean profile of all lines in the template. Accordingly, in addition to the radial velocity, the width yields the stellar rotational velocity $v \sin i$, while the equivalent width provides a metallicity estimate if T_{eff} is known approximately (Mayor, 1980; Benz & Mayor, 1981; Queloz, 1995). In the absence of systematic line asymmetry, the underlying shape of the cross-correlation function is well approximated by a Gaussian, with asymmetry reflecting systematic structure in the individual spectral lines.

2.2.3 Deriving radial velocities from Doppler shifts

The measured Doppler shift, Equation 2.37, in practice includes effects other than the line-of-sight velocity of the target star's centre of mass. Contributions from the motion of the observer around the solar system barycentre, due to Earth rotation and orbital motion, are time varying, significant at levels of up to 0.5 and 30 km s^{-1} respectively, and must be accounted for.

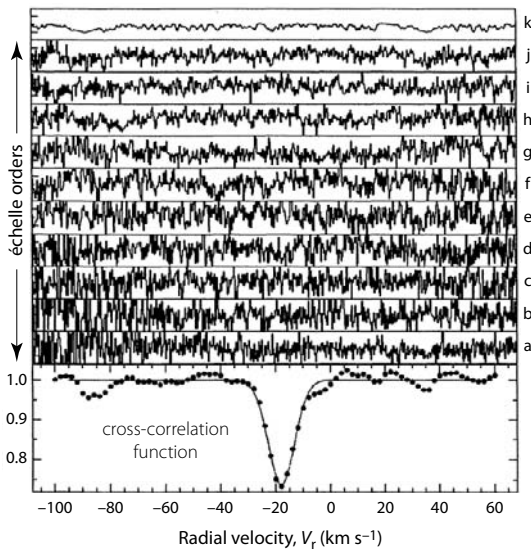


Figure 2.9: Example cross-correlation function for a K0 III star with $S/N \approx 1$. Observations at $R = 40000$ span 411–444 nm in ten orders, each covering 4 nm. Some 1000 lines match the template. The top trace is the same order as that immediately below, but at $S/N = 40$. The cross-correlation function is shown at bottom. From Queloz (1995, Figure 2).

Higher-order relativistic and secular projection effects, and additional spectroscopic line shifts introduced by gravitational redshift and stellar surface effects such as convective flow, pulsations, and star spots (Dravins, 1975; Nidever et al., 2002; Lindegren & Dravins, 2003) may be less critical in determining the relative radial velocities central to exoplanet detection. But there are situations where their contributions must be considered, and all must be accounted for in any *absolute* determination of radial space motion at the m s^{-1} level. The magnitude of these effects is as follows.

Effects of Earth motion To detect changes in a star's radial velocity due to an orbiting planet, measured velocities must be referred to a non-moving frame, or at least one with a constant rectilinear space motion, and the solar system barycentre (as opposed to the heliocentre) is consequently adopted as reference.

The time-varying motion of the Earth around the barycentre, which includes effects due to gravitational perturbations from the other planets, is described by the solar system ephemerides provided by JPL (e.g. Konopliv et al., 2006) and IMCCE (Fienga et al., 2008, 2009). By adjusting for known effects, residual velocity terms can be brought below the 1 m s^{-1} level (cf. Stumpff, 1980).

Line shifts The contribution from the star's *gravitational redshift* is given by (e.g. Misner et al. 1973, eqn 25.26N; Lang 1980, eqn 2–234)

$$v_r \approx \frac{GM_\star}{R_\star c} . \quad (2.41)$$

This is valid in the Newtonian limit, $R_\star \gg R_S$, where the Schwarzschild radius $R_S \equiv 2GM_\star/c^2$. The contribution amounts to 636 m s^{-1} for the Sun, and ranges from $\sim 680 \text{ m s}^{-1}$ at F5V to $\sim 500 \text{ m s}^{-1}$ at M5V, and several tens of km s^{-1} for white dwarfs.

Convective motions in the photospheres of cool stars lead to spectral line asymmetries (§2.2.7). A net blueshift results from the contribution of photons from the larger and brighter photospheric granules compared with the downward motion in the darker and cooler intergranular lanes. This effect is of order -0.5 km s^{-1} for the Sun, and ranges from about -1000 m s^{-1} at F5V to about -200 m s^{-1} at K0V (Dravins et al., 1981). Effects due to pressure broadening are below about 100 m s^{-1} for main sequence stars (Dravins, 1999).

Stellar rotation imposes small radial velocity effects (Gray, 1999), while the contribution of variable meridional flows may be comparable (Beckers, 2007). More critical for high-accuracy relative velocities are effects of surface features such as star spots, which can induce a radial velocity amplitude of a few m s^{-1} (e.g. Saar & Donahue, 1997; Hatzes, 2002), resulting in periodic modulation at the stellar rotation period (see §2.2.6).

Stellar space motion Effects of a star's radial motion on the secular evolution of its measured parallax and proper motion are treated by Dravins et al. (1999), and are encapsulated in Equations 3.13. Equivalently, a constant space motion results in a changing radial velocity with time due to the changing projection geometry. For Barnard's star, observed with VLT–UVES over five years (Figure 2.10), Kürster et al. (2006) measured a secular radial velocity acceleration consistent with the predicted value of $4.50 \text{ m s}^{-1} \text{ yr}^{-1}$ based on the Hipparcos proper motion and parallax combined with the absolute radial velocity of -110.5 km s^{-1} from Nidever et al. (2002).

With knowledge of the star's parallax ϖ , proper motion μ , and systemic radial velocity γ , it is possible to predict this contribution to any secular evolution of the γ velocity which might otherwise be attributed to long-period planetary orbits.

Zero point As a result of these astrophysical as well as instrumental effects, establishing the zero point for absolute radial motions at $\lesssim 50 \text{ m s}^{-1}$ remains notoriously difficult. Nidever et al. (2002) determined barycentric radial velocities for 889 late-type stars observed at Keck and Lick with typical accuracies of 0.3 km s^{-1} , and found a difference in zero point of 53 m s^{-1} compared with the system of the 38 stable FGK stars adopted as standards by the Geneva group (Udry et al., 1999b,a).

2.2.4 Wavelength calibration

Accurate wavelength calibration is a prerequisite for reaching high radial velocity accuracy. In the early 1970s,

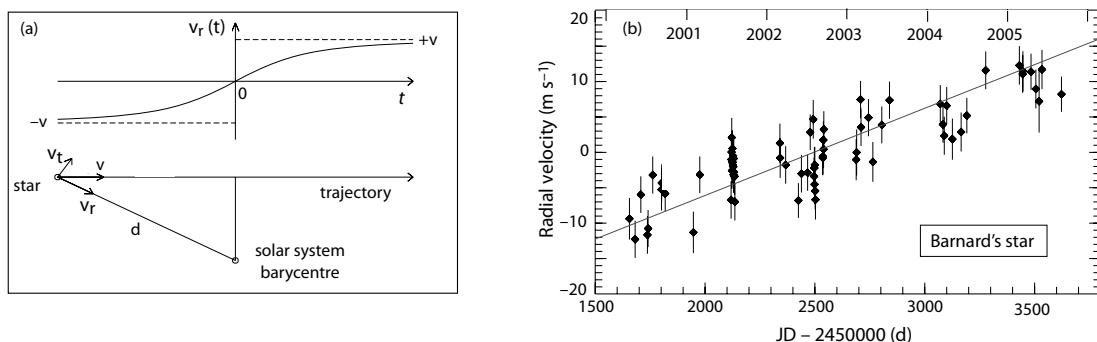


Figure 2.10: (a, bottom) geometry of the space motion of a nearby star: v_t and v_r are, respectively, the transverse and radial components of the space velocity v , and d is the current distance; (a, top) absolute radial velocity as a function of time and its asymptotic limits $\pm v$. (b) barycentric radial velocity measurements of Barnard's star over five years, compared with the predicted secular acceleration (solid line). Adapted from Kürster et al. (2003, Figure 1) and Kürster et al. (2006, Figure 3).

accuracies of around 1 km s^{-1} were limited by photographic plate technology, and by guiding errors at the spectrograph slit which introduced shifts in the stellar spectrum relative to comparison arc lines.

Wavelength calibration was significantly improved by the use of telluric (atmospheric) water vapour lines (Griffin, 1973; Griffin & Griffin, 1973; Walker et al., 1973; Gray & Brown, 2006), largely eliminating errors caused by the different optical paths of the stellar beam and the calibration lamp. Disadvantages include limited spectral ranges where telluric lines of suitable strength are found, in addition to various systematic errors arising from variable path length as a function of zenith distance, producing variations of 20 m s^{-1} per air mass, and from significant winds, producing systematic shifts of 20 m s^{-1} (Gray & Brown, 2006).

Gas cells The use of captive gases to provide a dense and accurate wavelength reference, superimposed on the stellar spectral lines, started with the use of hydrogen fluoride (HF). Although toxic and corrosive, its 3–0 vibration band gave a well-spaced line distribution, with no isotopic confusion, and of similar natural width to those in typical stellar spectra (Campbell & Walker, 1979).

An alternative, iodine (I_2), also mononuclidic, was used by Beckers (1976) and Koch & Woehl (1984) for solar observations, and later by Marcy & Butler (1992) for their precision radial velocity programme at the Lick Observatory 3-m telescope (a retrospective is given by Beckers, 2005). It has a strong line absorption coefficient, and requires a path length of only a few cm. Accuracies improved accordingly, to around 25 m s^{-1} by the early 1990s, and to some 3 m s^{-1} just a few years later (Butler et al., 1996).

The gas cell is placed in the telescope light path, just before the spectrograph slit. Sharp absorption lines of known wavelength are superimposed on the stellar spectrum, thereby providing calibration of both wavelength and the spectrograph point-spread function. For

an échelle, the latter is complex with many degrees of freedom (Valenti et al., 1995).

The spectrum through the absorption cell is then modeled as (Marcy & Butler, 1992, and their figure 1)

$$I_{\text{obs}}(\lambda) \propto [I_{\star}(\lambda + \Delta\lambda_{\star}) T_{\text{I}_2}(\lambda + \Delta\lambda_{\text{I}_2})] \otimes \text{PSF}, \quad (2.42)$$

where $\Delta\lambda_{\star}$ and $\Delta\lambda_{\text{I}_2}$ are the shifts of the star spectrum and iodine transmission function, determined by least-squares fitting to the composite spectrum, I_{obs} ; and $\otimes$ PSF represents convolution with the spectrograph point-spread function. $I_{\star}$, via $I_{\star} \otimes \text{PSF}$, is obtained *a priori* by observing each star without the iodine cell in place. The iodine cell transmission function, T_{I_2} , is obtained from external measurements using a high-resolution Fourier transform spectrometer. The final corrected Doppler shift is then given by $\Delta\lambda = \Delta\lambda_{\star} - \Delta\lambda_{\text{I}_2}$.

Modified calibration sequences have been developed. Iterating from an existing spectral template of a similar star can eliminate the need for an observed stellar template spectrum (Johnson et al., 2006c). Modified calibration sequences to measure binary stars, and in particular double-lined spectroscopic binaries, are described by Konacki (2005b).

The iodine gas cell is now standard in many planet search instruments (including AAT-UCLES, HET-HRS, Keck-HIRES, Lick 3 m, Magellan-MIKE, OAO-HIDES, Subaru-HDS, Tautenburg, TNG-SARG, and VLT-UVES), being of particular relevance for spectrographs which are not intrinsically highly stabilised.

The advantages of the iodine absorption cell are the large number (several thousand) of absorption lines, and the common path of the starlight and the iodine absorption. The measured iodine lines therefore simultaneously track changes in the instrument point-spread function. The disadvantages are the 20–30% loss of light, and the clustering of the absorption line bands in the visible (500–620 nm), making wavelength calibration for redder M dwarfs more problematic.

Thorium–argon calibration HARPS (Mayor et al., 2003), and its predecessor ELODIE (Baranne et al., 1996) employ a thorium–argon emission lamp as the reference wavelength spectrum. In its practical implementation, two optical fibres are used to transfer light to the spectrograph, one collecting the stellar light, the other simultaneously recording either a thorium–argon reference spectrum, or the background sky.

The advantages of the thorium–argon lamp for wavelength calibration are the large numbers of strong emission lines over a wide optical to infrared range (Palmer & Engleman, 1983; Hinkle et al., 2001; Lovis & Pepe, 2007), and the improvement in throughput due to the absence of the iodine absorption cell.

Fibre-optic image scrambling When implemented, a fibre feed from telescope focus to spectrograph offers two further advantages. Firstly, it addresses the variable light illumination of the spectrograph slit, and the associated complications of wavelength calibration. This is because multiple internal reflections within the fibre cause scrambling of the image structure between input and output, decoupling the illumination of the spectrograph optics and detector from guiding errors and calibration source misalignments (Heacox, 1986; Hunter & Ramsey, 1992; Walker et al., 2003b). In addition, the spectrograph can be placed away from the telescope, in a mechanically and thermally stable environment.

The fibre-feed approach, combined with the use of the thorium–argon calibration lamp, was adopted for ELODIE at the OHP 1.9-m telescope (Baranne et al., 1996), and used for the detection of the first exoplanet discovered around a main sequence star, 51 Peg (Mayor & Queloz, 1995). The same concept was subsequently incorporated into HARPS.

Infrared wavelength calibration One issue that has limited high-accuracy radial velocities in the near-infrared is the lack of a suitable wavelength calibrator. Iodine and thorium–argon lines used in the visible, for example, are sparse in the near infrared.

Calibration using imprinted atmospheric lines reaches $10\text{--}20\text{ m s}^{-1}$. For the McMath Fourier transform spectrometer at Kitt Peak, Deming et al. (1987) found variations of order 20 m s^{-1} based on telluric methane. For the $R = 100\,000$ Nasmyth-mounted cryogenic échelle spectrograph CRIRES at the VLT, designed for high-resolution spectroscopy between $0.96\text{--}5.2\text{ }\mu\text{m}$, Seifahrt & Käufel (2008) reported telluric N_2O lines near $4.1\text{ }\mu\text{m}$ stable to around $10\text{--}20\text{ m s}^{-1}$. Both studies used external N_2O gas cells for wavelength calibration.

The use of near-infrared gas cells is in its infancy. Mahadevan & Ge (2009) considered a number of options, and concluded that $\text{H}^{13}\text{C}^{14}\text{N}$, $^{12}\text{C}_2\text{H}_2$, ^{12}CO , and ^{13}CO together could provide useful calibration in the $1.65\text{ }\mu\text{m}$ H band. Experiments using an NH_3 cell for VLT–

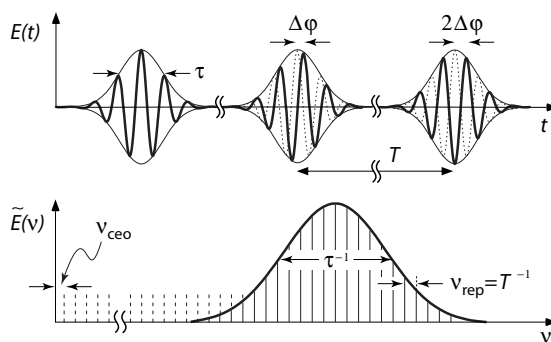


Figure 2.11: Principle of the laser frequency comb. A pulse train (top) produces the optical frequency ‘comb’ in Fourier space (bottom), with shorter pulse envelopes τ producing broader frequency combs. Within the laser cavity, the round-trip pulse time, T , determines the repetition frequency, $\nu_{\text{rep}} = 1/T$. Dispersive elements cause a difference in the group and phase velocities, shifting the carrier wave with respect to the envelope (top) by $\Delta\phi$ per pulse. In the frequency domain, the comb shifts by $\nu_{\text{ceo}} = \Delta\phi/2\pi T$. From Murphy et al. (2007, Figure 1), © John Wiley & Sons, Inc.

CRIRES in the $2.2\text{ }\mu\text{m}$ K band reach a precision of $3\text{--}5\text{ m s}^{-1}$ over weeks or months (Bean et al., 2010a).

Laser frequency combs The I_2 and Th–Ar wavelength calibration sources depart somewhat from an ideal radial velocity standard. A more robust calibrator would cover the entire optical/infrared range with individually unresolved lines, of uniform spacing and intensity, and with accurately known wavelengths determined by fundamental physics. Laser frequency ‘combs’, generated from mode-locked femtosec-pulsed lasers (Reichert et al., 1999; Jones et al., 2000a; Udem et al., 2002), seem to offer this possibility (Figure 2.11).

Their application to the calibration of spectrographs for exoplanet detection and other applications, at around the 0.01 m s^{-1} level, has been evaluated (Murphy et al., 2007), with laboratory prototypes demonstrated (Steinmetz et al., 2008; Li et al., 2008a).

The principle is based on the storage of a single laser pulse, maintained on a repetitive path, and circulating in a cavity as a carrier wave. After each circuit, a copy of the pulse is emitted through an output mirror, resulting in an indefinite pulse train, while the energy lost is replenished by stimulated emission in the lasing medium.

The absolute frequencies of the comb lines are set by

$$\nu = \nu_{\text{ceo}} + n \nu_{\text{rep}}, \quad (2.43)$$

where $\nu_{\text{rep}} = 1/T$ is the repetition rate of the laser (with T being the round-trip travel time), ν_{ceo} is a carrier-envelope offset frequency which must be calibrated (resulting from non-commensurability of the carrier-wave and repetition frequencies), and n is an integer, with modes given by $n \sim 10^5\text{--}10^6$. Both ν_{rep} and ν_{ceo} can be synchronised by reference to atomic clocks.

Table 2.1: Estimated radial velocity jitter, σ'_{rv} , for various spectral types and luminosity classes.

Type	Jitter (m s^{-1})	Comment	Reference
FV	7–30	$\nu_{\text{rot}} = 8–10 \text{ km s}^{-1}$	Saar et al. (1998)
"	2–7	inactive stars	Wright (2005)
GV	20–45	$\nu_{\text{rot}} = 8–10 \text{ km s}^{-1}$	Saar et al. (1998)
GKV	4.6 ± 1.8	$\nu_{\text{rot}} \leq 2 \text{ km s}^{-1}$	Saar et al. (1998)
"	2–5	inactive stars	Wright (2005)
MV	2–7	inactive stars	Wright (2005)
GKIV	5–10	subgiants	Wright (2005)
KIII	20	giants	Hekker et al. (2006)

A laser frequency comb under development at CfA Harvard has been used to calibrate the TRES spectrograph at the Whipple Observatory, and is being further developed for use with HARPS-north (Li et al., 2008a; Latham, 2008b). The prototype uses a mode-locked titanium-sapphire laser operating at $\sim 800 \text{ nm}$, providing $\nu_{\text{rep}} \sim 1 \text{ GHz}$. A Fabry-Pérot cavity is introduced to increase the intrinsic line spacing to a more usable 40 GHz . Using GPS time referencing, the comb frequencies are expected to have long-term fractional stability and accuracy better than 10^{-12} , corresponding to a velocity variation below 0.01 m s^{-1} . By providing a common absolute frequency standard, external time referencing should also allow the direct comparison of measurements made at different observatories.

2.2.5 Exposure metering

Starting with ELODIE (Baranne et al., 1996), many instruments including Keck-HIRES, the Lick Observatory Hamilton Spectrometer, and the Lick Automated Planet Finder (APF) 2.4-m telescope, employ an integral exposure meter to monitor ongoing exposures. This is used to optimise exposure times for a required signal-to-noise, and to calculate the photon-weighted midpoint of each exposure for barycentric correction (Kibrick et al., 2006). HIRES, for example, uses a propellor mirror behind the spectrometer slit to extract a few percent of the light directed to a dedicated photomultiplier tube. APF employs a semi-transparent pellicle mirror, operating at 1 Hz , for autoguiding, exposure metering, and barycentric photon weighting.

2.2.6 Accuracy limits

Improvements in detector technology, wavelength calibration, and various other aspects of instrument design brought the state-of-the-art radial velocity accuracy to some $3\text{--}5 \text{ m s}^{-1}$ by the mid-1990s (e.g. Butler et al., 1996), and to around $0.3\text{--}0.5 \text{ m s}^{-1}$ today (Pepe & Lovis, 2008). Such accuracy represents a displacement of a few nm at the CCD detector, demanding a combination of optical and algorithmic techniques to maintain the required metrology over several years.

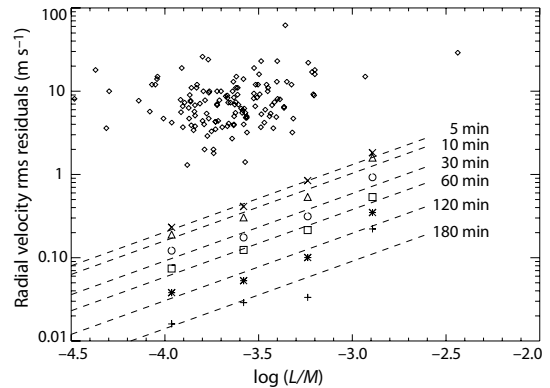


Figure 2.12: Oscillation jitter of AAT-UCLES asteroseismology targets as a function of $\log_{10}(L/M)$ (luminosity-mass ratio) for various simulated integration times. Power laws derived by O'Toole et al. (2008, their eqn 2) are shown as dashed lines. Residual rms values for the sample of known planets from Butler et al. (2006) are shown as small diamonds. From O'Toole et al. (2008, Figure 4), © John Wiley & Sons, Inc.

Although the radial velocity amplitude is independent of the distance to the star, signal-to-noise considerations limit observations to the brighter stars, typically $V \lesssim 8\text{--}10 \text{ mag}$. Equation 2.26 indicates that radial velocity measurements implicitly favour the detection of massive planets, those with small a (i.e. small P) and, with adequate temporal sampling, large e (but see §2.1.1 for detectability in practice).

Error sources presently imposing practical limits on achievable accuracy are of three main types: instrumental terms (currently dominated by a combination of mechanical and thermal stability, and wavelength calibration), stellar noise, and photon noise.

Stellar noise Activity in the stellar atmosphere, stellar oscillations, surface granulation, unrecognised planetary companions, and systematic errors may all contribute 'jitter' to the radial velocity measurements.

Jitter due to inhomogeneities of the stellar atmosphere (spots, plages) is often significant, frequently imposes a limit to the accuracy of radial velocity measurements, and is expected to vary on time scales comparable to the stellar rotation period. It is correlated with stellar chromospheric activity (e.g. Saar & Donahue, 1997; Saar et al., 1998; Butler & Marcy, 1998; Saar & Cuntz, 2001; Tinney et al., 2002; Shkolnik et al., 2005; Jenkins et al., 2006; Desort et al., 2007; Saar, 2009). Emission in the core of the Ca H/K lines is a useful proxy, although active stars seen pole-on can have a high activity index but low radial velocity jitter.

The effects of stellar oscillations on Doppler measurements are typically smaller than those produced by activity, and are most significant for giants and subgiants. O'Toole et al. (2008) made an investigation using data from asteroseismological investigations using AAT-

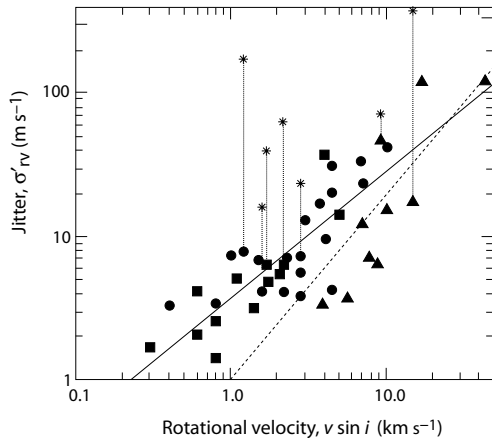


Figure 2.13: Early results from the Lick survey, showing the dependency of astrophysical jitter, σ'_{rv} (Equation 2.44) on stellar rotational velocity, $v \sin i$ for main sequence stars, indicated as F (triangles), G (circles) and K (squares). Stars with planets are plotted before (asterisks) and after (connected by vertical dashed lines) accounting for the planet's contribution. The power-law fits have exponent 0.9 for G and K stars (solid line), and 1.3 for F stars (dashed). From Saar et al. (1998, Figure 2a), reproduced by permission of the AAS.

UCLES. They found a power-law relation between measured noise and both the luminosity-to-mass ratio of the target stars and integration times (Figure 2.12), showing that it is advantageous to average over several p-mode oscillation frequencies (O'Toole et al., 2009b). HARPS results have also shown that 15-min integrations are sufficient to damp resulting radial velocity variations to below 0.2 m s^{-1} (Udry & Mayor, 2008).

Surface granulation can induce variability of order 1 m s^{-1} for solar type stars (Kjeldsen et al., 2005), which may demand measurements over several hours to damp.

The overall effects of jitter can be characterised as an excess in the radial velocity standard error as (Saar et al., 1998; Wright, 2005)

$$\sigma'_{rv} = (\sigma_{rv}^2 - \sigma_{\text{meas}}^2)^{1/2}, \quad (2.44)$$

where σ_{meas} is the contribution of the measurement error to the radial velocity standard error, σ_{rv} , for a given star. σ'_{rv} includes contributions from all of the effects noted above. Astrophysical contributions are dependent on a number of variables, including stellar rotational velocity, age, activity; some indicative estimates are given in Table 2.1 and Figure 2.13.

Photon noise Achievable radial velocity accuracies scale with signal-to-noise. For the HARPS instrument at the ESO 3.6-m telescope, a photon-limited precision of $\sim 1 \text{ m s}^{-1}$ is reached for a $V = 7.5$ mag G dwarf in around 60 s. In the absence of other noise sources, a single measurement accuracy of 0.1 m s^{-1} requires exposure times $\gtrsim 100$ min. Photon-noise limits for CORALIE and HARPS were given by Bouchy et al. (2001).

Earth-mass planets in the habitable zone The sun spot and plage properties of the Sun over one solar cycle, between 1993–2003, have been used to infer the radial velocity curve of a solar-type star exhibiting such features (Lagrange et al., 2010; Meunier et al., 2010). The radial velocity amplitude varies, in a complex way, from a few tenths up to $\sim 5 \text{ m s}^{-1}$. Assuming radial velocity accuracies in the range 0.01 – 0.1 m s^{-1} for a Sun-like star at 10 pc, detection of a $1M_{\oplus}$ planet in the habitable zone, between 0.8 – 1.2 AU , would require weekly sampling over several years, with frequent temporal sampling being crucial. Variations in the Sun's total solar irradiance during cycle 23, as measured by SOHO–VIRGO, is reported by Lanza et al. (2003).

2.2.7 Excluding other sources of periodicity

Effects other than an orbiting exoplanet can result in periodic stellar radial velocity variations, including most of the noise terms considered in the previous section.

Star effects Surface activity such as spots and plagues, as well as inhomogeneous convection, can lead to periodic radial velocity variations: a spot with a filling factor of a few percent can induce a radial velocity amplitude of a few m s^{-1} (Saar & Donahue, 1997). Activity may be discounted if the radial velocity period is distinct from the stellar rotation period, or if the star shows only very low level photometric variability at the radial velocity period (although spots are a time-dependent phenomenon). Spectroscopic measurements of the Ca II H and K lines provide a proxy for surface (magnetic) activity (Baliunas et al., 1995), and may reveal rotational modulation attributable to stellar rotation. For close-in planets, magnetic interactions may cause additional stellar activity, as in CoRoT-7 (Lanza et al., 2010).

Stellar pulsations may also cause radial velocity variations, but tend to be discounted if their period and amplitude do not correspond to known excitation mechanisms for the relevant spectral type.

Binary companions Stellar companions can mimic exoplanet signatures, and can be difficult to rule out (see, e.g., HD 41004: Santos et al., 2002b; Zucker et al., 2003, 2004). Hipparcos catalogue astrometry, both the double star annex and the goodness-of-fit statistics for the single-star model (catalogue fields H29–H30), provides a first level of discrimination.

The presence of a lower-level stellar spectrum can also be investigated using TODCOR, a cross-correlation algorithm developed to detect and measure radial velocities of two components of a spectroscopic binary (Mazeh & Zucker, 1994). Its application to HD 41004 is described by Zucker et al. (2003). An extension to triple stellar systems TRICOR/TRIMOR is described, and applied to HD 188753, by Mazeh et al. (2009b).

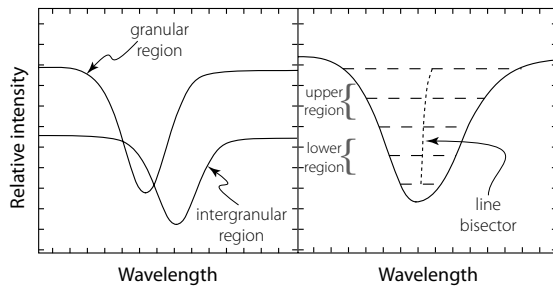


Figure 2.14: Schematic of line profile asymmetry caused by convection, and the resulting line bisector. Left: line profiles resolving individual convection cells would show blueshifted lines from the upflow regions, with redshifted lines from the intergranular downflow lanes. Right: the resulting line profile at low spatial resolution, with a consequently asymmetric line bisector. Adapted from Dravins et al. (1981, Figure 1), reproduced with permission © ESO.

A particularly confusing configuration may arise when a tight binary system orbits the target star. Schneider & Cabrera (2006) suggested that the binary wobble around its own centre of mass can result in the same modulation of the three basic observables (radial velocity, astrometric position, and time of arrival of a periodic signal) as for an orbiting planet. A more detailed analysis by Morais & Correia (2008) showed that the binary actually mimics two planets, rather than a single orbiting planet, but with very similar orbital periods.

Bisector analysis The *spectral line bisector* is the locus of median points midway between equal intensities on either side of a spectral line, thereby dividing it into two halves of equal equivalent width (Figure 2.14). Line bisectors are used to quantify stellar line profile asymmetries, and are important in identifying the underlying cause of certain types of radial velocity variation.

Since the work of Voigt (1956), line bisectors have been used to describe the nature and strength of convection in the Sun, and in star photospheres on the cool side of the granulation boundary where deep convective motion is significant (Dravins, 1975; Dravins et al., 1981; Gray, 1982, 1983; Nowak & Niedzielski, 2008). In such photospheres, local upward motion in the brighter (hotter) convective cells is balanced by a downflow in the darker (cooler) intergranular lanes. In the absence of spatial resolution, averaging over many granules, an overall convective blueshift, of order -0.5 km s^{-1} , results from the dominant contribution of blueshifted photons originating from the larger and brighter granules.

Bisectors due to solar-like granulation are shaped like a distorted ‘C’, and like the top-half of the letter ‘C’ for stars somewhat hotter or more luminous than the Sun. They show reversed curvature on the hot side of the granulation boundary, due to the structure of the surface convection zone (Gray, 1989; Gray et al., 2008).

The shape of the spectral line bisector can be quan-

tified in various related ways: (a) the *bisector velocity span* (or 7% span) is constructed as the difference in bisector velocity between upper and lower regions of the line (Figure 2.14), avoiding the wings and cores (Toner & Gray, 1988; Hatzes, 1996; Queloz et al., 2001); (b) the *bisector inverse slope* (Queloz et al., 2001; Santos et al., 2002b) is $v_t - v_b$, where v_t is the mean bisector velocity between 10–40% of the line depth (top), and v_b is that between 55–90% (bottom); (c) the *bisector curvature* is the difference in velocity span between the upper and lower halves of the bisector (e.g. Hatzes, 1996; Nowak & Niedzielski, 2008), for example $(v_3 - v_2) - (v_2 - v_1)$, where v_1, v_2, v_3 are mean velocities at 20–30%, 40–55%, and 75–100% of the line depth (Dall et al., 2006). Various other forms are used (e.g. Dall et al., 2006).

Bisector analysis is usually made directly on the mean line profile given by the cross-correlation function, and is frequently used as an aid in distinguishing planetary signatures from other types of radial velocity modulation (Povich et al., 2001; Martínez Fiorenzano et al., 2005). For a signal of planetary origin the bisector span is expected to be independent of radial velocity, whereas a distinct correlation arises for blended systems (e.g. for HD 41004: Santos et al., 2002b, Figure 3), or for periodic variations resulting from star spots (e.g. for HD 166435: Queloz et al., 2001, Figure 7).

Dall et al. (2006) used HARPS observations of bright stars, both active and inactive, to derive the mean bisectors from individual lines, as well as from the cross-correlation function (Figure 2.15). They showed that different spectral lines can show different bisector shapes, even between lines of the same element, calling for caution in deriving global stellar properties from the bisector derived from the cross-correlation function alone.

2.3 Instrument programmes

A list of radial velocity instruments applied to exoplanet detection and characterisation is given in Table 2.2. It includes both large successful survey instruments, which now typically make use of many tens of nights on each of many telescopes throughout the world, as well as an incomplete list of more modest and less certain plans; it is intended to give a flavour of the activity in this field.

Most instruments are échelle spectrographs; specific ‘names’ are given when commonly used. The division between ‘early’ and ‘ongoing’ is a little arbitrary, but intended to recognise the radial velocity surveys that were being undertaken in advance of the first more secure exoplanet detection announced in 1995.

2.3.1 State-of-the-art in échelle spectroscopy

Of the numerous high-performance radial velocity instruments now operational, this section gives a short description of just two exemplars: HARPS at the ESO 3.6-m

Table 2.2: Overview of spectroscopic instruments applied to radial velocity exoplanet searches. Resolving power, $R = \lambda/\Delta\lambda$, and accuracy σ (in m s^{-1}), are indicative, the latter particularly dependent on magnitude (... $\rightarrow$... indicates that accuracies have improved accordingly over the project's development life). References cover a mix of early and more recent results, and instrumental descriptions, intended to provide an entry to the more detailed literature. The East-Asia network comprises OAO/BOAO/Xinglong (Izumiura, 2005). * instruments use Michelson interferometry and external dispersion (§2.3.2).

Telescope-instrument	Lead institute	R	σ	Start	Reference
Early:					
CFHT 3.6-m	échelle DAO/UBC	100 000	15	1980	Campbell et al. (1988), Walker et al. (1995)
KPNO 0.9-m	échelle Kitt Peak	74 000	4	1987	McMillan et al. (1990), McMillan et al. (1994)
HJS 2.7-m	échelle U. Texas/McDonald	60 000	4–7	1988	Cochran & Hatzes (1994), Endl et al. (2008)
Lick 3-m	échelle Lick Observatory	40 000	10 $\rightarrow$ 3	1992	Marcy & Butler (1992), Cumming et al. (1999)
ESO CAT 1.4-m	CES ESO planet search	100 000	20	1992	Hatzes et al. (1996), Kürster et al. (1999)
OHP 1.9-m	ELODIE France–Switzerland	42 000	15	1993	Mayor & Queloz (1995), Baranne et al. (1996)
Whipple 1.5-m	AFOE SAO–HAO	70 000	5	1994	Brown et al. (1994), Noyes et al. (1997)
Ongoing:					
Keck 10-m	HIRES Keck	85 000	3 $\rightarrow$ 1	1996	Vogt et al. (1994), Vogt et al. (2000)
AAT 3.9-m	UCLES Anglo–Australian	80 000	3	1998	Tinney et al. (2001)
Euler 1.2-m	CORALIE Geneva	50 000	7 $\rightarrow$ 3	1998	Queloz et al. (2000b), Tamuz et al. (2008)
HET 9.2-m	HRS U. Texas/McDonald	60 000	3	1998	Tull (1998), Cochran et al. (2007)
ESO VLT 8-m	UVES ESO	47 000	30	2000	Deckert et al. (2000), Joergens (2008)
TNG 3.6-m	SARG INAF, Italy	46 000	3	2001	Gratton et al. (2004)
Tautenburg 2-m	échelle Thüringer LS	67 000	3–10	2001	Hatzes et al. (2003b) Esposito et al. (2006)
OAO 1.9-m	HIDES Okayama	70 000	6	2001	Sato et al. (2005b), Sato et al. (2008a)
ESO 3.6-m	HARPS Geneva	115 000	0.3–1	2003	Mayor et al. (2003), Mayor et al. (2009a)
MPG 2.2-m	FEROS Max Planck–ESO	48 000	10	2003	Kaufer et al. (1999), Setiawan et al. (2008b)
Magellan 6.5-m	MIKE Magellan	50 000	5	2004	López-Morales et al. (2008), Minniti et al. (2009)
Subaru 8.2-m	HDS NAOJ	60 000	4–5	2004	Noguchi et al. (2002), Sato et al. (2009)
BOAO 1.8-m	BOES Bohyunsan, Korea	90 000	6	2004	Izumiura (2005), Kim et al. (2006)
Xinglong 2.2-m	CES China	44 000	30	2005	Izumiura (2005)
OHP 1.9-m	SOPHIE OHP, replaces ELODIE	75 000	3	2006	Perruchot et al. (2008)
Sloan 2.5-m	MARVELS* SDSS/Keck	11 000	3–20	2008	van Eyken et al. (2007), Ge et al. (2009)
Magellan 6.5-m	PFS Carnegie	38 000	1	2009	Crane et al. (2008)
Mt Abu 1.2-m	échelle PARAS, India	70 000	3–5	2009	Chakraborty (2008)
APF 2.4-m	APF Lick	80 000	1	2009	Kibrick et al. (2006)
WHT 4.2-m	HARPS–N SAO–Geneva	115 000	0.3–1	2010	Latham (2008b)
Near infrared:					
ESO VLT 8-m	CRIRES ESO (1–5 μm)	100 000	10–20	2007	Käufl et al. (2004), Bean et al. (2010a)
Palomar 5-m	TEDI* Berkeley (0.9–2.4 μm)	20 000	5	2007	Edelstein et al. (2007)
Gemini 8.1-m	PRVS Gemini (0.9–1.7 μm)	50 000	10	study	Ramsey et al. (2008), Jones et al. (2008c)
Future:					
ESO VLT 4×8-m	ESPRESSO ESO	140 000	0.1	2014?	Liske et al. (2009), Pasquini et al. (2009)
E–ELT 42-m	CODEX ESO	150 000	0.01	2016?	Pepe & Lovis (2008), Pasquini et al. (2008a)
TMT 30-m	MIRES TMT (2.5–10 μm)	120 000	–	2018?	Tokunaga et al. (2006)
GMT 7×8.4-m	SHARPS SAO	150 000	–	2018?	Latham (2008b), Johns (2008a)

telescope at La Silla, and HIRES at the Keck I 10-m telescope in Hawaii.

HARPS HARPS (High Accuracy Radial Velocity Planet Searcher) built on experience gained with ELODIE and CORALIE, and was designed to reach accuracies of around 1 m s^{-1} (Mayor et al., 2003). It has been operating at the ESO 3.6-m telescope, La Silla, since 2003 (Rupprecht et al., 2004). It is an échelle spectrograph with $R = 115\,000$, and operates over a spectral range 378–691 nm distributed over échelle orders $N = 89 - 161$. The detector consists of two CCDs (totaling $4\text{k} \times 4\text{k}$, $15 \mu\text{m}$ pixels), with one spectral order ($N = 115$, 530–533 nm) lost in the gap between the two.

The spectrograph is housed in a vacuum vessel, thermally controlled to a few mK to minimise spectral drifts due to temperature and air pressure variations. It is fed by two fibres, one collecting the stellar light, the other simultaneously recording either a thorium–argon reference spectrum, or the background sky. The fibres have an aperture on the sky of 1 arcsec, with image scrambling providing a uniform spectrograph pupil illumination independent of decentering.

In 60 s integration it produces a signal-to-noise ratio of 110 per pixel at 550 nm for a $V = 6$ mag G2V star, and a photon noise error of about 0.9 m s^{-1} . Errors introduced by guiding, focus, and instrumental uncertain-

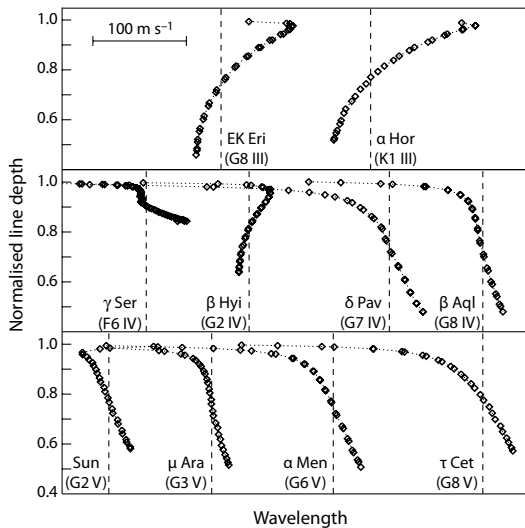


Figure 2.15: Mean bisectors, constructed from the cross-correlation function, of various bright stars according to luminosity class, III (top row) to V (bottom). Vertical dashed lines indicate the radial velocity. Error bars are the size of the plot symbols or smaller. From Dall et al. (2006, Figure 2), reproduced with permission © ESO.

ties lead to typical long-term radial velocity accuracies of about 1 m s^{-1} rms for spectral types later than G, and for non-rotating stars with $v \sin i < 2 \text{ km s}^{-1}$. Short-term precision of 0.2 m s^{-1} and long-term precision of $0.3\text{--}0.6 \text{ m s}^{-1}$ have been achieved (Pepe & Lovis, 2008).

Lovis et al. (2008) summarise the results from the first three years of operations, and detail the global error budget taking account of guiding accuracy and photon noise. The HARPS guaranteed time high-precision sample is presented in Sousa et al. (2008).

A copy of HARPS (HARPS-north) is now installed at the WHT 4.2-m telescope at La Palma (Latham, 2008b).

Keck-HIRES Another example of high-accuracy radial velocity instrumentation is the Keck I 10-m telescope on Mauna Kea, Hawaii, equipped with the HIRES échelle spectrometer. HIRES was one of the five first-light instruments for the Keck I telescope, designed and built at UCO/Lick Observatory (Vogt et al., 1994). The project started in 1988, and achieved first light in 1993. While described as a conventional échelle spectrometer, it was designed to have a relatively large order separation, at the expense of overall wavelength coverage, to allow both for accurate sky subtraction (for faint objects), and for image slicing (for bright objects).

In the configuration described by Vogt et al. (2000), the resolving power is $R = 80\,000$, and the spectra span a wavelength range from 390–620 nm. Wavelength calibration uses an iodine absorption cell.

Given the number and quality of the exoplanet discoveries made by Keck-HIRES (Table 2.3), it is interest-

ing to note that amongst the first-light science originally foreseen (including quasar absorption lines, beryllium in the early universe, lithium abundances, and asteroseismology), exoplanet detection and characterisation did not figure (Vogt et al., 1994).

2.3.2 Externally dispersed interferometry

Background Various interferometric-based techniques have been applied to spectroscopy and radial velocity measurements in the past, although not for exoplanets. In *Fourier transform spectroscopy*, for example, light enters a Michelson interferometer through a beam-splitter. The resulting interference pattern, in frequency space, depends on the path delay within the interferometer, and the recorded intensity is a measure of the light's temporal coherence. One mirror is moved to introduce a variable path length, the temporal coherence is measured as a function of this path delay, $I(\delta)$, and the spectrum is recovered by Fourier transform of $I(\delta)$. Fourier transform spectroscopy provides for very high spectral resolution, but only for very bright sources due to the inefficient use of photons which are sampled only sequentially at each path delay setting.

Fixed delay interferometers with (very) narrow band passes isolating a single spectral line have used this principle for solar measurements since the 1980s, reaching radial velocity accuracies of around 3 m s^{-1} (e.g. Kozhevnikov et al., 1995, and references). Other approaches have employed a more extended optical delay path range within the interferometer, without resorting to physical scanning, for example using grisms or holographic gratings (*holographic heterodyne spectroscopy*) in place of the mirrors (e.g. Connes, 1985; Frandsen et al., 1993; Douglas, 1997).

The combination of a Michelson interferometer with post-disperser was suggested for spectroscopic applications already by Edser & Butler (1898). It entered the exoplanet literature with the work of Erskine & Ge (2000), where it was described as a *fringing spectrometer*. Subsequent developments have been described under the epithets of a *dispersed fixed-delay interferometer* or an *externally dispersed interferometer* (Ge, 2002; Ge et al., 2002b; Erskine, 2003; Erskine et al., 2003; van Eyken et al., 2004; Mahadevan et al., 2008b).

Principles In this approach, the high-resolution échelle spectrograph central to other radial velocity instruments (§2.2) is replaced by a fixed-delay 'field widened' (or field compensated) Michelson interferometer, allowing the useful solid angle accepted by the interferometer to be greatly enlarged (see, e.g., Hilliard & Shepherd, 1966). Spatial fringes at a given interference order result from the different path lengths of the input beam through the interferometer. The interferometer output enters a low- to medium-resolution spectrograph with $R \sim 10\text{--}30\,000$. Doppler measurements are

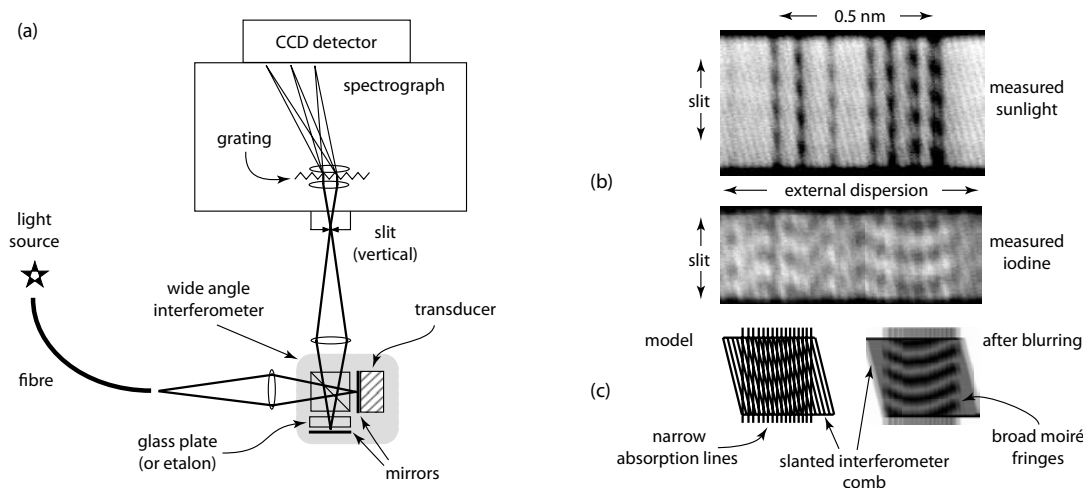


Figure 2.16: Principle of an externally-dispersed interferometer: (a) a wide-angle Michelson interferometer with a fixed internal delay generates spatial fringes which are input to a low-resolution spectrograph which disperses the light in the orthogonal direction, creating a fringing spectrum at the CCD detector. The glass plate, or etalon, serves as the field widening element (e.g., Hilliard & Shepherd, 1966). The transducer steps the interferometer delay in four quarter-wave increments to remove non-fringing artefacts in multiple exposures; (b) fringe pattern resulting from illumination with sunlight (above) and iodine (below), in which the underlying phase-slanted interferometer comb is partially resolved; (c) schematics illustrating the formation of moiré patterns, in which the high detail spectral features are heterodyned to lower resolution where they survive the blurring of the external spectrograph. From Erskine (2003, Figures 1 and 5), reproduced by permission of University of Chicago Press.

then made by monitoring the phase shifts of fringes created spatially at the output of the Michelson interferometer, rather than the tracking of absorption line centroids as in échelle spectroscopy (Figure 2.16).

High Doppler sensitivity is achieved by optimising the fixed optical delay within the interferometer to maximise the combination of fringe contrast and the number of photons per fringe. Ge (2002) selected a total internal path delay of $d \sim 11$ mm, meaning that the interferometer is working (in a number of orders) around $m = d/\lambda \approx 20000$. Accordingly, a Doppler shift of 1 part in 20 000 results in one fringe (2π rad) of phase shift.

Sensitivity is further and substantially enhanced by the simultaneous measurement of numerous fringes over a broad wavelength range, which is achieved by coupling the interferometer output to a low- to medium-resolution spectrometer, which disperses the light orthogonal to the fringes. The wavelength range of 34 nm in the system used at Lick by Ge et al. (2002b) is dictated by the post-disperser resolution and CCD dimensions. The function of the disperser being simply to separate the otherwise superimposed fringes, the Doppler error is essentially independent of its resolution.

The output is recorded on a CCD, with fringes sampled across a restricted number of pixels in the direction of the spectrometer slit, and the spectrum sampled in the dispersion direction. In the design described by Ge et al. (2002b), for example, one fringe is sampled by

20 pixels in the spatial direction, and four pixels in the dispersion direction. The interferometric orders themselves provide a frequency comb which extends over the entire bandwidth, thus generating frequency markers analogous to the fiducial lines of an absorption cell, but uniform in spacing, shape, and amplitude.

Operation in ‘phase uniform’ mode involves stepping the selected interferometer delay in four quarter-wave increments to remove non-fringing artefacts in multiple exposures. In ‘phase slanted’ mode, the fringes are slanted with respect to the dispersion direction by tilting one of the interferometer mirrors so that the delay has a linear component transverse to the dispersion direction (Erskine, 2003). Moiré fringes then result from the underlying fringe comb generated by the interferometer, and the emission/absorption lines of the source. The combination effectively heterodynes (down converts) the high spatial frequencies of the spectrum to low spatial frequencies resolved by the disperser.

The accuracy of measured Doppler shifts appears competitive with existing high-resolution spectrometers, but with the potential of reducing the manufacturing complexity of the échelle spectrograph, improving mechanical stability, simplifying the point-spread function, improving throughput, and extending the useful wavelength range. Furthermore, by replacing the multi-order cross-dispersed échelle spectrum with a single lower resolution spectrum spread along just one dimen-

sion of the CCD (cf. Figure 2.9), the orthogonal dimension can be used for spectra of other objects. These can be fibre-fed to the (wide-angle) interferometer entrance, resulting in a multiplexing multi-object capability.

Instrument developments A prototype instrument (exoplanet tracker) was used to confirm the known exoplanet, 51 Peg, at the KPNO 2.1-m by van Eyken et al. (2004), and as the basis of the W.M. Keck exoplanet tracker instrument (Mahadevan et al., 2008a,b). The first new planet discovered using this technique was HD 102195 (ET1), made at the KPNO 0.9-m, and confirmed with the KPNO 2.1-m and the HET-HRS (Ge et al., 2006). The planet has $P = 4.1$ d and $M_p \sin i = 0.49 M_J$.

The concept also underlies the MARVELS survey at Apache Point Observatory (Multi-object APO Radial-Velocity Exoplanet Large-area Survey). MARVELS forms part of the six-year Sloan Digital Sky Survey SDSS III extension, operating between 2008–14. Its scientific goal is to monitor 11 000 $V = 8 - 12$ stars over 800 sq. deg. for six years. The survey targets 90% F8 and later main sequence stars and subgiants, and 10% G and K giants with $V = 7.6 - 12$ (van Eyken et al., 2007; Ge & Eisenstein, 2009; Ge et al., 2009). The instrument acquires radial velocities, at $3 - 20 \text{ m s}^{-1}$ depending on V magnitude, for 60 objects simultaneously in the 3° field, by lining up the distinct low-resolution spectra across the CCD (Wan et al., 2006). A trial survey of 420 $V = 8 - 12$ mag solar-type stars in 2006 was reported by Ge et al. (2007).

An All-Sky Extrasolar Planet Survey (ASEPS), using wide-field telescopes to monitor stars to $V = 12 - 13$ mag, might increase the number of exoplanets by a factor of 10–100 (Ge, 2007; Ge et al., 2007). Observing strategies for large-scale multi-object surveys have been simulated by Kane et al. (2007).

Extension to the infrared The same technique can be used at longer wavelengths. The TEDI instrument (TripleSpec Exoplanet Discovery Instrument) extends the concept to the near infrared ($0.9 - 2.4 \mu\text{m}$) by coupling the wide-angle Michelson interferometer to the existing $R = 3000$ TripleSpec échelle spectrometer at the Palomar 5-m telescope, whose spectral resolution alone is insufficient for exoplanet radial velocity measurement (Edelstein et al., 2007, 2008). The instrument is used in phase uniform mode, and both interferometer outputs are relayed to the spectrograph.

A similar development is underway at APO.

Dispersed FTS Broadly similar principles are used in the *dispersed Fourier transform spectrometer* under development at the USNO (Hajian et al., 2007). But rather than operating at a fixed delay position, the interferogram of the Fourier transform spectrometer is coarsely sampled over a wide range of delay positions, so that a complete high-resolution broad-band spectrum can

be reconstructed. The addition of a post-disperser essentially creates several thousand separate narrow-band Fourier transform spectrometers operating in parallel.

2.3.3 Future developments

Planned instruments CODEX is an instrument concept for the European-ELT (Extremely Large Telescope). It aims to detect the expansion of the Universe directly, by measuring the Doppler shift of high-redshift quasar Ly- α absorption lines as a function of time. The experiment targets a Doppler accuracy of around 0.02 m s^{-1} maintained over several decades, which would also have direct application to exoplanet studies (Pepe & Lovis, 2008; Pasquini et al., 2008a,b).

The design has developed from HARPS, and incorporates high stability, slanted volume phase holographic gratings with $R \gtrsim 120\,000$, a wavelength range 370–720 nm, fibre-optic scrambling to reduce the impact of guiding errors (where 0.01 m s^{-1} corresponds to $10 \mu\text{as}$ on the sky), and wavelength calibration using laser frequency combs. The schedule is tied to that of E-ELT, with first light currently envisaged for 2016.

SHARPS is a spectrograph designed for the proposed Giant Magellan Telescope, planned for completion in 2018, with similar scientific objectives.

SAO are developing new instruments for follow-up of the Kepler mission, including the multi-object Hectochelle (Szentgyorgyi & Furész, 2007).

Alternative approaches A different approach to determining radial velocity variations is to measure accelerations directly. Connes (1985, 1994) proposed a system in which the stellar light, and light from a white light source, are passed alternately through a Fabry-Pérot etalon. A feedback loop adjusts the etalon spacing to achieve coincidence. A tunable laser tracks the Fabry-Pérot spacing, and a measurable beat signal is generated from a stabilised laser. The objective is to lock onto the stellar lines, and to monitor fluctuation using terrestrial frequency standards, providing the absolute acceleration of the source with respect to the observer.

Connes (1985) estimated that for a $P = 10$ yr orbit, a 1 m telescope observing a $V = 10$ mag star for 10 yr and an observing time of 10×1 hr per year, would reach an acceleration amplitude of $2 \times 10^{-5} \text{ m s}^{-2}$, corresponding to a radial velocity error of 0.1 m s^{-1} . A laboratory system was reportedly built and tested (Schmitt, 1997; Bouchy et al., 1999).

Riaud & Schneider (2007) have proposed extracting the planet's radial velocity signal from the residual stellar flux halo remaining in coronagraphic adaptive optics imaging in combination with the planned extremely large telescopes. The radial velocity signal is estimated to be detectable within the residual contrast ratio of $10^3 - 10^4$, compared to the contrast ratio of 10^{10} in the absence of a coronagraph.

2.4 Results to date

2.4.1 The first radial velocity exoplanets

Early radial velocity surveys, on a relatively small number of stars, were primarily aimed at characterising the sub-stellar/brown dwarf mass function by searching for binary companions of main-sequence stars with masses below $1 M_{\odot}$ (Campbell et al., 1988; Marcy & Moore, 1989; Marcy & Benitz, 1989; McMillan et al., 1990; Duquennoy & Mayor, 1991; Tokovinin, 1992). Some were part of programmes to establish improved radial velocity standards for the IAU (Latham et al., 1989).

As accuracies improved towards plausible planetary signals of around $10\text{--}20 \text{ m s}^{-1}$, existing groups intensified their efforts, and others started new observing programmes, leading to the monitoring of many more stars over a number of years.⁹

The first radial velocity detections were announced cautiously, and only substantively confirmed some years later. Thus Campbell et al. (1988) identified a possible $P = 2.7 \text{ yr}$, $1.7 M_J$ object around $\gamma \text{ Cep}$, parameters which were subsequently questioned (Walker et al., 1992), but which were eventually confirmed by the 1981–2002 study of Hatzes et al. (2003a). Latham et al. (1989) reported a $P = 84 \text{ d}$, $11 M_J$ companion to HD 114762, which they suggested was a probable brown dwarf. These values were confirmed by Cochran et al. (1991), and further refined by Butler et al. (2006). Hatzes & Cochran (1993) reported a possible $P = 558 \text{ d}$, $2.9 M_J$ companion to the K giant HD 62509 ($\beta \text{ Gem}$), parameters again substantially confirmed in the 25-year baseline study by Hatzes et al. (2006).

The discovery of 51 Peg b The discovery of a very short-period $P = 4.2 \text{ d}$ ($a = 0.05 \text{ AU}$) $0.47 M_J$ planet surrounding the star 51 Peg, was announced by Mayor & Queloz (1995). The discovery was promptly confirmed by the Lick Observatory group, who were also quickly able to report two new planets around stars that they had been monitoring: 70 Vir (Marcy & Butler, 1996) and 47 UMa (Butler & Marcy, 1996). The compelling realisation that planetary mass objects existed around main sequence stars marked the start of a substantive and world-wide acceleration in exoplanet research.

The reality of 51 Peg b was the subject of some early and intense controversy. In part this was motivated by its unexpectedly short orbital period and close proximity to the parent star. But an alternative explanation – that the radial velocity shifts arose from non-radial os-

cillations – was also put forward to explain possible distortions in the absorption line bisector. Studies that followed (Gray, 1997; Hatzes et al., 1997; Marcy et al., 1997; Gray & Hatzes, 1997; Willems et al., 1997; Brown et al., 1998a,b; Gray, 1998; Hatzes et al., 1998a,b) finally resulted in a consensus that the planet hypothesis was the most reasonable.

2.4.2 The present radial velocity census

Apart from the extreme planetary system around the pulsar PSR B1257+12, and until the first transiting planet discovered by photometry in 2003 (OGLE–TR–56: Konacki et al., 2003a), almost the first hundred planets around normal main-sequence stars were discovered using radial velocity techniques. The number of radial velocity discoveries is still rising strongly, as enlarged surveys, higher measurement accuracies, and longer temporal baselines take effect (Table 2.3).

The various measurement groups have now surveyed ~ 2500 stars with masses in the range $0.3\text{--}2.5 M_{\odot}$, and brighter than $V = 12 \text{ mag}$ (Marcy et al., 2008), while many more are being added to new survey programmes. As described in subsequent sections, and summarised in Table 2.4, the initial focus on main sequence stars has progressively expanded to encompass surveys which include various other stellar types and environments.

Overall results As of 2010 November 1, there were 494 known exoplanets, of which 461 in 390 systems (45 multiple) had measured radial velocity orbits, and 358 had been discovered by radial velocity measurements (Appendix C). Figure 2.17 shows the planets discovered from radial velocity measurements by year, as a function both of M_p (ordinate) and semi-major axis a (circle size). The decreasing lower mass bound with time reflects the improving instrumental accuracy leading to smaller detectable values of K . For a given planet mass, $K \propto P^{-1/3}$ (Equation 2.26), which explains their preferentially smaller a .

Figure 2.18 illustrates the region probed by the radial velocity discoveries, compared with transit and astrometric detections at current and projected accuracies.

Figure 2.19 shows the number of exoplanets discovered to date by radial velocity monitoring as a function of spectral type.

2.4.3 On-line compilations

There are numerous on-line resources and compilations related to exoplanet surveys and results (cf. Table 1.3). Two are particularly useful in the context of radial velocity exoplanet detections, since they maintain up-to-date records of new discoveries, compilations of related parameters, and a variety of tools for their inspection and analysis, viz:

⁹Walker (2008) provides a recent perspective on the radial velocity exoplanet searches carried out in the 1970s–1980s, and states ‘It is quite hard nowadays to realise the atmosphere of scepticism and indifference in the 1980s to proposed searches for extra-solar planets. Some people felt that such an undertaking was not even a legitimate part of astronomy’.

Table 2.3: Number of planets discovered by specific radial velocity instruments, based exclusively on the compilation at exoplanets.org as of 2010–11–01. This listed 317 planets actually discovered by radial velocity measurements (i.e. an incomplete compilation), compared to 461 with measured radial velocities and 494 discovered by all methods given by exoplanet.eu at the same date. A planet discovered by observations at N observatories, is counted for $1/N$ discoveries by each, and final numbers are rounded.

Observatory/instrument	Number
Keck–HIRES	101
ESO 3.6–HARPS	50
Euler–CORALIE	34
Lick (Shane/APF)	32
AAT–UCLES	28
OHP–ELODIE	15
HET–HRS	13
Okayama–HIDES	10
Magellan–MIKE	7
Tautenburg	6
Subaru–HDS	5
OHP–SOPHIE	3
McDonald HJS	3
Whipple–AFOE	2
Bohunsan–BOES	2
CFHT (from 1988)	1
ESO–CES	1
ESO–FEROS	1
KPNO–ET	1
Oak Ridge+Coravel (from 1989)	1
Xinglong–CES	1

<http://exoplanet.eu>, the Exoplanet Encyclopedia, is a compilation of all exoplanet announcements, data, and bibliography, maintained by Jean Schneider at the Observatoire de Paris–Meudon. The data base can be interrogated, for example, according to radial velocity measurements alone.

<http://exoplanets.org>, the Exoplanet Orbit Database, includes results from radial velocity and transit surveys for stars within 200 pc, aiming to incorporate the most secure spectroscopically-measured orbital parameters. It updates the Catalogue of Nearby Exoplanets (Butler et al., 2006; Jones et al., 2008a), and is maintained by Jason Wright and Geoff Marcy.

2.4.4 Main sequence stars

G and K dwarfs Most of the early Doppler planet searches concentrated on G and K main sequence stars (dwarfs), in the mass range $\sim 0.7 - 1.3M_{\odot}$. They have numerous absorption lines, are relatively bright, are relatively slow rotators with low rotational broadening, and have relatively stable atmospheres, with photometric jitter for inactive G dwarfs extending down to $2-3 \text{ m s}^{-1}$ (Table 2.1). Early reviews of these radial velocity searches were given by Latham (1997); Butler & Marcy (1998); Latham et al. (1998); Marcy & Butler (1998b); Nelson & Angel (1998); and Marcy & Butler (2000).

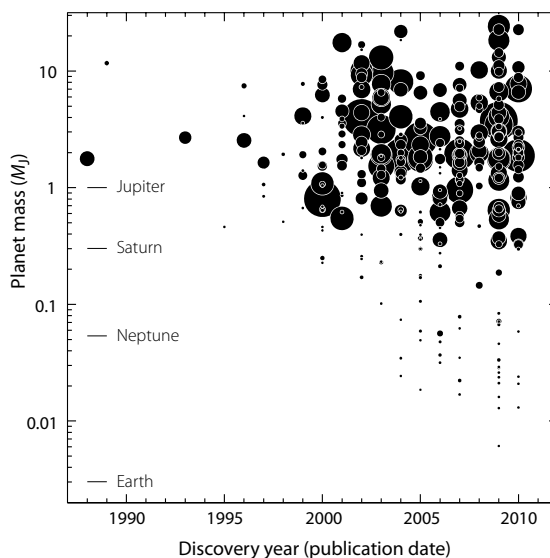


Figure 2.17: Planets discovered by radial velocity measurements, according to mass and year of discovery. Circle sizes are proportional to the semi-major axis a . Data are for 383 planets from exoplanets.org, 2010–11–01.

Programmes have collectively expanded to cover most late-type main sequence stars brighter than $V \sim 7.5 - 8.5$ mag in a systematic manner. Of the larger programmes (Table 2.4), CORALIE is surveying about 1600 stars in the southern hemisphere (Tamuz et al., 2008). In the north, the ELODIE survey of more than 1000 targets focused on metal-rich stars (e.g. da Silva et al., 2006, 2007). ELODIE was replaced by SOPHIE (following the HARPS design) in 2006, and the same programme has continued under the SOPHIE exoplanet consortium (e.g. Santos et al., 2008). In parallel, the California & Carnegie search program uses the Keck, Lick, and AAT telescopes to survey about 1000 stars in the north and south (Valenti & Fischer, 2005).

The N2K consortium is using the Keck, Magellan, and Subaru telescopes to survey the ‘next 2000’ stars (Fischer et al., 2005; López-Morales et al., 2008). Estimates of T_{eff} , $[\text{Fe}/\text{H}]$, and binarity were constructed from a starting list of more than 100 000 FGK dwarfs (Ammons et al., 2006; Robinson et al., 2006b). Their primary target list was then constructed from the more than 14 000 main sequence and sub-giant stars with $d < 110$ pc, $V < 10.5$ and $0.4 < B - V < 1.2$. Their final selection of 2000 stars was also biased towards higher metallicity targets, using broad-band photometry to define a subset with $[\text{Fe}/\text{H}] > 0.1$.

A large survey, of ~ 10 000 F8 and later dwarfs, started with the Sloan SDSS III MARVELS programme in 2008 (Ge et al., 2009). The first sub-stellar mass companions, including the brown dwarf MARVELS–1 b, have been reported (Fleming et al., 2010; Lee et al., 2011).

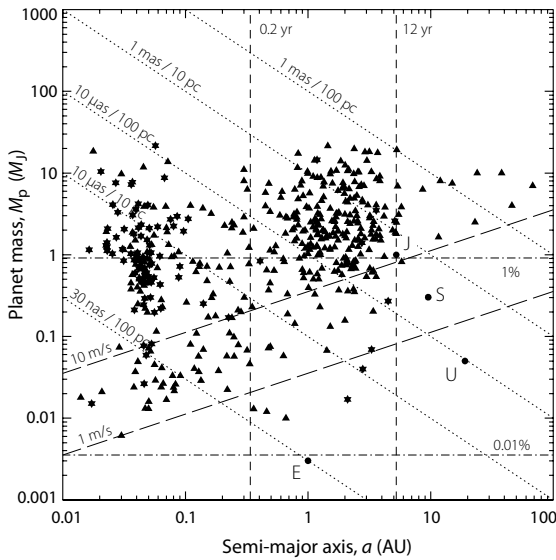


Figure 2.18: Detection domains based on orbital motion. Lines from top left to bottom right show astrometric signatures of 1 mas and 10 μ as at distances of 10 and 100 pc (Equation 3.1, assuming $M_{\star} = M_{\odot}$). Short- and long-periods cannot be detected by planned space missions: vertical lines show limits at $P = 0.2$ and 12 yr. Lines from top right to bottom left show radial velocity semi-amplitudes of $K = 10$ and $K = 1 \text{ m s}^{-1}$. Horizontal lines are transit thresholds of 1% and 0.01%, corresponding to $\sim 1R_J$ and $\sim 1R_{\oplus}$ respectively. Positions of E(earth), (J)upiter, (S)aturn and (U)ranus are shown, as are planets discovered from radial velocity ($M_p \sin i$, $\blacktriangle$) and transit photometry ($\star$). Data are from exoplanet.eu, 2010–11–01.

Examples Out of several hundred planets now detected in this category, a small selection of graphical results is shown in Figures 2.20–2.23 as examples of the variety, measurement quality, and orbit reconstruction possibilities. These figures relate to the following systems (and where all radial velocity derived mass estimates are implicitly uncertain by the factor $\sin i$):

HD 4113 (G5V): Tamuz et al. (2008) used CORALIE to discover a $1.63 M_J$ planet in a highly eccentric ($e = 0.90$) $a = 1.28 \text{ AU}$, $P = 526\text{-d}$ orbit. The radial velocity curve shows a sharp, strongly asymmetric form arising from the eccentricity (Figure 2.20), and an underlying linear trend attributable to an outer, longer-period planet.

55 Cnc (G8V): Fischer et al. (2008) used Lick and Keck observations (1989.1–2007.4), to characterise five orbiting planets. Planets were identified by periodogram analysis of residuals to successive Keplerian fits (Figure 2.21). $\{M_p \sin i, P\}$ (M_J , d) are $\{0.024, 2.8\}$, $\{0.83, 14.6\}$, $\{0.17, 44.4\}$, $\{0.15, 261\}$, $\{3.9, 5400\}$.

HD 40307 (K2.5V): Mayor et al. (2009b) used HARPS to characterise the orbits of three ‘super-Earth’ planets (Figure 2.22; the total radial velocity semi-amplitude of the combined three-planet signal is only $\sim 6\text{--}7 \text{ m s}^{-1}$). $\{M_p \sin i, P\}$ ($M_{\oplus}$, d) are $\{4.2, 4.3\}$, $\{6.8, 9.6\}$, $\{9.2, 20.5\}$.

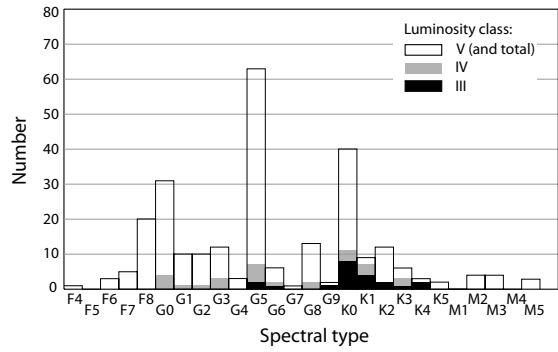


Figure 2.19: The number of exoplanets discovered from radial velocity monitoring versus spectral type (other planets discovered from transit photometry, and confirmed by radial velocity, are not included). Data, from exoplanets.org on 2010–11–01, includes 261 systems (stars host to a multiple planetary system are counted only once). The majority of exoplanets found from radial velocity measurements orbit G and K dwarfs.

61 Vir (G5V): Vogt et al. (2010b) used 4.6 years of Keck–HIRES and AAT data to characterise this ‘super-Earth and two Neptune’ planetary system (Figure 2.23). The planets are all in low-eccentricity orbits; their $\{M_p \sin i, P\}$ ($M_{\oplus}$, d) are $\{5.1, 4.2\}$, $\{18.2, 38\}$, $\{24.0, 124\}$.

M dwarfs, and infrared surveys Towards lower stellar masses, late K and M dwarfs are faint, and high signal-to-noise spectra tend to be limited by telescope size. But special attention has been given to M dwarfs by numerous groups (Table 2.4), since their lower mass implies lower detectable planetary masses for the same radial velocity semi-amplitude (Equation 2.26). At the same time, being of lower luminosity, the corresponding habitable zone defined by the possible presence of liquid water is closer to the host star, with semi-major axes more accessible to radial velocity surveys.

Programmes range from the pioneering survey of 65 M dwarfs started at the Mount Wilson 2.5-m telescope in 1984 (Marcy et al., 1986) to the survey of 100 M dwarfs started in 2007 at the Palomar 5-m using an ‘externally-dispersed interferometer’ (§2.3.2) in the form of TEDI, a Michelson interferometer coupled to the TripleSpec near-infrared spectrometer (Edelstein et al., 2007).

Results to date suggest that giant gaseous planets occur less frequently around M dwarfs (Endl et al., 2006; Bonfils et al., 2007), a conclusion already reached by Marcy et al. (1986), although a number of lower mass planets, in the super-Earth to Neptune-mass range, have been found. Discoveries include GJ 581 (Bonfils et al., 2005b; Udry et al., 2007; Mayor et al., 2009a), GJ 674 (Bonfils et al., 2007), and GJ 176 (Forveille et al., 2009). More than half are below $25 M_{\oplus}$, with Endl et al. (2008) quoting a 1.3% detection rate for Jovian-type planets at $a < 1 \text{ AU}$ earlier than M5V. Despite a low overall detection rate, multiplicity rates nevertheless appear high.

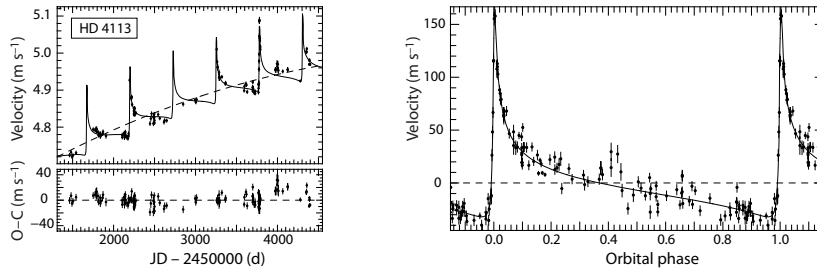


Figure 2.20: Left: radial velocity measurements for HD 4113, observed with CORALIE, with the best-fit Keplerian solution including a linear drift (residuals below). Right: the same measurements folded at $P = 526.58$ d, yielding $K = 97.7$ m s $^{-1}$, $a = 1.28$ AU, $e = 0.90$, and $M_p \sin i = 1.63 M_J$. The planet–star separation varies between 0.12–2.4 AU. The long-term drift, and its curvature, may imply a brown dwarf companion at 8–20 AU, with $P \sim 20$ –90 yr. From Tamuz et al. (2008, Figure 1), reproduced with permission © ESO.

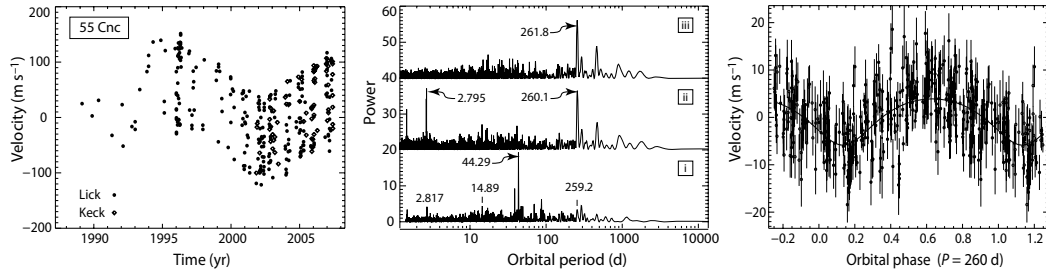


Figure 2.21: The five-planet system 55 Cnc. Left: velocities from Lick/Keck (1989.1–2007.4). Middle: periodograms of residuals to successive Keplerian models (peaks in days): (i) with respect to the 2-planet (14.65 and 5200 d) model: the peak at 44.3 d is due to the third planet; (ii) with respect to the resulting 3-planet model (other major peaks are aliases); (iii) with respect to the 4-planet model (2.8, 14.7, 44.3, and 5200 d); the peak marked is due to the fifth planet (the other major peak is an alias). Right: periodicity of the fifth planet in the Keck data, after subtracting the effect of the other planets. From Fischer et al. (2008, Figures 1–4, 8), reproduced by permission of the AAS.

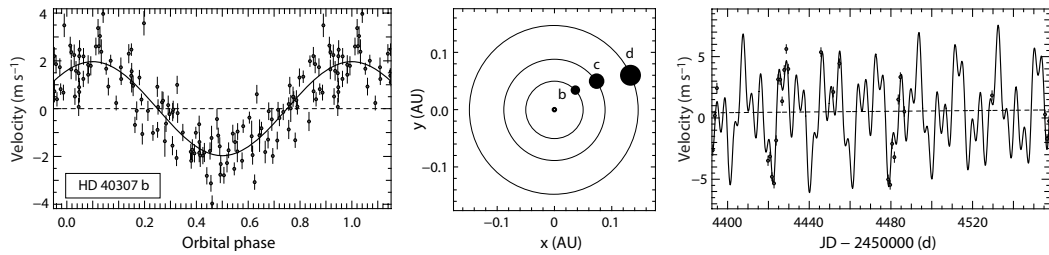


Figure 2.22: The ‘three super-Earth’ system HD 40307, observed with HARPS. Left: phase-folded velocities and Keplerian curve for the lowest mass planet b, after correction for the other two planets. Middle: pole-on view of the orbits; planet size is proportional to mass. Right: measured velocities with the three Keplerian model superimposed. From Mayor et al. (2009b, Figures 2–4), reproduced with permission © ESO.

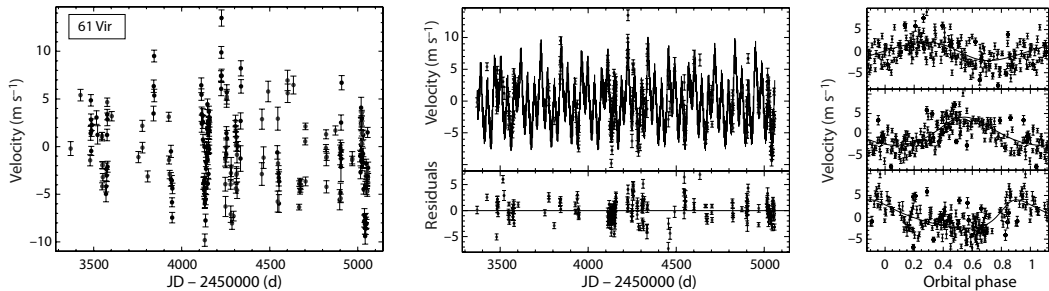


Figure 2.23: The ‘super-Earth and two Neptune’ system 61 Vir, observed with Keck and AAT. Left: relative radial velocities. Middle: observed velocities, and the three-planet model fit (residuals below). Right: modeled contributions from the three planets b–d (top to bottom, with $e = 0.1, 0.14, 0.35$ respectively). From Vogt et al. (2010b, Figures 2, 6–7), reproduced by permission of the AAS.

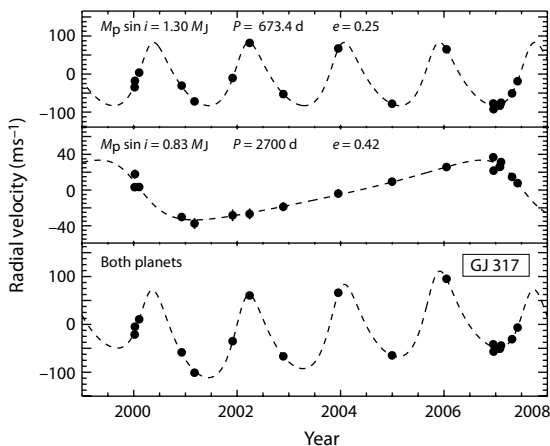


Figure 2.24: The M dwarf GJ 317, and the best-fitting two-planet model. The top two panels show the single-planet fits with the other component removed. Adding an outer planet with $P \approx 2700$ d decreases the scatter of the residuals from 12.5 to 6.32 m s^{-1} , and the reduced χ^2 from 2.02 to 1.23. From Johnson et al. (2007a, Figure 4), reproduced by permission of the AAS.

By late 2010, only seven Doppler-detected giant planets ($M > 0.3 M_J$) were known around six M dwarfs (Johnson et al., 2010c). These include GJ 317 of $1.2 M_J$ (Johnson et al., 2007a, Figure 2.24), and one of the most massive at $2.1 M_J$ around HIP 79431 (Apps et al., 2010). Of particular interest is GJ 876, with its two resonant Jovian-mass planets and an inner super-Earth (Rivera et al., 2005), and a fourth outer Uranus-mass planet in a Laplace resonance configuration, with periods 30.4, 61.1, and 126 d (Rivera et al., 2010).

Johnson et al. (2010c) estimate that $3.4^{+2.2}_{-0.9}\%$ of stars with $M_\star < 0.6 M_\odot$ host planets with $M \sin i > 0.3 M_J$ and $a < 2.5 \text{ AU}$. Restricted to metal-rich stars with $[\text{Fe}/\text{H}] > +0.2$, the occurrence rate rises to $10.7^{+5.9}_{-4.2}\%$.

Early-type dwarfs Observationally, radial velocity measures for higher mass main sequence stars (A–F spectral type) are complicated by three effects: a smaller number of spectral lines due to higher surface temperatures; line broadening due to higher rotational velocities, of order $v \sin i \approx 100 - 200 \text{ km s}^{-1}$ for A-type stars (Galland et al., 2005a); and large atmospheric ‘jitter’ due to higher surface inhomogeneities and pulsation, of as much as 50 m s^{-1} or more for mid-F stars (Saar et al., 1998, see also Table 2.1). Together, these limit the applicability of the basic cross-correlation method (Griffin et al., 2000).

Chelli (2000) considered cross-correlation in Fourier space, which allows enhanced suppression of high-frequency noise and low-frequency continuum variations. The correlation is made between each spectrum of the target star and a reference spectrum specific to that star constructed from the sum of all the available spectra. When applied to early-type stars (Galland

et al., 2005a), it suggested improved detection possibilities: for example, reaching the planetary domain with ELODIE for A-type main sequence stars with $v \sin i$ up to 100 km s^{-1} for $P < 10$ d, and for all A and F-type stars with HARPS, even for large $v \sin i$ (Figure 2.25).

The resulting search for a sample of A–F main sequence stars, with ELODIE/SOPHIE in the north, and HARPS in the south, led to the first detection of a $9.1 M_J$ companion around the F6V star HD 33564 (Galland et al., 2005b), a $25 M_J$ brown dwarf in orbit around HD 180777, a pulsating A9 dwarf with a high rotation of $v \sin i = 50 \text{ km s}^{-1}$ (Galland et al., 2006b) and a series of subsequent discoveries from the southern survey (Desort et al., 2009, 2010, and references).

2.4.5 Evolved stars

Observational complications inherent to high-mass dwarfs can be somewhat circumvented by observing their counterparts which have evolved away from the main sequence. After main sequence hydrogen burning, radii expand and atmospheres cool, leading to an increase in metal lines. Angular momentum loss as stars cross the subgiant branch results from a coupling of stellar winds to rotationally-generated magnetic fields (Gray & Nagar, 1985; Schrijver & Pols, 1993; do Nascimento et al., 2000). The combination of cooler atmospheres and smaller rotational broadening makes evolved stars well suited for precision radial velocity surveys.

Subgiants Subgiants offer a reasonably favourable region of the HR diagram for radial velocity searches, with low rotational velocities of order $v \sin i \lesssim 5 \text{ km s}^{-1}$, and reasonably low photometric jitter of around $5 - 10 \text{ m s}^{-1}$ (Wright, 2005, see also Table 2.1). Surveys at Keck and Lick are being made around 150 subgiants, selected from Hipparcos according to $V < 7.6 \text{ mag}$, $2 < M_V < 3.5$, and $0.55 < B - V < 1.0$ (Johnson et al., 2006b). Numerous planets are being reported (Johnson et al., 2007b; Johnson, 2008; Johnson et al., 2010a,b, 2011; Harakawa et al., 2010), with 30 luminosity class IV listed at exoplanet.eu in late 2010.

G and K giants Radial velocity surveys around G and/or K giants have been made by several groups (Table 2.4): at Lick (Frink et al., 2002; Nidever et al., 2002; Hekker et al., 2006, 2008); Okayama–HIDES and Subaru–HDS (Sato et al., 2003, 2005b, 2008a,b, 2010); La Silla–FEROS (Setiawan et al., 2003a,b, 2004); CORALIE and HARPS (Lovis & Mayor, 2007; Tautenburg (Hatzes et al., 2005; Döllinger et al., 2007, 2009); HET–HRS (Niedzielski et al., 2007; Niedzielski & Wolszczan, 2008; Nowak & Niedzielski, 2008; Niedzielski et al., 2009a,b), and BOAO–BOES (Han et al., 2010). A large survey, of some 1000 GK giants, started with the Sloan SDSS III MARVELS programme in 2008 (Ge et al., 2009).

Table 2.4: Radial velocity surveys focusing on particular stellar types (excluding brown dwarfs). Surveys, instruments, and descriptions evolve with time, so the table is indicative rather than definitive. References describe the sample or more recent global results. Listings are chronological in survey start date within each category, and there is often overlap between some categories.

Selection	Telescope/instrument	N(stars)	Start	Reference
Main sequence:				
F8–M5	ESO–CES [ended 1998]	37	1992	Endl et al. (2002)
GK, $V < 7.65$, $v \sin i < 5 \text{ km s}^{-1}$	OHP–ELODIE	324	1994	Queloz et al. (2001)
FGKM (California–Carnegie)	Keck+Lick+AAT	1040	1995	Valenti & Fischer (2005)
FGK	Lick–Hamilton	107	1995	Marcy & Butler (1998a)
F7–M5, $< 50 \text{ pc}$	Keck–HIRES	530	1996	Vogt et al. (2000)
FGKM	Lick+Keck	844	1998	Nidever et al. (2002)
FGK IV–V, $V < 8$, $\delta < -20^\circ$	AAT–UCLES	300	1998	Jones et al. (2002)
Hipparcos parallax-limited	La Silla–CORALIE	1650	1998	Tamuz et al. (2008)
F–M, metal-rich	AAT–UCLES	20	1999	Tinney et al. (2003)
F–M (N2K), metal-rich	Keck+Magellan+Subaru	2000	2004	Fischer et al. (2005, 2007)
metal-rich, Hipparcos-based	OHP–ELODIE/SOPHIE	1061	2004	da Silva et al. (2006)
low activity (Rocky Planet)	AAT–AAPS	55	2007	O’Toole et al. (2009b,a)
metal-rich (Calan–Hertfordshire)	ESO–HARPS	350	2008	Jenkins et al. (2009a)
GKM (Eta–Earth)	Keck–HIRES	230	2008	Howard et al. (2009, 2011a,b)
F8 and later	SDSS III MARVELS	10 000	2008	Ge et al. (2009)
M dwarfs:				
M2–M5, $V < 11.5$	Mt Wilson 2.5-m	65	1984	Marcy et al. (1986)
Lick/Keck	Lick+Keck	24	1994	Marcy et al. (1998)
$V < 7.5$	AAT–UCLES	7	1998	Jones et al. (2002)
CNS3, $d < 9 \text{ pc}$, $V < 15$	OHP–ELODIE/SOPHIE (N)	127	1995	Delfosse et al. (1999a)
" $d < 12 \text{ pc}$	La Silla–FEROS (S)	200	2000	Bonfils et al. (2004)
ESO survey	VLT–UVES	40	2000	Zechmeister et al. (2009)
"	VLT–UVES	26	2002	Guenther & Wuchterl (2003)
McDonald survey	HET–HRS+HJS	100	2002	Endl et al. (2003)
$d < 11 \text{ pc}$	HARPS	200	2003	Mayor et al. (2009a)
Palomar	Palomar–TEDI	100	2007	Edelstein et al. (2007)
K–M (M2K)	Keck	1600	2009	Apps et al. (2010)
Early-type dwarfs:				
early-type, B8–F7	ELODIE/SOPHIE+HARPS	185	2004	Desort et al. (2009)
Subgiants:				
subgiants	Lick+Keck	159	2006	Johnson et al. (2006b, 2010b)
Giants:				
G/K/clump giants	Lick+Keck	45	1999	Nidever et al. (2002)
"	Lick–CAT	179	1999	Hekker et al. (2006, 2008)
"	Okayama–HIDES/Subaru	300	2001	Sato et al. (2008a,b, 2010)
"	La Silla–FEROS	83	2001	Setiawan et al. (2003a,b)
" (in 13 open clusters)	HARPS	115	2003	Lovis & Mayor (2007)
" (Penn State/Torun)	HET–HRS	1000	2004	Niedzielski & Wolszczan (2008)
"	Tautenburg 2-m	62	2004	Döllinger et al. (2007, 2009)
"	SDSS III–MARVELS	1000	2008	Ge et al. (2009)
"	BOA–BOES	55	2010	Han et al. (2010)
Open clusters:				
Hyades giants	Okayama–HIDES	4	2001	Sato et al. (2007)
Hyades dwarfs	Keck+others	94	2002	Cochran et al. (2002)
13 clusters	CORALIE+HARPS	115	2003	Lovis & Mayor (2007)
Abundance-based:				
metal-poor	FEROS	70	2008	Setiawan et al. (2010)
metal-poor	Keck–HIRES	200	2003	Sozzetti et al. (2006, 2009)
metal-poor	HARPS	105	2003	Santos et al. (2007)
Young stars:				
young stars, Cha I cloud	VLT–UVES	12	2000	Joergens (2006)
young stars, 1–100 Myr	FEROS+HARPS	200	2003	Setiawan et al. (2008a,b)
Binary and multiple stars:				
CORAVEL single-lined (SB1)	ELODIE+CORALIE	101	2001	Eggenberger (2010)
wide binaries, $a = 100 - 1000 \text{ AU}$	TNG–SARG	100	2003	Gratton et al. (2003)
binary/multiple	Keck–HIRES/HET–HRS	450	2003	Konacki (2005a,b)
visual binaries	Okayama–HIDES	9	2004	Toyota et al. (2009)

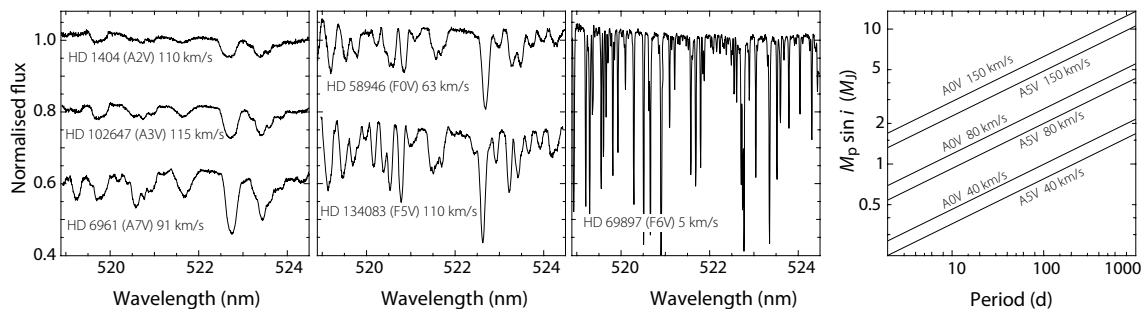


Figure 2.25: Left three panels: spectra of early-type stars acquired with ELODIE, showing the progression in both the number and the broadening of spectral lines for representative examples of A2V, A3V, A7V, F0V, F5V and F6V spectral types. Right: simulated mass detection limits for A-type stars observed with HARPS. From Galland et al. (2005a, Figures 1 and 10), reproduced with permission © ESO.

By 2008, nine planets around stars with $M \gtrsim 1.5 M_\odot$ had been reported by Johnson (2008), with 20 luminosity class III listed at exoplanet.eu in late 2010.

2.4.6 Other star categories

Open clusters The determination of stellar masses for giant stars is complicated by the fact that evolutionary tracks for stars covering a wide range of masses converge to the same region of the Hertzsprung–Russell diagram. This consideration motivated Sato et al. (2007) and Lovis & Mayor (2007, 115 stars in 13 clusters) to target giants in intermediate-age (0.2–2 Gyr) open clusters, where cluster membership combined with the estimated cluster age provides a more secure estimate of the host star masses compared to that of field giants. In contrast to the absence of transiting planets from searches in open clusters (§6.2.3), a few cluster planets have already been discovered from these radial velocity surveys. A $7.6 M_J$ planet orbiting the Hyades giant ϵ Tau was reported by Sato et al. (2007), while two other massive planets (10.6 and $19.8 M_J$ respectively) have been reported around giants in the open clusters NGC 2423 and NGC 4349 by Lovis & Mayor (2007).

Observations of 94 main sequence Hyades stars yielded no close-in giant planets (Cochran et al., 2002; Paulson et al., 2002, 2003, 2004a,b).

Metal-poor stars The observed correlation between exoplanet occurrence and host star metallicity suggests that radial velocity searches around metal-poor stars should have relatively low success rates. Until 2008, the low-metallicity record was held jointly by HD 155358 (Cochran et al., 2007), and HD 47536 (Setiawan et al., 2003a), both with $[\text{Fe}/\text{H}] = -0.68$.

Specific searches around low-metallicity stars have been targeted since 2003, for example with HARPS (Santos et al., 2007), FEROS (Setiawan et al., 2010), and with Keck–HIRES (Sozzetti et al., 2006, 2009). The results confirm that giant planets around low-metallicity stars

are less common, with only one ($[\text{Fe}/\text{H}] = -0.49$) having been detected in five years from the 105 stars surveyed by Santos et al. (2007).

Nevertheless, planets around stars of still lower metallicity have been discovered. These include HIP 13044, a member of a Galactic halo stream (Helmi et al., 1999) with $[\text{Fe}/\text{H}] = -2.0$ (Setiawan et al., 2010).

Young stars While there is a general consensus that planets form from the disks of dust and gas around newly-formed stars, details of their early formation, including the time scale of planet formation, remain unclear. Searches for planets around young stars with protoplanetary disks, of age 1–100 Myr, are being targeted by FEROS/HARPS (Setiawan et al., 2008a). Setiawan et al. (2007) reported a $6.1 M_J$ planet around HD 70573, a GV star with an age of about 100 Myr. Setiawan et al. (2008a) reported a planet of mass $9.8 M_J$ around TW Hya, a nearby young star with an age of 8–10 Myr that is surrounded by a well-studied circumstellar disk. The planet orbits the star with $P = 3.56$ d ($a = 0.04$ AU), inside the inner rim of the disk, perhaps demonstrating that planets can form before the disk has been dissipated by stellar winds and radiation.

Binary and multiple stars Results and specific programmes aimed at characterising planets around binary and multiple stars are considered in §2.7.

2.5 Properties of the radial velocity planets

As discoveries proceed, more significant statistical analyses of the planetary population are becoming possible, with statistically distinct properties and correlations becoming evident through principal component or hierarchical clustering analyses (e.g. Marchi, 2007). Various reviews of exoplanet properties have been given as discoveries have advanced (e.g. Marcy et al., 2006; Udry & Santos, 2007; Santos, 2008; Johnson, 2009).

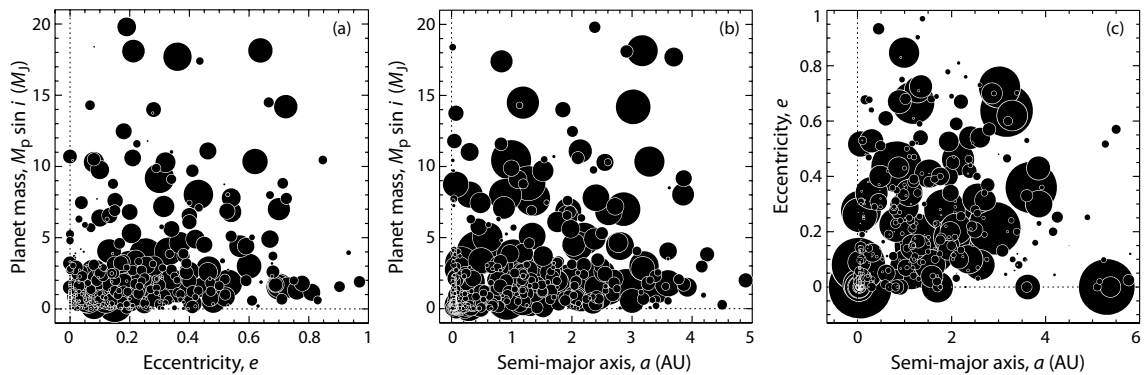


Figure 2.26: Properties of the radial velocity planets: (a) planet mass $M_p \sin i$ versus eccentricity e , with circle sizes proportional to a ; (b) planet mass $M_p \sin i$ versus semi-major axis a , with circle sizes proportional to e ; (c) eccentricity e versus semi-major axis a , with circle sizes proportional to $M_p \sin i$. Data are from *exoplanet.eu*, 2010–11–01.

2.5.1 Frequency of massive planets

For the target list of 1200 FGKM dwarfs in the solar neighbourhood monitored by the California and Carnegie Planet Search programme (Wright et al., 2004), Marcy et al. (2008) report that 87% of stars observed for more than a decade show no Doppler variations at a 3σ limit of 10 m s^{-1} . These limits largely exclude the existence of a significant number of undiscovered Jupiter-mass planets within 3 AU, and Saturn-mass planets within 1 AU.

At least 6–7% have giant planets, with $M_p > 0.5 M_J$ and $a < 5 \text{ AU}$, a similar fraction also being reported by the Geneva group (Udry & Mayor, 2008). Some 15% of these fall into the category of ‘hot Jupiters’, with $P \lesssim 10 \text{ d}$, and orbiting very close to their host stars.

The remaining 6% of stars show long-term radial velocity trends, often with significant curvature, indicating sub-stellar, brown dwarf, or planetary companions with orbital periods of a decade or more (Patel et al., 2007).

2.5.2 Mass distribution

Various distributions of the radial velocity planet population are shown in Figures 2.26–2.27. In the following discussions, it is recalled that for any individual object the measured mass is uncertain by the factor $\sin i$. Similarly, the underlying distribution of M_p (rather than that of the minimum mass $M_p \sin i$) cannot be derived unambiguously (cf. Jorissen et al., 2001). For a random orientation of orbital planes, $\langle M_p \rangle = (\pi/2) M_p \sin i$.

Figure 2.27 shows that the distribution of minimum masses continues to rise at least as far as Saturnian masses. A power-law fit gives $dN/dM \propto M^{-1.15}$ (Marcy et al., 2008), with a cut-off above $\sim 12 M_J$. Incompleteness at smaller masses, due to the increased difficulty of planet detection for decreasing mass, indicates that the mass distribution probably rises even more steeply towards lower masses.

Brown dwarf desert A prominent feature of the early radial velocity exoplanet discoveries was the general absence of substellar objects with masses in the range $10 - 80 M_J$, a phenomenon which was referred to as the ‘brown dwarf desert’. This clear mass separation between stellar and planetary companions to solar-type stars has continued, and is taken to signify evidence of a different formation mechanism for these two populations. Doppler surveys of several thousand stars have yielded only a small number of brown dwarf candidates in this mass range. Similar statistics result from comprehensive infrared coronagraphic searches (McCarthy & Zuckerman, 2004). The majority of these may be hydrogen-burning stars with low orbital inclination (Halbwachs et al., 2000; Udry et al., 2000), and this absence of brown dwarfs renders exoplanets distinguishable by their high occurrence at low masses, irrespective of the $\sin i$ uncertainty.

Low-mass planets Below $\sim 0.1 M_J$, and despite the progressively smaller radial velocity variations and observational incompleteness, which together render the detection of lower mass planets more difficult, the distribution of masses appears to rise (Udry & Mayor, 2008).

As of late 2010, eight planets are known with $M \sin i < 0.02 M_J$ ($\sim 5.4 M_\oplus$), of which two were transit discoveries (CoRoT-7 b and GJ 1214 b). The six others are close-in planets ($a < 0.07 \text{ AU}$, $P < 5 \text{ d}$ except for GJ 581 c), with two of the lowest orbiting the low-mass M dwarf GJ 581. The are, in order of mass, GJ 581 e ($1.9 M_\oplus$, HARPS; Mayor et al., 2009a), HD 156668 b ($4.1 M_\oplus$, Keck; Howard et al., 2011a), HD 40307 b ($4.2 M_\oplus$, HARPS; Mayor et al., 2009b), 61 Vir b ($5.1 M_\oplus$, Keck/AAT; Vogt et al., 2010b), GJ 581 c ($5.4 M_\oplus$, HARPS; Udry et al., 2007), and HD 215497 b ($5.4 M_\oplus$, HARPS; Lo Curto et al., 2010).

Low-mass planets are being frequently found as members of multiple planet systems, and may indeed be found preferentially in multiple systems (Lo Curto

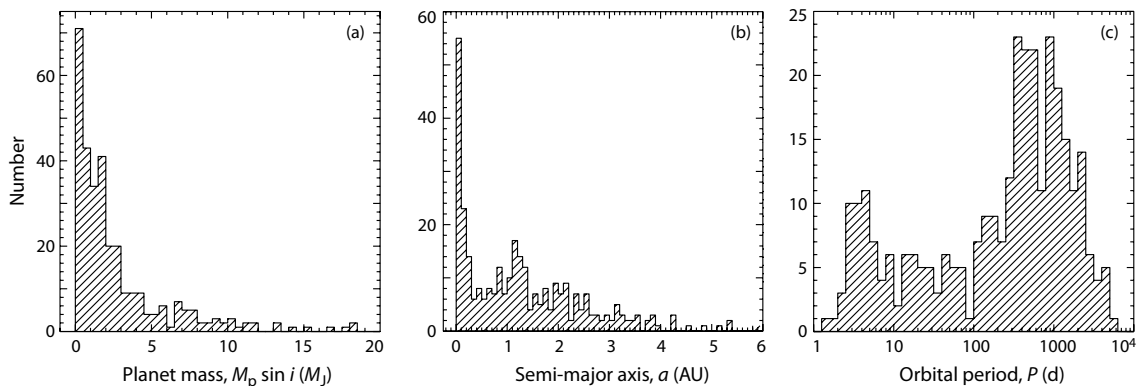


Figure 2.27: Histograms for the planets discovered from radial velocity surveys: (a) planet masses, $M_p \sin i$ (M_J); (b) semi-major axes, a (AU); (c) orbital periods (log scale), P (d). Data are for 383 planets from exoplanets.org, 2010–11–01.

et al., 2010). Amongst these are the $7.5M_{\oplus}$ GJ 876 d in the four-planet system (Rivera et al., 2005), the Neptune-mass three-planet system around HD 69830 (Lovis et al., 2006; Ji et al., 2007), the three super-Earth system HD 40307 (Mayor et al., 2009b), and the two-planet system HD 215497 (Lo Curto et al., 2010). The M dwarf GJ 581 hosts as many as six planets $\leq 7M_{\oplus}$ (Udry et al., 2007; Vogt et al., 2010a), with two or three within or at the edges of its habitable zone (§11.9.2).

The discovery of so many low-mass planets, close to the detection threshold and over a relatively short period of the high-precision Doppler surveys, suggests the existence of a large population of low-mass planets, perhaps reaching 30% for G and K dwarfs (Mayor et al., 2009b). Some of the scatter in radial velocity measurements for individual objects is probably attributable to undetected, low-amplitude, multi-planet systems.

Bimodality in the mass distribution is seen in the homogeneous HARPS results (Udry & Mayor, 2008), and is predicted by Monte Carlo simulations of accretion-based formation models (Mordasini et al., 2008).

2.5.3 Orbits

Period distribution The distribution of orbital periods is shown in Figure 2.27c. There is a ‘pile-up’ of planets at $P \sim 3$ d ($a \sim 0.05$ AU), together with a possible minimum in the interval $P \sim 10 - 100$ d ($a \sim 0.1 - 0.4$ AU).

The significant numbers of giant planets with $P \sim 3 - 10$ d ($a \sim 0.05 - 0.1$ AU), orbiting close to their host stars, was essentially unexpected before the early exoplanet discoveries.¹⁰ The early trend has continued: 50 planets have $P < 10$ d. As discussed elsewhere, these

‘hot Jupiters’ are believed to have formed much further out, before being scattered or migrating inwards to their present location, with some mechanism halting the migration before the planets fall onto their host stars.

The observed masses and orbits of giant planets within 5 AU are reproduced rather well by current theories of their formation and migration (Kley et al., 2005), including their dependence on stellar metallicity and mass (Ida & Lin, 2004b, 2005b; Armitage, 2007b).

The period distribution for $P \geq 1000$ d ($a \geq 2$ AU) is less certain, with the apparent decline beyond 3 AU probably largely attributable to the limited duration, of around 10 years, of the high-precision Doppler surveys (Kholshchikov & Kuznetsov, 2002). Some of the best constraints, out to some 8–10 AU, come from the almost 30-yr baseline for 17 objects included in the early CFHT and McDonald Observatory observations (Wittenmyer et al., 2006, 2007). Extrapolation out to 20 AU, comparable to the orbit of Uranus, is correspondingly uncertain, but even a flat extrapolation would approximately double the known rate (Marcy et al., 2005a). This suggests that a large population of still undetected Jupiter-mass planets may exist between 3–20 AU, perhaps implying that as many as 20% of Sun-like stars have a giant planet within 20 AU (Cumming et al., 2008).

Attempted adaptive optics imaging of giant planets around young stars from their thermal infrared emission using Gemini north–Altair (Lafrenière et al., 2007a) and VLT–NACO (Apai et al., 2008) has yielded independent limits on their occurrence, also suggesting an absence of giant planets beyond some 20–30 AU.

Current theories support the idea that gas giants may be rare beyond 20 AU. The declining densities of protoplanetary disks with increasing radius, combined with the longer dynamical time scales, together imply a longer time scale for planetary growth compared with protoplanetary disk lifetimes of order 3 Myr (Alibert et al., 2005a; Hubickyj et al., 2004).

¹⁰Although Struve (1952), in considering the timeliness of radial velocity searches, commented that ‘It is not unreasonable that a planet might exist at a distance of $1/50$ AU... Its period around a star of solar mass would then be about 1 day.’

Mass–period relation A correlation is seen between M_p and a (Figure 2.26b). There is a paucity of massive planets on short-period orbits (Zucker & Mazeh, 2002; Udry et al., 2002; Jiang et al., 2007). There is also a rise in the maximum planet mass with increasing a (Udry et al., 2003b). Since the more massive planets are presumably formed further out in the protoplanetary disk, where accretion material is abundant and orbital paths longer, the resulting larger masses are perhaps less easily displaced by whatever migration mechanism forces the inward movement.

Eccentricities Most pre-discovery theories of planetary formation suggested that exoplanets would be in circular orbits similar to those in the solar system (Boss, 1995; Lissauer, 1995). In practice (Figure 2.26), there is a significant correlation between a (or P) and e (Stepinski & Black, 2000). Close-in planets are in preferentially low eccentricity orbits, while exoplanets with $P \geq 6$ d have e spanning the range 0–0.93, with a median $e \sim 0.3$. A few, long-period, low-eccentricity orbits are found, representing a small sample of solar system analogues.

As discussed elsewhere, the origin of the eccentric orbits probably arises from several mechanisms: gravitational interaction between multiple giant planets; interaction between the giant planets and planetesimals in the early stages of planet formation; or the secular influence of an additional passing mass, either unbound or bound (with the four most extreme eccentricities being found orbiting components of binary systems). Furthermore, for small pericentre distances, tidal circularisation appears to be an important damping mechanism.

Solar system analogues The high frequency of eccentric orbits amongst giant planets with semi-major axis $a \gtrsim 1 - 2$ AU suggests that common perturbation mechanisms may cause non-circular orbits both for the giant planets themselves, and for any lower mass rocky planets closer in. The eccentric giant planets tend to eject lower-mass objects, or enhance orbital eccentricities.

Around solar-mass stars, analogues to Jupiter, both in terms of low eccentricity and comparable orbital semi-major axis, still seem to offer the best prospects for harbouring Earth-mass planets moving in circular orbits within the habitable zone. The $0.94 M_\odot$ star 55 Cnc has a low-eccentricity ($e = 0.06$) $3.8 M_J$ planet with $a = 5.8$ AU, not too dissimilar from Jupiter, although with four large planets orbiting inward of 1 AU.

A better Jupiter analogue orbits the $0.88 M_\odot$ star HD 154345 (Wright et al., 2007). With $M_p \sin i = 0.92 M_J$, $P = 9$ yr, $a = 4.2$ AU, and $e = 0.16$, the planet moves in a nearly circular orbit with no other giant planets lying inward of it. There may be additional planets with $P = 45 - 60$ d, while the star itself has a heavy element abundance close to solar, with $[\text{Fe}/\text{H}] = -0.10 \pm 0.04$ (Valenti & Fischer, 2005).

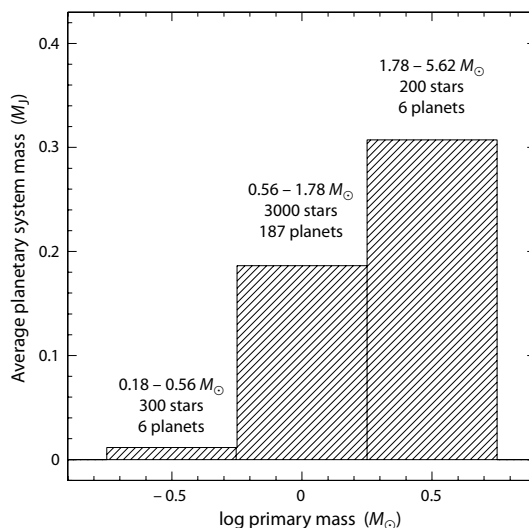


Figure 2.28: Average planetary system mass as a function of stellar mass, for all planets known as of February 2007. More massive stars seem to harbour more massive planetary systems. From Lovis & Mayor (2007, Figure 11), reproduced with permission © ESO.

2.5.4 Host star dependencies

Metallicity of host star A correlation between the presence of gas giant planets and high metallicity of the host star was noted in the early years of exoplanet discoveries (Gonzalez, 1997). More homogeneous studies have confirmed the trend, with investigations extending to the abundances of numerous elements in addition to iron. The consensus is that the effect is attributable to primordial effects rather than to self-enrichment by accretion, and is considered in further detail in §8.4. The correlation between occurrence and metallicity may not extend to the lowest mass planets (Udry et al., 2006), nor to those orbiting giants stars (Pasquini et al., 2007).

Mass of host star The results of the surveys of FGK main sequence stars in the mass range $0.3 - 1.4 M_\odot$, along with the few, typically low-mass planets detected around M dwarfs, and the more massive planets found from the evolved star surveys, suggest that the material trapped in the form of planets is strongly correlated with the mass of the primary star (Figure 2.28).

If the surface density of disk material furthermore scales with the central star mass, there should be an additional correlation between planet occurrence and stellar mass (Laughlin et al., 2004a; Ida & Lin, 2005b; Kennedy & Kenyon, 2008b).

2.6 Multiple planet systems

This section opens with some general considerations relevant for multiplanet systems, including an introduc-

tion to resonances, stability, and chaos, before considering a number of specific systems within five broad and non-exclusive categories (following Wright, 2010b): systems with three or more giants (§2.6.4), systems with two giants in mean motion resonance (§2.6.5), other systems in which planet–planet interactions are non-negligible (§2.6.6), non-interacting systems (§2.6.7), and systems containing only lower mass planets with $M_p \sin i < 20M_\oplus$ (§2.6.8).

2.6.1 General considerations

Frequency of multiple systems The first radial velocity target known to comprise multiple planets was the triple planetary system ν And (Butler et al., 1999). This was followed by the discovery of a second planet orbiting 47 UMa (Fischer et al., 2002), and a resonant pair orbiting GJ 876 (Marcy et al., 2001a).

Some 10–15% of all planetary systems are presently known to be multiple (Figure 2.29), with a further comparable fraction showing evidence for multiplicity in the form of long-term radial velocity trends.

An increasing number of multiple systems are being discovered, partly as a result of improved accuracies, and partly as a result of longer measurement baselines. The tenth triple system, HIP 14810, was announced by Wright et al. (2009a). μ Ara and GJ 876 each comprise four planets, 55 Cnc counts five, GJ 581 possibly has six, and HD 10180 up to seven.

Longer temporal coverage of known systems is leading to the discovery of outer companions with long orbital periods. Nine radial velocity planets have $P \geq 10$ yr (including μ Ara e, with 55 Cnc d and HD 134987 c at $P \sim 15$ yr, and $P \sim 40$ yr suggested for 47 UMa d). Improved radial velocity accuracy is leading to the discovery of numerous low-mass planets in multiple systems, with around 20 now known (in total) below $0.03M_J$ ($10M_\oplus$).

Multiple systems and theories of formation As developed in detail in Chapter 10, the core accretion model provides a compelling scenario for giant planet formation. In brief, from an embryonic disk of dust and gas, dust particles collide and grow to form progressively larger planetary cores. If a significant amount of gas remains in the disk, a sufficiently massive core can gravitationally accrete more gas, rapidly growing in mass. Such giant planets are likely to form beyond the snow line, at around 3 AU for solar-type stars, where ices can participate in the initial planetary cores. The fact that some 20% of known exoplanets orbit within 0.1 AU, where little ice is available, leads to the hypothesis that short-period planets formed far out and migrated inwards to their present locations.

The discovery of numerous systems in or near mean motion resonance lends support to the migration hypothesis, with hydrodynamical and N-body simulations

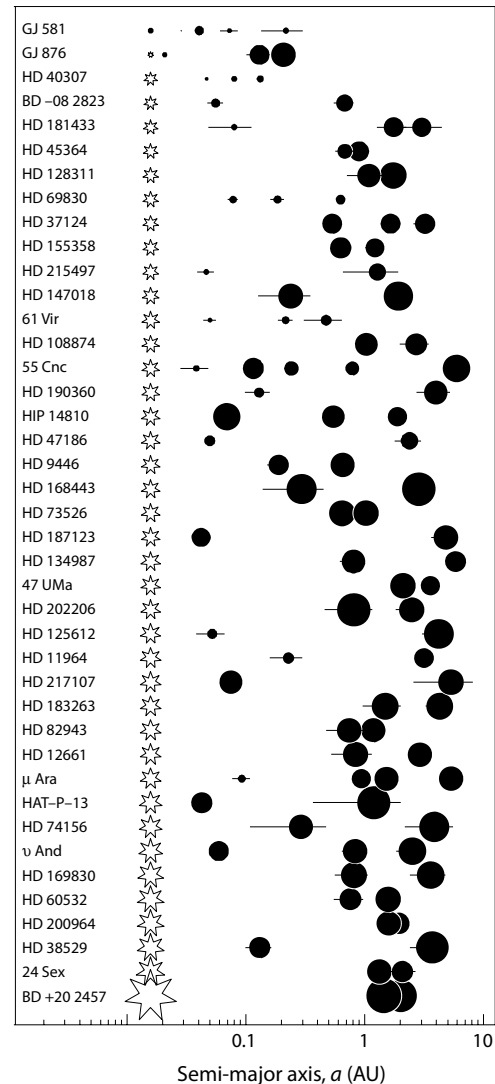


Figure 2.29: Multiple planet systems, ordered by host star mass (indicated at left with size proportional to $M_\star$, ranging from $0.31M_\odot$ for GJ 581 to $2.8M_\odot$ for BD +20° 2457). Each planet in the system is shown to the right, with sizes proportional to $\log M_p$ (ranging from about $0.01 - 20M_J$). Horizontal bars through the planets indicate maximum and minimum star-planet distance based on their eccentricities. Data are for 97 planets in 41 systems from exoplanets.org, 2010–11–01. From a concept by Marcy et al. (2008, Figure 13).

with externally applied damping suggesting that the 2:1 mean motion resonance (and others) appear naturally as the planets migrate inwards at different rates.

Various mechanisms may be responsible for the wide distribution of observed eccentricities, some driving eccentricities to larger values while damping others to low values. An early era of strong planet–planet scattering producing large values may be followed by a damping phase as a result of subsequent interactions

with remaining planetesimals. Evidence for these various mechanisms are detailed in §2.6.4 and §2.6.5.

Coplanarity The $\sin i$ uncertainty for each planet means that the extent to which multiple systems have coplanar orbits is not well constrained by present observations. Numerical simulations, such as those by Thommes & Lissauer (2003), Adams & Laughlin (2003), and others, suggest that a significant fraction of planetary systems involving giant planets may be substantially non-coplanar. Dynamical mechanisms that lead to fast amplification of the relative inclination are especially effective in the first-order resonance configurations (Thommes & Lissauer, 2003). Also, dynamical relaxation and collisional scattering of protoplanets may favour large relative inclinations, even if they initially emerge in a flat protoplanetary disk.

Constraints on the relative orbital inclinations of individual systems are frequently presented on the basis of long-term numerical orbit integrations, which may reveal islands of stability or instability for certain hypothesised relative inclinations. Direct observations, from astrometric measurements, are restricted to systems with large astrometric signatures for each individual planet (§3.1). v And, observed with HST-FGS, currently provides the most convincing example (§3.8.2), with the large relative inclination between planets c and d ($\Delta i_{cd} = 29^\circ.9 \pm 1^\circ$) lending preliminary support to the simulation results.

Statistics of multiple planet systems The number of multiple systems is presently still relatively small, while initially single systems are progressively becoming multiple systems with the discovery of new planets. Nevertheless some differences are evident between single and multiple systems.

Most striking is that the pile-up of hot Jupiters between 0.03–0.07 AU, and the discontinuity at ~ 1 AU observed in the distribution of single systems, are both absent from the more uniform distribution seen in multiplanet systems (Figure 2.30). Systems with multiple giant planets appear to lead to a suppressed occurrence of close-in planets. Both characteristics presumably reflect details of migration in the two types of system. Various other differences in the properties of single and multiple systems, in $M_p \sin i$, e , and $M_\star$, the implications for migration models, and the possibilities of selection effects in the present samples, are considered further by Wright et al. (2009b).

Dynamical modeling In the basic kinematic (or ‘Keplerian’) fitting approach (§2.1.1), the total radial velocity signal due to an orbiting system of n_p planets is assumed to result from the independent reflex motions due to each planet separately.

Many multiple planetary systems may, however, show effects of gravitational planet–planet interactions,

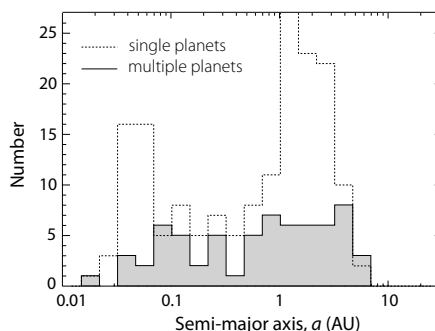


Figure 2.30: Distribution of semi-major axes for multiple planet systems (solid) and apparently single systems (dashed). The pile-up of hot Jupiters, and the jump in abundance beyond 1 AU seen in the single planet systems are not evident in the multiple planet systems. From Wright et al. (2009b, Figure 9), reproduced by permission of the AAS.

even over relatively short time scales. These can lead to observable evolution of the orbital parameters over periods of years, and to radial velocity variations of the central star that differ substantially from velocity variations derived assuming the planets are executing independent Keplerian motions. Dynamical analyses of these more complex systems are made using either of two approaches.

N-body numerical integration methods are used both for dynamical (or ‘Newtonian’) orbit fitting (see also §2.1.1), and for studying the detailed planetary motions over relatively short time intervals in the future, including testing the validity of analytical results. N-body integration rests on the basic laws of gravity and motion. Results may yield ‘deprojected’ planetary masses (without the $\sin i$ ambiguity) and, in favourable cases based on stability arguments, relative orbital inclinations.

Analytical methods make use of the *disturbing function*, the difference in gravitational potential due to a star alone, and that due to a star and other perturbing planets. Analytical theory successfully describes two principal phenomena seen in multiple systems: secular (non-periodic) evolution, and resonances.

In *secular theory*, terms that depend on the planet’s mean motion, n , as well as other higher-order orbit terms, are ignored, and the theory describes the system’s *secular evolution*, essentially predicting how the shape of an average orbit evolves with time. In most two planet systems, secular theory predicts that their eccentricities oscillate, with an increasing eccentricity of one planet (corresponding to an increase of orbital angular momentum) being accompanied by decreasing eccentricity of the other through conservation of angular momentum. Secular theory also predicts the general behaviour of the difference in the two longitudes of pericentre, $\Delta\omega$: depending on initial conditions, $\Delta\omega$ may oscillate around 0 (aligned libration), 2π (anti-aligned li-

bration), or circulate through 2π .

In *resonant theory*, terms that depend on the mean motions are included, but only those related to the resonance under study are considered.

2.6.2 Resonances

Orbital resonances arise when two orbiting bodies exert a regular, periodic gravitational influence on each other as a result of simple numerical relationship between frequencies or periods. The repetitive force between them does not average to zero over long time scales, and secular theory is therefore not applicable. The periods involved can be the rotation and orbit periods of a single body (spin-orbit coupling), the orbital periods of two or more bodies (orbit-orbit coupling), or other more complicated combinations of orbit parameters, such as eccentricity versus semi-major axis or orbit inclination (Kozai resonance).

Resonances can result in unstable interactions, in which the bodies exchange momentum and their mutual orbits evolve. In other circumstances a resonant system can be stable and self correcting, so that the bodies remain in resonance indefinitely. Some of the many manifestations of orbital resonances observed in the solar system (§12.6) have already been observed in the orbits of exoplanets (e.g. Kley, 2010).

Mean motion resonances A significant number of multiple exoplanet systems appear to have orbital periods which are related by commensurabilities of the form $P_1/P_2 \approx i/j$, where subscripts 1, 2 refer to the inner and outer planets, and i and j are small integers. These commensurabilities arise from dynamical considerations; as noted below, exact commensurability is not a resonance condition *per se*.

The concepts and analytical tools used to understand resonance phenomena in the solar system have been developed over more than a century (Peale, 1976). Murray & Dermott (2000) provide an exposition of secular and resonant theory, including a detailed treatment of mean motion resonances. The following summarises the concepts and terminologies most widely encountered in exoplanet studies to date.

Consider two planets moving in circular coplanar orbits which satisfy

$$\frac{n_2}{n_1} \approx \frac{p}{p+q}, \quad (2.45)$$

where $n_1 = 2\pi/P_1$ and $n_2 = 2\pi/P_2$ are the mean motions of the inner and outer planets, and p and q are integers. If the two planets are at *conjunction* at time $t = 0$ (in alignment, independently of where along the orbits this might occur), then the next conjunction occurs when

$$n_1 t - n_2 t = 2\pi, \quad (2.46)$$

and the time interval between successive conjunctions is therefore given by

$$\Delta T = \frac{2\pi}{n_1 - n_2} = \frac{p}{q} \frac{2\pi}{n_2} = \frac{p}{q} P_2 = \frac{p+q}{q} P_1, \quad (2.47)$$

from which

$$q \Delta T = p P_2 = (p+q) P_1. \quad (2.48)$$

For $q = 1$, each planet completes an integer number of orbits between successive conjunctions, with every conjunction occurring at the same longitude in inertial space. More generally, every q -th conjunction occurs at the same longitude, with q defining the *resonance order* of the *mean motion resonance*.

If the outer planet moves in an eccentric orbit with $e_2 \neq 0$ and $\dot{\omega}_2 \neq 0$, i.e. if the longitude of the pericentre precesses, then resonances can still occur if

$$\frac{n_2 - \dot{\omega}_2}{n_1 - \dot{\omega}_2} = \frac{p}{p+q}, \quad (2.49)$$

in which case

$$(p+q)n_2 - p n_1 - q \dot{\omega}_2 = 0. \quad (2.50)$$

Now, every q -th conjunction takes place at the same point in the outer planet's orbit, but no longer at the same longitude in inertial space. This relation encapsulates the dynamical significance of mean motion resonance. It also shows that the near (but not necessarily exact) commensurabilities in orbital periods are a consequence of (typically) small orbital precession.

Expressed in terms of the mean longitude, λ , and generalised to both orbits, corresponding *resonant arguments* (or resonant angles), ϕ , can be identified as

$$\phi = (p+q)\lambda_2 - p\lambda_1 - q\tilde{\omega}_{1,2}. \quad (2.51)$$

For the 2:1 and 3:1 resonances, for example, the resonant arguments take the form

$$\phi(2:1) = 2\lambda_2 - \lambda_1 - \tilde{\omega}_{1,2} \quad (p=1, q=1), \quad (2.52)$$

$$\phi(3:1) = 3\lambda_2 - \lambda_1 - 2\tilde{\omega}_{1,2} \quad (p=1, q=2). \quad (2.53)$$

The resonant argument measures the angular displacement of the two planets at their point of conjunction, with reference to one or other of the pericentres.

Resonant dynamics are important if ϕ varies slowly relative to the orbital motion. If, rather than circulating, the resonant argument is stationary, or oscillates (librates), then the planets are in resonance. *Exact resonance* describes the particular combination of mean motions and precession rates for which the time variation of a particular resonant argument is zero ($\dot{\phi} = 0$), while the term *deep resonance* is used to describe systems with small libration amplitudes. Outside of exact resonance, the orbital elements $a, e, \tilde{\omega}$ and the resonant argument ϕ all evolve with time.

Physics of resonance: Peale (1976) gives a heuristic physical description of resonance, reproduced here in outline.

Consider a test particle of negligible mass moving in an eccentric outer orbit around a more massive object. Assume a fixed line of apsides, precise commensurability of the mean motions, and conjunctions occurring close to, but not at, apocentre.

Net tangential forces increase the orbital angular velocity of the test particle, moving the successive conjunctions closer to apocentre. Similarly, conjunctions occurring after apocentre move successive conjunctions back towards apocentre. Conjunctions thus librate stably about the apocentre, preserving commensurability. A secular variation in the pericentre longitude $\tilde{\omega}$ does not alter this fundamental libration. Similarly, pericentre conjunctions correspond to an unstable equilibrium configuration (Figure 2.31).

For conjunctions occurring precisely at the apocentre of the outer test particle, radial forces accelerate it towards the primary, resulting in a trajectory slightly inside its nominal orbit. The test particle accordingly reaches its closest approach to the primary slightly earlier, and the line of apsides has rotated in a retrograde sense as a result. This regressive motion of the line of apsides can reduce, or even dominate, the normal prograde motion resulting from the oblateness of the primary. This type of orbital resonance also leads to a secular increase of the eccentricity.

The more general case In the more general analysis of orbital resonances, the time derivatives of the orbital elements are expressed in terms of partial derivatives of the disturbing function. The general form of the argument which then appears in the expansion of this disturbing function is (Peale 1976, eqn 7; Murray & Dermott 2000, eqn 8.18)

$$\phi = j_1 \lambda_1 + j_2 \lambda_2 + k_1 \tilde{\omega}_1 + k_2 \tilde{\omega}_2 + i_1 \Omega_1 + i_2 \Omega_2, \quad (2.54)$$

where the variables λ , ω , Ω are confined to the arguments (angular dependencies) of the two orbits. Equation 2.51 is a special case of this more general form. More complex resonances have different numbers of resonant arguments involving various permutations of these terms, classified according to the relevance of the coefficients i, j, k (e.g. Peale, 1976, Table 2). The majority of resonance configurations are characterised by only one librating resonant argument (Michtchenko et al., 2008a,b).

Motion near the 2:1 (and other first-order) resonances depends on initial conditions, and includes libration (in a , e , and ϕ), apocentric libration, and inner and outer circulation (e.g. Murray & Dermott, 2000, Figure 8.16). The phase space boundary which separates librating (or oscillating) from circulating regimes (as in a pendulum) is determined by the total energy, and is referred to as the *separatrix*.

Examples of specific resonant exoplanet systems, and some inferences that can be made from them, are considered in §2.6.5. From the complexity of the mean motion resonance phenomenon it should be apparent

that very diverse systems, and complex behaviour in the long-term orbit integrations, can be expected.

Resonance capture and migration The stability of a system once in resonance is understood. The reason for the existence of so many multiple exoplanets in mean motion resonance was, initially, somewhat less evident.

The origin of orbital commensurabilities in the solar system is attributed either to dissipative processes early on in its formation, or to the slow differential increase in the semi-major axes of satellite orbits as a result of tidal transfer of angular momentum from the primary (Goldreich, 1965; Peale, 1976). Goldreich (1965) showed that a pair of satellites under the influence of their planet's tidal forces, initially with non-commensurable mean motions, will change their semi-major axes, the inner spiraling outwards until their periods become commensurable, thereafter maintaining this state as the orbits continue to evolve.

The present consensus is that the observed exoplanet orbital resonances could not have simply formed *in situ*. Rather, the evidence from both hydrodynamic and N-body simulations is that some planets have become trapped in specific resonances as a result of differential convergent migration in which dissipative processes steadily altered their semi-major axes and eccentricities (e.g. Bryden et al., 2000; Kley, 2000; Snellgrove et al., 2001; Lee & Peale, 2002; Nelson & Papaloizou, 2002; Papaloizou, 2003; Thommes & Lissauer, 2003; Kley et al., 2004, 2005; Lee, 2004). The occurrence of orbital resonances constitutes strong evidence for such migration having occurred.

As shown for example by Kley et al. (2004), a sufficiently slow migration of two giant planets embedded in a protoplanetary disk can end with either a 3:1 or 2:1 resonant system, depending on the rate of migration. For masses in the range $1 - 4 M_{\oplus}$, Papaloizou & Szuszkiewicz (2005) found somewhat similar behaviour using both hydrodynamic and N-body simulations. Slow convergent migration results in period ratios given by first-order commensurabilities $p+1:p$ with p as small as 1–2, while more rapid migration (typically characterised by larger mass ratios) results in higher p commensurabilities such as 4:3 or 5:4.

Various studies of the migration of individual planets have characterised the motion by a migration rate, $\dot{a}/a$, and an eccentricity damping rate, $\dot{e}/e$. Corresponding e-folding times are given by

$$\tau_a = -\left(\frac{\dot{a}}{a}\right)^{-1} \quad \text{and} \quad \tau_e = -\left(\frac{\dot{e}}{e}\right)^{-1}. \quad (2.55)$$

For GJ 876, for example, Lee & Peale (2002) found that for a sufficiently slow migration, the final state depends only on the ratio of the e-folding times $K = \tau_a/\tau_e$. Resonant capture is suppressed for high drift or migration

rates, but depends on the detailed form of the resonant argument (Quillen, 2006b; Quillen & Faber, 2006).

From hydrodynamical calculations it has been proposed that K assumes values close to unity, reflecting the mass and viscosity of the protoplanetary disk (Kley et al., 2004). If such a system is subject to adiabatic migration, a given value of K results in specific values of the final eccentricities (Lee & Peale, 2002).

More on the observational evidence for resonance capture is given under the description of specific resonance systems (§2.6.5).

Apsidal motion Within the class of 2:1 resonances, for example, configurations of stability or instability are determined by the eccentricities of the inner and outer planets, while their apsidal alignment may also display different behaviour in different systems (Figure 2.31).

As first observed for the two planets GJ 876 b and c, both of the lowest-order eccentricity-type mean motion resonance variables (Equation 2.52)

$$\begin{aligned}\phi_1 &= 2\lambda_2 - \lambda_1 - \bar{\omega}_1 \quad \text{and} \\ \phi_2 &= 2\lambda_2 - \lambda_1 - \bar{\omega}_2\end{aligned}\quad (2.56)$$

librate around 0° . So therefore does the (apsidal resonance variable) quantity

$$\phi' = \bar{\omega}_1 - \bar{\omega}_2 \equiv \phi_1 - \phi_2, \quad (2.57)$$

meaning that the pericentres remain nearly aligned while librating, with conjunctions occurring (in this case) when both planets are near their pericentres. The deep resonance in GJ 876 is not only an inference from dynamical modeling, but the associated retrograde pericentre precession has been observed for many orbital periods as a consequence of their relatively close-in orbits ($P_b \sim 61$ d, $P_c \sim 30$ d).

A number of the 2:1 resonant exoplanet systems are observed to be in such a state of apsidal alignment, with their apocentres either aligned (e.g. GJ 876 and HD 37124) or anti-aligned (e.g. HD 60532, HD 73526, HD 155358, HIP 14810). The limited statistics suggest that alignment and anti-alignment might occur with approximately equal frequency (Zhou & Sun, 2003; Barnes & Greenberg, 2006a, 2007a; Barnes, 2008).

When it does occur, small-amplitude *apsidal libration* may manifest itself around equilibrium angles of 0° (aligned, or symmetric) or 180° (anti-aligned, or antisymmetric), but any other (asymmetric) angle can in principle serve as a stable equilibrium (Beaugé et al., 2003; Ji et al., 2003c,d; Lee, 2004; Voyatzis & Hadjidemetriou, 2005; Beaugé et al., 2006; Libert & Henrard, 2006; Marzari et al., 2006).

Not all theoretically-possible 2:1 resonance configurations can be achieved through migration alone, and their existence would provide further constraints on the system's evolution (Lee, 2004).

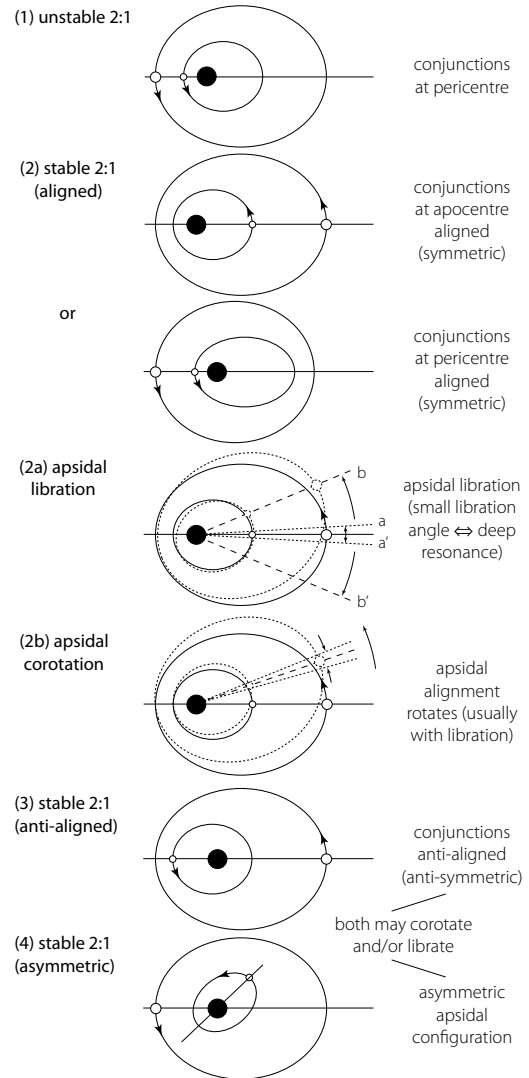


Figure 2.31: Schematic of the 2:1 mean motion resonance, and associated apsidal motion: (1) conjunctions at close approach (here at periastris) are unstable; (2) for stable resonances, the apsidal alignment may in practice librate over many orbits (2a), and orbital periods will not be precisely commensurate. Small libration angles (of a few degrees) correspond to deep resonance. Libration angles may be many tens of degrees, or the alignment angle may continuously rotate (2b), with additional libration superimposed; (3) conjunctions may also be anti-aligned and, again, either circulating and/or librating or, in theory, they may be asymmetric (4).

One mechanism proposed for establishing a significant apsidal libration in a pair of planets initially on nearly circular orbits invokes an impulsive generation of eccentricity, perhaps caused by the ejection of a planet (Malhotra, 2002). The model can match the aligned orbits observed in ν And (Chiang et al., 2001). An alternative mechanism, invoking adiabatic perturbations,

Hill radius and Hill stability: The *Hill radius* is the radius within which the gravity of one object dominates that of other bodies within the system. For a planet in a circular orbit of radius r , its Hill radius depends on the planet and central star mass as (e.g. Hamilton & Burns, 1992)

$$R_H = r \left(\frac{M_p}{3M_\star} \right)^{1/3}. \quad (2.58)$$

The concept of *Hill stability* is based on the more precise construct of zero-velocity surfaces introduced by Hill (1878) in his treatment of the lunar motion, and has since been used extensively in analytical considerations of both the restricted three-body problem, and the problem of general three-body motions. Although, in general, it is only possible to follow such close encounters in detail by numerical orbit integration of all three masses, the concept of Hill stability provides analytic limits on the range of the system's orbital elements if it is to remain stable, i.e. avoiding either disruption, collision, or positional exchange.

The parameter p^2E , where E is the energy and p is the angular momentum of the system, controls the topology of the zero-velocity surfaces, and determines the regions of allowed motions since the bodies may not cross the zero-velocity surfaces. The binary system is considered to be *Hill stable* against disruption or exchange if the value of $p^2|E|$ is greater or equal to the critical value derived for the corresponding three-body Hill surface defined by the position of the collinear Lagrange points. The condition is sufficient for stability, but is not necessary, such that exchange cannot occur if the condition is satisfied, but might not occur even if the condition is violated.

The theory was originally applied to coplanar systems in bound orbits. It was extended by Szebehely & Zare (1977) and Marchal & Bozis (1982) to bound stellar systems with mutually inclined orbits, by Gladman (1993) to the stability of planets on circular orbits, and by Veras & Armitage (2004a) and Donnison (2009) to the case of two massive planets with mutually inclined orbits. For further details, see, e.g. Murray & Dermott (2000); Veras & Armitage (2004a); Donnison (2006).

Conservative stability criteria based on multiples of their combined Hill radii (e.g. Hayes & Tremaine, 1998; Cuntz & Yeager, 2009) are also encountered.

could follow from tidal torques in a massive primordial disk providing the eccentricity excitation for the outer planet (Chiang & Murray, 2002). Malhotra (2002) suggested that a large apsidal libration amplitude would favour the impulsive mechanism, whereas small amplitudes (as for GJ 876) would favour the adiabatic.

In systems initially in a 3:1 mean motion resonance, dynamical studies show that apsidal librations are also common. Maps of dynamical stability nevertheless show a complicated structure in phase space, with chaos and order coexisting and alternating as the initial eccentricities or apsidal phases evolve (Voyatzis, 2008).

2.6.3 Long-term integration and system stability

Dynamical stability as applied to exoplanet systems can carry at least two somewhat different meanings (Barnes

& Greenberg, 2006c; Barnes, 2008): *Lagrange stability* in which no planet is ever ejected and the semi-major axes remain bounded for all time; and *Hill stability* or *hierarchical stability* in which the outermost planet might escape but not the inner, implying that the ordering of the bodies remains constant (see box).¹¹

The derivation of an orbit solution which matches the radial velocity observables for a multiple planetary system, as derived from either kinematic or dynamical fitting (§2.1.1), does not in itself guarantee that the resulting orbits are stable over long periods of time. The formal best-fit solution might, for example, describe an unstable system which would swiftly disintegrate through ejection of one of the planets, or by causing two of the planets to collide. If such unstable behaviour is predicted to occur on short time scales, the solution is typically rejected as implausible.¹² Conversely stability, even verified over long time periods, cannot prove the correctness of the solution, although it may be considered as a necessary condition for its validity.

The development of efficient numerical methods for exploring parameter space, or for studying long-term orbit evolution, has been important in understanding the large-scale and long-term dynamics of the solar system. These tools are now routinely applied to studies of the stability and evolution of multiple planets (e.g. Barnes & Quinn, 2004). Of particular importance has been the use of *symplectic integrators*, developed for numerical integration of systems governed by Hamilton's equations within a symplectic geometry (see box on page 44).

Wright et al. (2009b) describe their algorithmic approach to assessing system stability which proceeds as follows: orbits are assumed to be coplanar and edge-on, with stellar masses established independently (Takeda et al., 2007a). The radial velocity parameters are converted into initial conditions in a Jacobi coordinate system (Lissauer & Rivera, 2001; Lee & Peale, 2003). Systems are then integrated for $\geq 10^8$ yr using the hybrid integrator *Mercury* (Chambers, 1999). Solutions are classified as unstable if two planets are predicted to collide, or if a planet falls onto the star ($d < 0.005$ AU), or if it is ejected from the system ($d > 100$ AU).

¹¹ *Hierarchical* has also been used to describe exoplanet systems in which there is a significant ratio of their semi-major axes, $a_1/a_2 \leq 0.3$ (Lee & Peale, 2002; Goździewski & Konacki, 2004; Wright, 2010b). With these two distinct definitions, a hierarchical planetary system is not necessarily hierarchically stable. In multiple star terminology, hierarchical structuring describes orbits nested within significantly larger orbits.

¹² An example is the two-planet system HD 82943, with an age of 2.9 Gyr. Despite the genetic search algorithm leading from almost all initial values to the solution given by Mayor et al. (2004), Ferraz-Mello et al. (2005) have shown that the published orbits correspond to a fate in which one of the planets is expelled, or collides with the star, in less than 100 000 yr.

Symplectic integrators: Central to many long-term orbit studies is the use of *symplectic integrators*. In describing the motion of dynamical systems, a sequence of deterministic algebraic steps are executed to follow its evolution from an initial to some final state. Such *algebraic mappings* include the classical (non-symplectic) fourth-order Runge–Kutta iteration for the approximate solution of ordinary differential equations (e.g. Press et al., 2007). These conventional integrators carry the penalty of ‘numerical dissipation’, which results in a secular change of energy even in a conservative system. The consequence is that positional errors then grow quadratically with time.

Symplectic integrators were developed as a specific numerical integration for the solution of Hamilton’s equations, i.e. for particular problems occurring in classical mechanics within the appropriate (symplectic) geometry. Wisdom & Holman (1991), who called their method an ‘N-body map’, and Kinoshita et al. (1991), who called their method a ‘modified symplectic integrator’, developed such a mapping for the N-body problem, inspired by efforts to understand the very long-term stability of solar system orbits. Essentially, the Hamiltonian is split into an unperturbed Keplerian part, and an additional contribution describing smaller mutual gravitational interactions; each part can then be integrated separately. Separation involves the introduction of a new system of variables transformed from standard Cartesian to Jacobi coordinates, in which (hierarchically) the masses of bodies m_i and m_j are replaced by a virtual body of mass $m_i + m_j$, and their individual positions replaced by their relative positions and that of their centre of mass. For a step-by-step description see, e.g., Murray & Dermott (2000, Section 9.5.4).

The defining feature of symplectic integrators is that they preserve symplecticity; practically this means that the total energy and angular momentum do not exhibit secular evolution, with the consequent advantage that positional errors grow only linearly with time.

Use of symplectic integrators for the N-body problem is now widespread. Higher-order gravitational effects (e.g. quadrupole moments, general relativity, and stellar mass loss) can be included. Other developments include reducing long-term errors (Saha & Tremaine, 1992), introducing separate time steps for each body (Saha & Tremaine, 1994), improved treatment of close approaches (Levison & Duncan, 1994), allowing for dissipative terms such as drag, inelastic collisions, radiation forces, and tidal distortions (Malhotra, 1994), and special considerations for binary star systems (Chambers et al., 2002).

Symplectic integrators used for exoplanet studies include SWIFT (Levison & Duncan, 1994), the multi-time step SyMBA (Duncan et al., 1998), the hybrid integrator Mercury (Chambers, 1999), SABA4 (Laskar & Robutel, 2001), and HNBODY (Rauch & Hamilton, 2002). A hybrid N-body technique allowing for radial zones with distinct time steps has been developed by McNeil & Nelson (2009).

The effects of resonances and chaos can mean that stable and unstable solutions are only marginally separated in parameter space. The evolution of the system must therefore be explored for a range of parameters in the vicinity of the χ^2 minimum to ensure the solution’s stability. HD 183263 is an example of a two-planet

system where the χ^2 minimum falls close to a stability boundary, such that many solutions consistent with the data are unstable (Figure 2.32).

Dynamical classification Barnes (2008) has classified multiple planet systems according to the type of dynamical interaction that dominates. A system is considered tidally dominated if the semi-major axis $a \lesssim 0.1$ AU (Rasio et al., 1996), resonantly dominated if one or more resonant arguments librates, and secularly dominated otherwise (as is the case for the solar system gas giants).

The general dynamical properties of multiple exoplanet systems, and a summary of their dynamical state, is given by Barnes (2008), where a tabulation provides, for each pair: the resonance, if applicable; the apsidal motion, viz. circulation or (anti-)aligned libration; proximity to the apsidal separatrix (Barnes & Greenberg, 2006b); proximity to the Hill stability boundary (Barnes & Greenberg, 2007b); and whether the orbit evolution is dominated by tidal, resonant, or secular interactions.

Of the 34 planet pairs classified by Barnes (2008), the numbers in each category were 10, 8 and 16 respectively.

Dynamical packing Many planetary systems are found to be dynamically ‘full’, in the sense that no additional companions can survive in between the observed planets, with most pairs of planets lying close to dynamical instability. This finding has led to the *packed planetary systems* hypothesis (Barnes & Quinn, 2004; Barnes & Raymond, 2004; Raymond & Barnes, 2005; Raymond et al., 2006a), and to general studies of orbital stability for system that are closely spaced (Smith & Lissauer, 2009).

Various searches have been made for regions of Lagrange stability between pairs of planets that are more widely separated (Menou & Tabachnik, 2003; Dvorak et al., 2003b; Barnes & Raymond, 2004; Funk et al., 2004; Raymond & Barnes, 2005; Raymond et al., 2006a; Rivera & Haghighipour, 2007; Barnes & Greenberg, 2007b). In a few systems (for example, in HD 74156 b–c, HD 38529 b–c, and 55 Cnc c–d) there are gaps which are large enough to support additional Saturn-mass planets.

A new Saturn-mass planet in the HD 74156 system was reported by Bean et al. (2008b). It lies between planets b and c, close to the stable orbit identified in the numerical simulations of Raymond & Barnes (2005). The discovery provides some support for the understanding of dynamical stability of multiple systems gained so far.

Long-term dynamical studies of the solar system (§12.6) demonstrate that, in its outer parts, it is not only very stable but also dynamically full (Gladman & Duncan, 1990; Holman & Wisdom, 1993; Levison & Duncan, 1993). The inner solar system is significantly less stable, but it too is dynamically full (Sussman & Wisdom, 1992) and Laskar (1994).

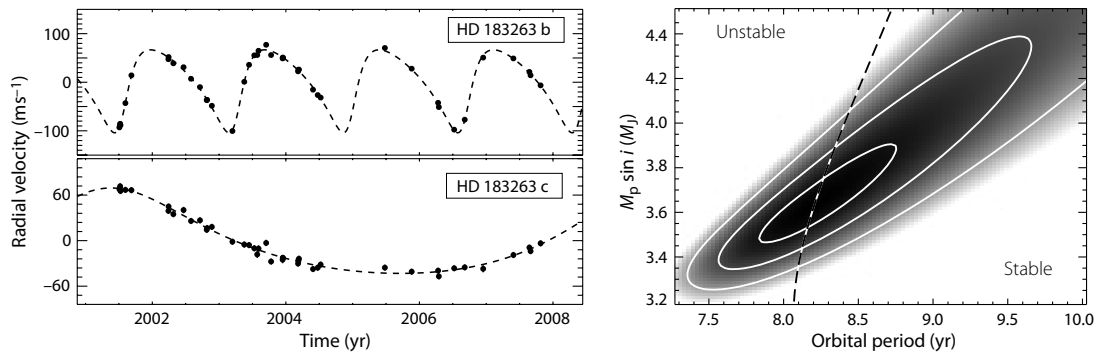


Figure 2.32: HD 183263 observed at the Keck Observatory. Left: the inner planet (b) with $P = 1.7$ yr, $M_p \sin i = 3.7 M_J$ (top) and the outer planet (c) with $P \sim 8$ yr, $M_p \sin i = 3.6 M_J$ (bottom); in each case, the effect of the other planet has been removed. Right: contours of χ^2 , with solid contour lines increasing by factors 1, 4, 9 from the minimum. The dashed curve divides stable from unstable orbital solutions, as determined from an ensemble of N -body integrations. From Wright et al. (2009b, Figures 2–3), reproduced by permission of the AAS.

Chaotic orbits An appreciation of the role of chaos has been an important development in the understanding of the solar system's dynamical evolution (Wisdom, 1987a,b; Murray, 1998; Murray & Holman, 2001), and the same principles are directly applicable to the study of multiple exoplanet orbits. In the solar system, chaos is apparent in phenomena such as the 3:1 Kirkwood gap (Wisdom, 1983), the tumbling motion of Saturn's moon Hyperion (Wisdom et al., 1984), and in elements of Pluto's orbit (Sussman & Wisdom, 1988), and has been attributed to result from overlapping resonances (Lecar et al., 2001).

Given a knowledge of initial conditions and forces acting upon it, the evolution of a conservative N -body system follows deterministic laws, in principle allowing the calculation of its past and future state with arbitrary accuracy. An orbit is loosely described as *chaotic* (as opposed to *regular*) if its dynamical state at some future time is sensitively dependent on the initial conditions.

Chaotic orbital motion may only become apparent after long times, for example only after 10^8 yr in the case of Pluto (Sussman & Wisdom, 1988), so that efficient methods of orbit integration, and procedures to quantify chaotic motion, are pivotal to its consideration.

The tendency to chaos can be quantified. The divergence of two trajectories in *phase space* (the six-dimensional mathematical space defined by its position and velocity vectors) with initial infinitesimal vectorial separation $\delta \mathbf{Z}_0$ can be written

$$|\delta \mathbf{Z}(t)| \approx e^{\Lambda t} |\delta \mathbf{Z}_0|, \quad (2.59)$$

where Λ is the *Lyapunov exponent*, or Lyapunov characteristic exponent (e.g. Lyapunov, 1892; Hénon & Heiles, 1964; Oseledec, 1968; Lichtenberg & Lieberman, 1983; Froeschlé, 1984; Murray & Dermott, 2000). The rate of separation can be different for different orientations of

the initial separation vector, and the number of Lyapunov exponents is therefore given by the dimensionality of the phase space. Reference is normally made only to the *maximal Lyapunov exponent* constructed from Equation 2.59

$$\Lambda = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|\delta \mathbf{Z}(t)|}{|\delta \mathbf{Z}_0|}, \quad (2.60)$$

since that dictates the predictability of the system.

The evolution of Λ , determined from numerical integration of the orbits, typically shows a marked difference between regular and chaotic behaviour: regular orbits tend to a slope of -1 in $\log \Lambda$ versus $\log t$, and chaotic orbits tend to some positive value of Λ . The corresponding *Lyapunov time*, the time taken for the displacement to increase by a factor e , is given by $1/\Lambda$.

As specific examples, both Sussman & Wisdom (1992) and Laskar (1996) found that the inner solar system is chaotic, with a Lyapunov exponent of $1/(5 \text{ Myr})$. The chaotic behaviour is traced to two secular resonances, one related to the perihelia and nodes of Mars and Earth, the other related to those of Mercury, Venus and Jupiter. As a consequence, an error of only 15 m in the Earth's initial position leads to an error of 150 m after 100 Myr, growing to 150×10^6 km after 100 Myr. It is thus conceivable to construct solar system ephemerides over 10 Myr, but practically impossible to predict planetary motions beyond 100 Myr. Similarly, the orbit of Pluto is found to be chaotic, with a Lyapunov exponent of $1/(20 \text{ Myr})$ (Sussman & Wisdom, 1988), although its small mass means that its chaotic orbit has negligible consequences for the other planets.

A system can be stable, at the same time as being chaotic, so long as its future evolution is restricted to within certain bounds. The absence of a clear pattern in the evolution of the orbital elements with time does not in itself imply that the orbit is chaotic, but nei-

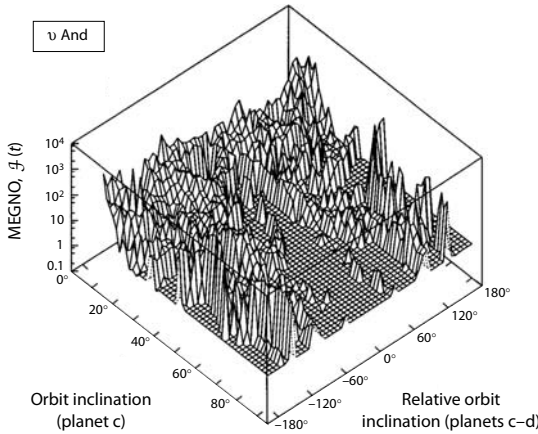


Figure 2.33: Stability analysis for ν And carried out with MEGNO using the Lick Observatory orbit data from Butler et al. (1999). The parameter $\mathcal{J}$ (Equation 2.64) develops along the vertical axis, as a function of both the orbit inclination of planet c, and the relative orbit inclinations of planets c and d. The flat areas indicate quasi-periodic zones, while the peaks indicate chaotic regions. From Goździewski et al. (2001, Figure 12), reproduced with permission © ESO.

ther does a chaotic orbit necessarily imply that the system will be catastrophically disrupted. There is no general relation, for example, between the Lyapunov time and the event time when a macroscopic instability of a planetary system occurs (Soper et al., 1990; Lecar et al., 2001; Michtchenko & Ferraz-Mello, 2001b). Neither does the Lyapunov time provide a measure of the ‘degree’ of chaos, since a system either is, or is not, chaotic.

Indicators of chaos Practical methods for rapidly distinguishing between regular and chaotic orbits include various types of frequency analysis (e.g. Laskar, 1990, 1993) and various types of fast Lyapunov indicators (e.g. Froeschlé et al., 1997; Guzzo et al., 2002).

For the four-planet system μ Ara, for example, Pepe et al. (2007) used the SABA4 symplectic integrator (Laskar & Robutel, 2001) to integrate over 2000 years with a step size of 0.02 yr. Orbit stability was measured using a simplified frequency analysis based on the mean motion of a given planet over two consecutive intervals of $t = 1000$ yr, n_p and n'_p . They used the parameter

$$D = \frac{|n_p - n'_p|}{t} \quad \text{deg yr}^{-1} \quad (2.61)$$

to provide a simplified measure of the orbit’s evolution, with values close to zero implying a regular solution, and with larger values indicating more chaotic motion.

Within the class of fast indicators is MEGNO (Mean Exponential Growth of Nearby Orbits; Cincotta & Simó, 2000), originally developed to study the global dynamics of non-axisymmetric galactic potentials. It has been used to investigate stability and resonant structure in

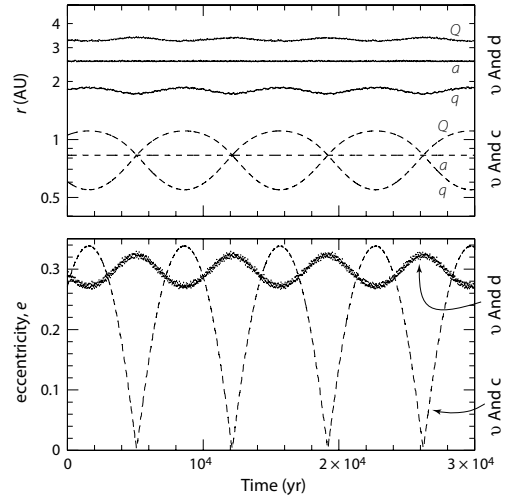


Figure 2.34: Secular evolution of the planetary system around ν And. Top: the semi-major axes, a (central lines), as well as the pericentre and apocentre distances (q and Q respectively), for the outer two planets, c and d. Bottom: evolution of the orbital eccentricities. Both planet c (lower, dashed) and planet d (upper, dotted) have a significant eccentricity at the present time ($t = 0$), although the eccentricity of c periodically returns to very small values near zero. From Ford et al. (2005, Figure 1), by permission from Macmillan Publishers Ltd, Nature, ©2005.

a number of multiple exoplanet systems, and typically performs over a relatively small number ($\sim 10^4$) of orbital periods of the outermost planet. Its essence is to amplify the effects of instability, such that signs of chaotic behaviour can be detected much earlier. This is seen by rewriting Equation 2.60 in integral form as

$$\Lambda = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \frac{\delta \dot{\mathbf{Z}}(t)}{\delta \mathbf{Z}(t)} dt, \quad (2.62)$$

then introducing the time-amplified *mean exponential growth of nearby orbits* $\mathcal{J}(T)$ (Cincotta & Simó, 2000, eqn 22) as

$$\mathcal{J}(T) \equiv \frac{2}{T} \int_0^T \frac{\delta \dot{\mathbf{Z}}(t)}{\delta \mathbf{Z}(t)} t dt, \quad (2.63)$$

with its evolving mean value

$$\bar{\mathcal{J}}(T) \equiv \frac{1}{T} \int_0^T \mathcal{J}(t) dt. \quad (2.64)$$

For regular orbits $\bar{\mathcal{J}}(T) \rightarrow 2$, while for chaotic orbits it grows, proportionally to the Lyapunov exponent, as $\bar{\mathcal{J}} \rightarrow (\Lambda/2)T$. Practical implementation requires the integration of two (straightforward) differential equations in addition to the underlying equations of motion.

Various multiple planet systems have been studied with this indicator, including the triple system ν And (Goździewski et al., 2001, and considered further below), HD 12661 (Goździewski, 2003b; Goździewski &

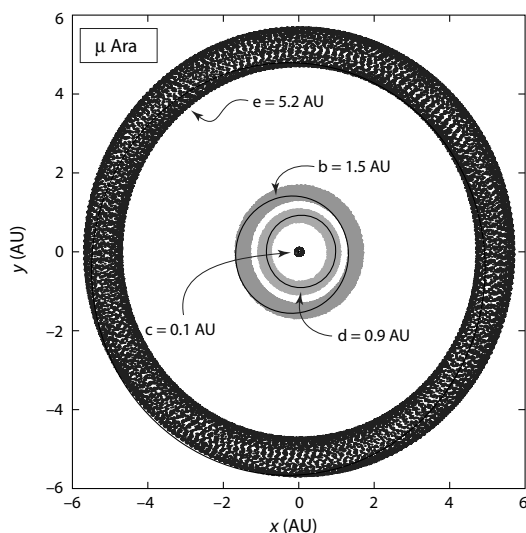


Figure 2.35: Face-on view of μ Ara (HD 160691), illustrating orbital evolution of the four planets over 1 Myr. Present orbits are shown as solid lines, while dots (indistinguishable here for the inner three planets) correspond to positions every 50 years in the dynamical integration. The semi-major axes are constant, while the eccentricities undergo small variations in the range 0.09–0.13, 0.16–0.21, 0–0.19, and 0.08–0.11 for planets b–e respectively. From Pepe et al. (2007, Figure 6), reproduced with permission © ESO.

Maciejewski, 2003), HD 37124 (Goździewski, 2003a; Goździewski et al., 2008a), HD 160691 $\equiv \mu$ Ara (Bois et al., 2003; Goździewski et al., 2003, 2005a), HD 169830 (Goździewski & Konacki, 2004), the quintuple 55 Cnc (Gayon et al., 2008), and the possibly retrograde resonance system HD 73526 (Gayon & Bois, 2008a).

A related chaos-detection method is through the use of *relative Lyapunov indicators*, which measure the difference between the convergence of the finite time Lyapunov indicators to the maximal Lyapunov characteristic exponent of two initially very close orbits (Sándor et al., 2000, 2004). The method is considered to be very fast in determining the ordered or chaotic nature of individual orbits, even those with very small Lyapunov characteristic exponents, with results over a few hundred times the period of the longest orbit. Examination of the stability of various exoplanet systems using this indicator has been variously described (e.g. Érdi et al., 2004; Sándor et al., 2004, 2007b).

2.6.4 Systems with three or more giant planets

ν And The first exoplanet system discovered to be multiple was ν And. In addition to the $0.6 M_J$ object in a low-eccentricity 4.6-d orbit originally detected by Butler et al. (1997), two more distant planets were identified from subsequent radial velocity observations, with $M_p \sin i$ of 2.0 and $4.1 M_J$, a of 0.82 and 2.5 AU, and large e of 0.23 and 0.36 respectively (Butler et al., 1999).

The innermost planet, with $e \sim 0$, significantly exceeds the minimum stability requirement given by the Hill radius criterion, suggesting little interaction with the outer two companions. For the two outer planets (c and d) in contrast, the system stability depends strongly on the planet masses, and hence their relative orbital inclinations, with certain combinations implying chaotic or unstable orbits.

A number of numerical studies have been carried out to assess the system stability over long time scales (Holman et al., 1997; Krymowski & Mazeh, 1999; Laughlin & Adams, 1999; Lissauer, 1999; Rivera & Lissauer, 2000; Stepinski et al., 2000; Barnes & Quinn, 2001; Chiang et al., 2001; Jiang & Ip, 2001; Lissauer & Rivera, 2001; Chiang & Murray, 2002; Michtchenko & Malhotra, 2004; Michtchenko et al., 2006). Mazeh et al. (1999), for example, derived a mass for the outer planet of $10.1 \pm 4.7 M_J$ using Hipparcos astrometry, implying an orbital inclination of 156° , and masses of the inner two planets of 1.8 ± 0.8 and $4.9 \pm 2.3 M_J$ if the orbits are coplanar.

Goździewski et al. (2001) used the fast orbit indicator MEGNO to derive $\langle \mathcal{J} \rangle(t)$ as a function of trial values of the inclination of planet c, and the relative inclinations of planets c and d (Equation 3.17). Quasi-periodic and chaotic zones show up prominently in various regions of the i_c, i_{c-d} parameter space (Figure 2.33).

From direct numerical integration of the orbits, using Mercury, Ford et al. (2005) found that for a coplanar configuration the system becomes dynamically unstable when $\sin i < 0.5$ while, if the coplanarity condition is relaxed, dynamical instability resulted from relative inclinations exceeding $\sim 40^\circ$. They found that the long-term secular evolution of the orbits (Figure 2.34) was best modeled as resulting from an impulsive perturbation to the outer planet, ν And d, in which a sudden change in its eccentricity was produced by a close encounter with another planet. After a brief period of chaotic evolution lasting $\sim 10^3$ yr, the perturbing planet would have been ejected, leaving the remaining two outer planets in a stable configuration resembling that observed today.

HD 37124 HD 37124 is an example of dynamical stability modeling leading to the discovery of additional planets. Vogt et al. (2000) detected a 150-d Jupiter-mass planet from Keck-HIRES observations. Butler et al. (2003) identified a second planet with $P \sim 6$ -yr. The two-planet solution was shown to be unstable by Goździewski (2003a), with further Keck data identifying a third planet (Vogt et al., 2005). $\{M_p \sin i (M_J), P(d), e\}$ for planets b, c, d are respectively $\{0.638, 154, 0.055\}$, $\{0.697, 2300, 0.2\}$, $\{0.624, 844, 0.14\}$. Goździewski et al. (2006) showed that the orbit of the outer planet is still poorly constrained, while planet–planet interactions are strong enough that kinematic modeling is insufficient to describe the system adequately.

Titius–Bode law: As noted by Johann Titius in 1766, and as reformulated and popularised by Johann Bode (1747–1826), the mean orbital distances of the six planets known at the time (Mercury to Saturn) are well approximated by

$$a = 0.4 + 0.3 (2)^i, \quad \text{where } i = -\infty, 0, 1, 2, 4, 5, \quad (2.65)$$

and a is the Sun–planet distance in AU. The discovery of Uranus in 1781 at 19.2 AU ($i = 6$), and the first asteroid Ceres in 1801 at 2.8 AU ($i = 3$) were considered as predictions confirming an underlying structural law of the solar system. As a ‘law’, however, it has two shortcomings: it has, currently, no physical basis, and it has too many exceptions (notably the presence of Neptune, and the absence of planets between Mercury and Venus). The topic has nevertheless generated considerable debate (Nieto, 1972), which continues to the present, both in numerical and dynamical form (e.g. Lynch, 2003; Neslušan, 2004; Bass & del Popolo, 2005; Ortiz et al., 2007; Bohr & Olsen, 2010), or perhaps as the ‘psychological tendency to find patterns where none exist’ (Newman et al., 1994).

Murray & Dermott (2000, Section 1.5) assessed its significance by applying a two-parameter geometrical progression to the major satellites of Uranus (Miranda to Oberon), as a distinct instance of orbital periods also displaying a number of simple numerical relationships. Although generating excellent numerical fits, their Monte Carlo analysis demonstrated that almost any distribution of periods would fit a Titius–Bode-type law equally well. They concluded that ‘*There is no compelling evidence that the Uranian satellite system is obeying any relation similar to the Titius–Bode law beyond what would be expected by chance. This leads us to suggest that the ‘law’ as applied to other systems, including the planets themselves, is also without significance.*’

Hayes & Tremaine (1998) made a similar statistical analysis for the solar system, and concluded that the law’s significance is simply that stable planetary systems tend to be regularly spaced; they suggested that the conclusion could be strengthened by more rigorous rejection of unstable planetary systems based on long-term orbit integration.

Multiple exoplanet systems have opened an extended panorama for further conjecture (Poveda & Lara, 2008; Kotliarov, 2008, 2009; Panov, 2009; Chang, 2010). Poveda & Lara (2008) made a Titius–Bode type fit to the five planets orbiting 55 Cnc. They predict a further planet with $a \approx 2.08$ AU ($P \approx 1130$ d) in the gap between $a = 0.781 - 5.77$ AU (evident in Figure 2.29), and possibly another at $a \approx 15.3$. A critique is given by Kotliarov (2009).

An alternative structural law discussed by Kotliarov (2008), which he attributes to Butusov (1973), is related to symmetry. It is similarly phenomenological but claims fewer exceptions. It can be expressed as

$$a_Z^2 = a_{Z+k} \times a_{Z-k}, \quad (2.66)$$

where a is the star–planet distance, and Z is the planet defining the centre of symmetry, with $k \in \{1, 2, \dots, Z-1\}$. Applied to the solar system, it implies that there is a Jovian symmetry to planetary distances, counting Jupiter as $k = 6$ with Ceres as $k = 5$. Applied to 55 Cnc, Kotliarov (2008) predicts planets at different locations to Poveda & Lara (2008), viz. at $a \approx 2.5$ and $a \approx 16.1$.

HD 74156 The two planet system was originally reported by Naef et al. (2004) from ELODIE data. A third planet was detected from a combination of CORALIE, ELODIE and HET–HRS data by Bean et al. (2008b), with the properties $M_p \sin i = 0.40 \pm 0.02 M_J$, $P = 347$ d, and $e = 0.25 \pm 0.11$.

Raymond & Barnes (2005) had predicted the parameter space that such a third planet could have occupied in this and three other systems, by identifying stability regions for massless test particles defined by the previously known planets. For HD 74156 they had shown that a Saturn mass planet could exist within a broad region of $a = 0.9 - 1.4$ AU and $e \leq 0.15$. Barnes et al. (2008a) gave an alternative slightly improved orbit, and suggested that the confirmed prediction of the third planet points to planet formation being an efficient process, with planetary systems typically containing many planets.

μ Ara (HD 160691) The properties of this four-planet system were progressively revealed by AAT–UCLES (Butler et al., 2001; McCarthy et al., 2004) and HARPS (Santos et al., 2004a; Pepe et al., 2007) data. The fourth planet was also tentatively announced by Goździewski et al. (2007). Simulations demonstrating the system stability are shown in Figure 2.35. The semi-major axes

remain almost constant, while the eccentricities of all four planets undergo small variations. Various other dynamical analyses have also been made (Bois et al., 2003; Goździewski et al., 2003, 2005a).

55 Cnc The 55 Cnc system was the first known planetary system comprising five planets (Table 2.5). The first planet, 55 Cnc b, was reported by Butler et al. (1997), and the second (and possibly a third) also from Lick data by Marcy et al. (2002). 55 Cnc c was confirmed by McArthur et al. (2004) based on Lick, ELODIE and HET Doppler measurements and HST–FGS astrometry. They also announced 55 Cnc e with $P = 2.8$ -d.

Fischer et al. (2008) used 18 years of combined Lick and Keck data to confirm the four proposed planets, and to identify a fifth, 55 Cnc f, moving in the large empty zone between two other planets. All five planets reside in low-eccentricity orbits, four having $e < 0.1$. They adopted a multi-planet Keplerian fitting procedure (§2.1.1), using a numerical N-body simulation to show that the system is dynamically stable.

55 Cnc has some basic structural attributes found in our solar system: the orbits are rather circular and nearly coplanar, and a dominant gas giant lies at a distance of about 6 AU. Although planets b and c have a period ratio

Table 2.5: The planets of 55 Cnc, ordered by increasing semi-major axis. Data are from Fischer et al. (2008).

Planet	$M_p \sin i$ (M_J)	a (AU)	P (d)	e
e	0.02	0.038	2.80	0.26
b	0.83	0.115	14.65	0.02
c	0.17	0.240	44.38	0.05
f	0.15	0.789	260.67	0.00
d	3.90	5.888	5371.82	0.06

of 3.027:1.000, hinting at a possible mean motion resonance (Marcy et al., 2002; Ji et al., 2003b; Voyatzis & Hadjidemetriou, 2006), the 3:1 mean motion resonance was excluded by the N-body model of Fischer et al. (2008), as none of the relevant resonant arguments librate.

Other dynamical simulations were made by Gayon et al. (2008) using MEGNO, who found that about 15% of the systems resulting from the nominal orbital elements of the system are highly chaotic. Raymond et al. (2008b) evaluated the stability of the large region between planets f and d using N-body integrations that included an additional, yet-to-be-discovered planet g with a radial velocity amplitude of $\sim 5 \text{ m s}^{-1}$, i.e. $M_p \approx 0.5 - 1.2 M_{\text{Saturn}}$. They found a large stable zone extending from 0.9–3.8 AU with $e < 0.4$, which could contain 2–3 additional planets each of $M_p \sim 50 M_{\oplus}$. Any planets exterior to planet d must reside beyond 10 AU.

2.6.5 Systems in mean motion resonance

The 2:1 resonance system GJ 876 GJ 876 was the first known M-dwarf host and, from Hipparcos, the fortieth nearest stellar system to the Sun. The orbit elements are particularly well determined due to their short periods, and orbit modeling shows that the two most massive planets b and c are locked in a 2:1 resonance, the first to be discovered amongst exoplanet systems.

Marcy et al. (1998) using Keck, and Delfosse et al. (1998) using ELODIE/CORALIE, both reported a 61-d, $2 M_J$ planet. From further Keck observations, Marcy et al. (2001a) showed that the radial velocity signal was actually the combination of two planets in a 2:1 mean motion resonance, with the inner planet having $P = 30.1 \text{ d}$ and $0.6 M_J$. In the original data, the inner companion was indistinguishable from an additional orbital eccentricity of the outer planet. The axes of the two orbits are nearly aligned, and the orbital elements evolve significantly with time. A third, non-interacting, $7.5 M_{\oplus}$ mass planet in a 1.9 d orbit was detected by Rivera et al. (2005).

Laughlin & Chambers (2001), Rivera & Lissauer (2001), and Nauenberg (2002a) independently developed fitting procedures to account for the gravitational interactions between the two outer planets. Rivera & Lissauer (2001) found that most of their solutions are stable for at least 10^8 yr , while test particles orbiting between the two planets are lost in $\lesssim 300 \text{ yr}$.

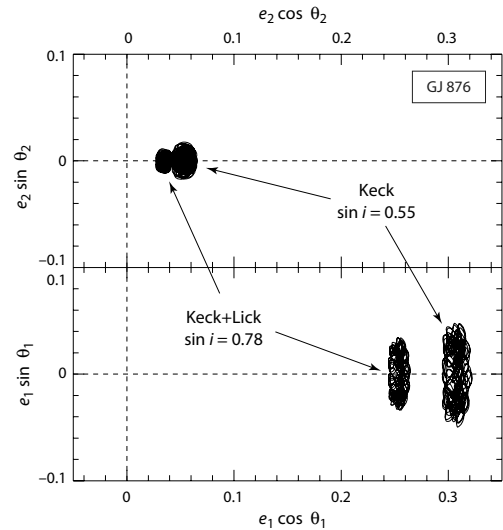


Figure 2.36: Small-amplitude librations, about 0° , of the two 2:1 mean-motion resonance variables for the GJ 876 planets. Trajectories for 3100 days (the average pericentre precession period for both planets) are shown as plots of $e_j \sin j$ versus $e_j \cos j$ ($j = 1, 2$) for the best-fit solutions to the Keck data alone, and for the combined Keck and Lick data, both from Laughlin & Chambers (2001). The values of $\sin i$ are obtained in the best-fit solution. From Lee & Peale (2002, Figure 1), reproduced by permission of the AAS.

Lee & Peale (2002) showed that both mean motion resonance variables, $\phi_1 = 2\lambda_2 - \lambda_1 - \tilde{\omega}_1$ and $\phi_2 = 2\lambda_2 - \lambda_1 - \tilde{\omega}_2$ (cf. Equation 2.52), perform very small amplitude librations around 0° (Figure 2.36). Their simultaneous libration about 0° implies that the variable $\phi_3 = \phi_1 - \phi_2 = \tilde{\omega}_1 - \tilde{\omega}_2$ also librates around 0° , and the planets are consequently in three resonances at the 2:1 mean motion commensurability. The small libration of ϕ_3 means that the lines of apsides of the two orbits are therefore nearly aligned, and conjunctions of the two planets occur very close to their longitudes of pericentre (Beaugé & Michtchenko, 2003).¹³ The existence of the mean motion resonance is taken as confirmation that the two orbits are essentially coplanar, with the deep resonance (i.e. with small libration amplitudes) implying that the system is stable indefinitely.

Laughlin et al. (2005a) incorporated additional velocity data obtained over 16 years, and confirmed the small libration amplitudes (Figure 2.37). Such a configuration can be explained by a slow differential migration resulting from an interaction of the planets with a protoplanetary disk (Snellgrove et al., 2001; Lee & Peale, 2002; Kley et al., 2005; Beaugé et al., 2006; Crida et al., 2010).

¹³The nature of the resonance contrasts with the case of the Io–Europa system, where the lines of apsides are anti-aligned, and where conjunctions therefore occur when Io is near its pericentre, and Europa is near its apocentre.

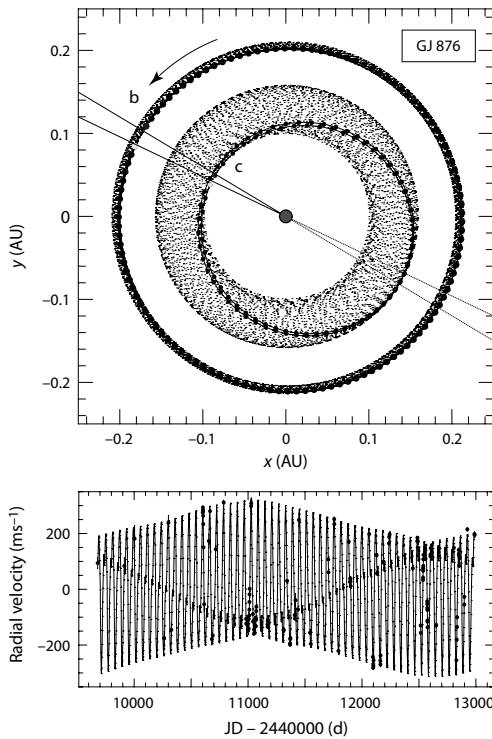


Figure 2.37: The 2:1 mean motion resonance in GJ 876. Top: the two clouds of small dots show the planet positions every 0.5 d over almost 10 yr, illustrating the precession in the line of apses of $-41^\circ \text{ yr}^{-1}$. Connected circles show the planet positions every 0.5 d for 60 d. The two lines radiating from the central star mark the longitudes of pericentre, $\bar{\omega}_b$ and $\bar{\omega}_c$, at JD 2449 710. They oscillate about alignment with a libration amplitude of 34° . Bottom: the measured stellar reflex velocities (circles), and the solution from the self-consistent, coplanar, three-body integration. From Laughlin et al. (2005a, Figures 1–2), reproduced by permission of the AAS.

Laughlin et al. (2005a) also showed that configurations with modest mutual inclinations are possible, which has an interesting corollary: for non-coplanar configurations the line of nodes of the inner planet precesses at about -4° yr^{-1} . The inner planet may therefore be observed to transit the host star at some time in the relatively near future, i.e. when either the ascending or descending node precesses through the line of sight, even though it is not transiting at the present time.

Laplace resonance in GJ 876 A fourth planet, GJ 876 e, was discovered from continued radial velocity monitoring by Rivera et al. (2010). N-body fits show that the four-planet system has an invariable plane with an inclination of $59^\circ 5'$, and is stable for more than 1 Gyr. Their model places the fourth planet in a three-body (Laplace) resonance with the two giant planets ($P_c = 30.4 \text{ d}$, $P_b = 61.1 \text{ d}$, $P_e = 126.6 \text{ d}$). Unlike the case of the Galilean satellites of Jupiter (Figure 2.38), the three planets come close to a triple conjunction once per orbit of planet e.

The critical argument for the Laplace resonance, $\phi = \lambda_c - 3\lambda_b + 2\lambda_e$, librates with an amplitude of $40^\circ \pm 13^\circ$.

The 2:1 resonance system HD 128311 The two-planet system HD 128311 offers an interesting insight into planetary migration along similar lines. The first planet was reported by Butler et al. (2003), and the second by Vogt et al. (2005). A Trojan 1:1 solution was suggested by (Goździewski & Konacki, 2006), while the most recent orbit solution is given by Wittenmyer et al. (2009).

Sándor & Kley (2006) performed a three-body numerical integration of the (Newtonian) orbital solution of Vogt et al. (2005), and examined the evolution of the difference in the longitudes of pericentre $\Delta\bar{\omega} = \bar{\omega}_2 - \bar{\omega}_1$, and the two appropriate resonant angles ϕ_1, ϕ_2 (Equation 2.52 corresponding to $\bar{\omega}_1, \bar{\omega}_2$ respectively). Of these, only ϕ_1 librates around 0° with an amplitude of $\sim 60^\circ$, while ϕ_2 and $\Delta\bar{\omega}$ rotate. At the same time, the eccentricities show large oscillations (Figure 2.39).

This behaviour is in contrast with the apsidal corotation predicted by adiabatic migration (Figure 2.39, top right). Instead, Sándor & Kley (2006) considered that the system's present state can be attributed to a strong scattering event in the past. Assuming that the two planets were once locked in a 2:1 resonance with apsidal corotation as a result of inward migration, they then studied two distinct perturbation scenarios which might have disrupted the corotation (Figure 2.39, bottom).

In the first, a sudden halting of migration was invoked. This might arise, for example, when the outer planet reaches some disk discontinuity with an empty region inside; such a model would be consistent with Spitzer observations of young stars which have shown that the inner part of protoplanetary disks may indeed contain only very little mass, possibly due to photo-evaporation by the central star (D'Alessio et al., 2005; Calvet et al., 2005). In the second scenario, an encounter with a low-mass $10M_\oplus$ planet approaching from inside or outside (similar to the perturbation proposed by Ford et al. 2005 for ν And) could also break the apsidal libration. The smaller planet would be ejected from the system, or thrown into a larger orbit, in the process.

Particularly in the case of a small planet approaching from the inside, the apsidal corotation of the giant planets is indeed broken. Consistent with the behaviour predicted from long-term orbit integration, ϕ_2 and $\Delta\bar{\omega}$ then circulate, while the giant planets remain in the 2:1 resonance, with ϕ_1 still librating around 0° .

The 3:1 antisymmetric resonance system HD 60532 Amongst systems in which a 3:1 mean motion resonance has been suggested, that of the two giant planets orbiting HD 60532 appears reasonably secure (Desort et al., 2008; Laskar & Correia, 2009; Sándor & Kley, 2010). The two orbital solutions found by (Laskar & Correia, 2009)

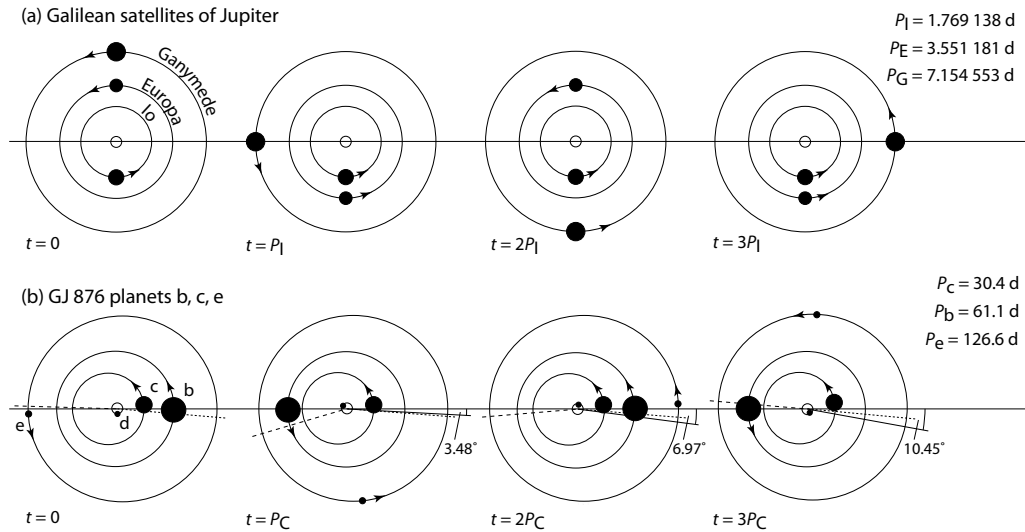


Figure 2.38: The two known examples of Laplace resonance, in which three (or more) orbiting bodies have a simple integer ratio between their orbital periods: (a) Jupiter's inner satellites Ganymede, Europa, and Io are in a 1:2:4 Laplace resonance, with Ganymede completing 1 orbit in the time that Europa makes 2, and Io makes 4. They never experience triple conjunctions. Their mean motions (Equation 2.8) are related by $n_I - 3n_E + 2n_G = 0$, satisfied to nine significant figures (Peale, 1976); (b) planets c, b, e of the M dwarf host star GJ 876 are an exoplanet triplet in a Laplace resonance. The reference frame rotates at the mean orbital precession of planet b, $-10^\circ.45$ over 90 d (solid lines). The apsidal line of b coincides with the x-axis, while those for planets c and e are shown with short-dash and long-dash lines respectively. In each system, the orbits and object radii (assuming $R \propto M^{1/3}$) are shown to scale. Lower figure is adapted from Rivera et al. (2010, Figure 7).

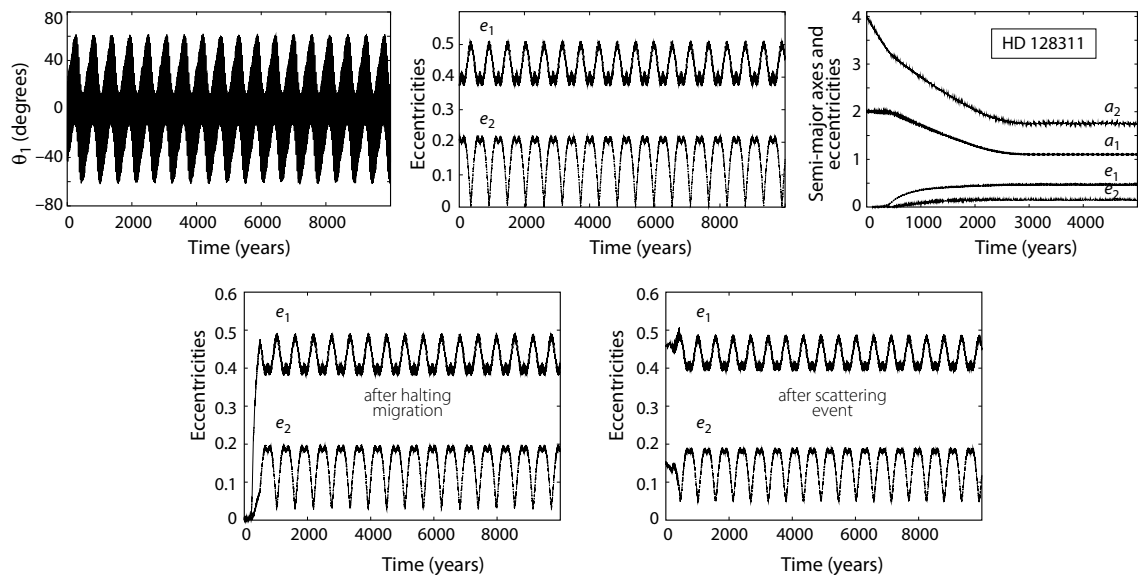


Figure 2.39: Resonant evolution of HD 128311. Evolution of the resonant angle ϕ_1 (top left) and eccentricities e_1 and e_2 (top middle) obtained by numerical integration of the orbital parameters of Vogt et al. (2005). Top right: predicted behaviour of the semi-major axes and eccentricities during adiabatic migration with an e-folding time of 2×10^3 yr; for the simulations, the ongoing migration is progressively stopped between $2 - 3 \times 10^3$ yr. Bottom left: evolution of the planetary eccentricities, both originally in circular orbits, following a sudden halting in the migration of the outer planet. Bottom right: evolution of the eccentricities following a scattering event with a low-mass planet migrating outwards. From Sándor & Kley (2006, Figures 1, 2, 3, 5), reproduced with permission © ESO.

Solar system 1:1 resonances and the Trojans: In the circular restricted three-body problem, three bodies move in circular coplanar orbits, with the mass of the third being negligible. In 1772, Lagrange proved the existence of five equilibrium points where the third particle has zero velocity and zero acceleration in the rotating frame. These are the three colinear Lagrangian equilibrium points L_1 , L_2 , L_3 , and two (leading and trailing) triangular equilibrium points L_4 , L_5 . Although L_1 – L_3 are unstable, special starting conditions of the third particle, in position and velocity, can nevertheless result in semi-stable periodic orbits in their vicinity. This is exploited by artificial satellites such as SOHO which observes the Sun from L_1 (Domingo et al., 1995), and others at (WMAP, Herschel, Planck) or destined for (e.g. Gaia) L_2 .

Stable orbits in the vicinity of L_4 or L_5 were first demonstrated by Routh (1875) for the special case of circular orbits, and by Danby (1964a,b) for elliptical orbits. They comprise a short-term component of motion with a period close to the orbital period, and a longer term periodic motion about the equilibrium point referred to as libration. Resulting orbits can be described as a short-period epicyclic motion around a long-period motion of the epicentre, the relative contributions of the two components determined by the starting conditions. The resulting elongated orbits around L_4 or L_5 are referred to as *tadpole orbits*. For an increased radial separation from L_4 or L_5 , the resulting *horseshoe orbits* can encompass both L_4 and L_5 .

Examples of solar system bodies in such stable 1:1 resonances are referred to as *Trojans*, and they include both asteroids and satellites. Librating mostly in tadpole orbits around the Sun–Jupiter triangular equilibrium points is the leading group around L_4 (the ‘Greeks’), including the first known (588) Achilles discovered by Max Wolf in 1906, and the trailing group around L_5 (the ‘Trojans’). Typical libration amplitudes are 15 – 20° (Shoemaker et al., 1989).

Other Trojan asteroids orbit the Sun–Mars system, with the first (5261) Eureka, discovered in 1990, librating about L_5 (Mikkola et al., 1994). Asteroid (3753) Cruithne is in a horseshoe orbit around the Sun–Earth system (Wiegert et al., 1997).

The dynamics of planet–satellite systems around their triangular equilibrium points are identical, and result in the *co-orbital* or *Trojan satellites*. Telesto and Calypso librate around the Saturn–Tethys system, and Helene and Polydeuces around the Saturn–Dione system. In the curious Janus–Epimetheus Saturnian system, Epimetheus moves in a horseshoe orbit, exchanging altitude with Janus as they approach in a four-year repeating cycle (Dermott & Murray, 1981a,b), and resulting in forced rotational libration (Tiscareno et al., 2009).

have very different inclinations, $i = 20^\circ$ or $i = 90^\circ$, with planetary masses differing by a factor $1/\sin i \sim 3$.

Sándor & Kley (2010) carried out detailed two-dimensional hydrodynamical simulations of appropriate thin disks with an embedded pair of massive planets, studying the effects of migration and resonant capture using N-body simulations.

For resonant capture, planet pairs must undergo convergent migration, such that if the inner planet migrates inwards at the same speed, or faster than the outer one, then no resonant capture occurs. Resonant capture through migration therefore depends on the inner planet opening a suitably large gap in the inner disk to slow down its inward migration rate. In their simulations, Sándor & Kley (2010) found that capture into the observed 3:1 resonance takes place only for higher planetary masses, thus favouring orbital solutions having the smaller inclination of $i = 20^\circ$ (Figure 2.40).

They also found that the inner disk, between the inner planet and the star, plays a key role in determining the final configuration of this, and by implication other, resonant systems. Specifically, fast inward migration of the outer planet may result in it crossing the 3:1 mean motion resonance without capture, ending the migration in the more robust 2:1 mean motion resonance. In the case of HD 60532, the damping effect of the disk on the inner planet’s eccentricity is responsible for an antisymmetric (antialigned) resonance configuration in which $\tilde{\omega}_2 - \tilde{\omega}_1$ oscillates around 180° , rather than the 0° in the case of aligned pericentres.

Other mean motion resonances Other systems believed to be in a 2:1 mean motion resonance include

HD 82943 (Goździewski & Maciejewski, 2001; Ji et al., 2003a; Mayor et al., 2004; Ferraz-Mello et al., 2005; Lee et al., 2006); HD 73526 (Tinney et al., 2003, 2006; Sándor et al., 2007a); and μ Ara d–b (Pepe et al., 2007).

Other possible 3:1 resonances include 14 Her (Goździewski et al., 2006, 2008b), and the contested case of 55 Cnc b–c (Marcy et al., 2002; Ji et al., 2003b; Fischer et al., 2008).

A number of other possible resonances have been identified: the 4:1 resonance of HD 108874 (Butler et al., 2003; Goździewski et al., 2006) and perhaps 14 Her (Wittenmyer et al., 2007); the 5:1 resonance of HD 202206 (Udry et al., 2002; Correia et al., 2005; Goździewski et al., 2006); a possible 3:2 resonance of HD 45364 (Correia et al., 2009; Rein et al., 2010); a possible 5:2 resonance of HD 37124 (Goździewski et al., 2006); a possible 6:1 or 11:2 resonance of HD 12661 (Goździewski, 2003b; Lee & Peale, 2003; Rodríguez & Gallardo, 2005; Zhang & Zhou, 2006; Veras & Ford, 2009); and a possible 5:2 or 7:3 resonance in 47 UMa (Laughlin et al., 2002).

Proximity to resonance Proximity to a resonance may still affect the secular motion, as in the case of the near 5:2 resonance between Jupiter and Saturn, Laplace’s ‘great inequality’ (e.g. Varadi et al., 1999; Michtchenko & Ferraz-Mello, 2001a). Such proximity has been suggested for HD 12661 and ν And (Libert & Henrard, 2007).

Inclination resonance In the absence of observational data (as opposed to theoretical dynamical constraints) on the mutual inclination of exoplanet orbits, most studies of mean motion resonances have focused on coplanar configurations. The extension to non-

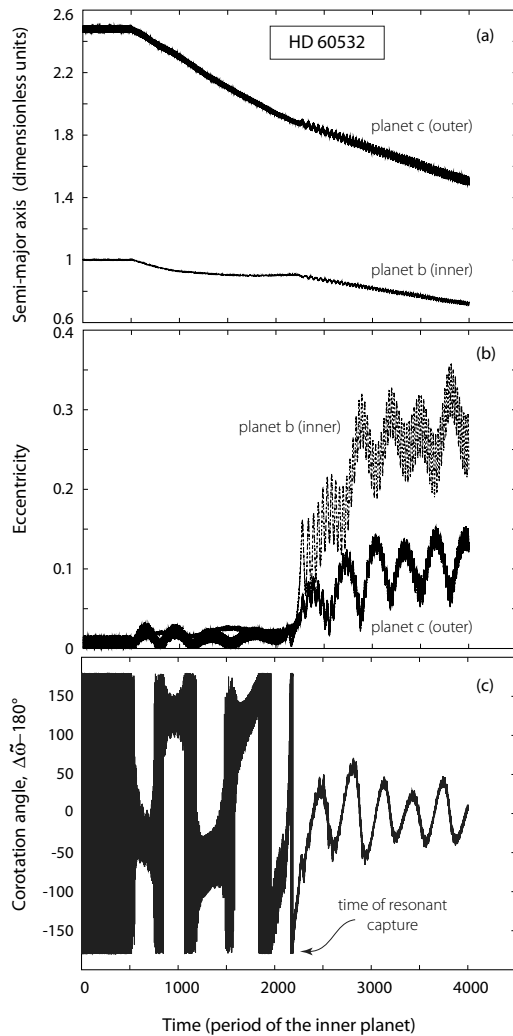


Figure 2.40: Dynamical simulation of the two giant planets orbiting HD 60532, embedded in a protoplanetary disk, showing (a) semi-major axes, (b) eccentricities, and (c) the evolution of the resonant angle $\Delta\tilde{\omega} = \tilde{\omega}_2 - \tilde{\omega}_1$. During the first 500 orbits of the inner planet, the planets are ‘fixed’ to obtain a steady state in the disk. Capture into the 3:1 mean motion resonance occurs after ~2300 periods of the inner planet, subsequent to which the orbits exhibit (antialigned) apsidal corotation. The planetary masses ($3.15M_J$ and $7.46M_J$) correspond to a (coplanar) system inclination of $i = 20^\circ$. From Sándor & Kley (2010, Figures 4 and 5b), reproduced with permission © ESO.

coplanar orbits is relevant for an understanding of eccentricity excitation and migration.

Thus Thommes & Lissauer (2003) found that, subsequent to a 2:1 eccentric resonance capture, a subsequent capture into a 4:2 *inclination resonance* is possible. This is the lowest-order inclination resonance at the 2:1 commensurability (since the mutual inclination appears first as $(\Delta i)^2$ in the series expansion of the disturbing function), and is characterised by the mean motion

resonance variables (cf. Equation 2.54)

$$\begin{aligned}\phi_{11} &= 2\lambda_1 - 4\lambda_2 + 2\Omega_1 \quad \text{and} \\ \phi_{22} &= 2\lambda_1 - 4\lambda_2 + 2\Omega_2,\end{aligned}\quad (2.67)$$

where Ω_j are the longitudes of the ascending nodes. The properties and subsequent evolution of this resonance are detailed by Lee & Thommes (2009).

Capture into higher-order resonances (such as the 3:1, 4:1, and 5:1) can also result in the excitation of inclinations, with mutual inclinations reaching $\Delta i \sim 20 - 70^\circ$ in the simulations of Libert & Tsiganis (2009b), at least for low-mass inner planets and for one or both planets developing eccentricities $e \gtrsim 0.4$. Amongst the current systems believed to be in higher-order resonance, HD 60532 (3:1), HD 108874 (4:1) and HD 102272 (4:1) do not satisfy these requirements, and are therefore likely to reside in coplanar resonances, an inference verified by numerical simulations.

Kozai resonance Large mutual inclinations between the orbits of multiple planets could be generated by (Kozai) resonant interactions, as well as more directly by planet–planet scattering and planetesimal-driven migration. Whether any of the known systems are in a stable Kozai resonant state, with mutual orbital inclinations of order $\Delta i \sim 40 - 60^\circ$, was investigated parametrically by Libert & Tsiganis (2009a). They found that four of the systems studied (ν And, HD 12661, HD 74156, and HD 169830) could in principle be in Kozai resonance, provided that their mutual inclination is at least 45° . Direct astrometric determination of the mutual inclinations in ν And by HST-FGS (§3.8.2, McArthur et al., 2010) gave $\Delta i_{\text{cd}} = 29.9 \pm 1^\circ$.

The 1:1 resonance A number of objects in the solar system move in 1:1 resonance orbits. They are examples of Lagrange’s celebrated solution to the restricted three-body problem, and generally satisfy the condition that the mass of the smallest is negligible compared with that of the other two (see box).

For the more general three-body problem, and in particular for two equal mass planets which share a time-averaged orbital period, Laughlin & Chambers (2002) identified a wide parameter space in which stable coorbital configurations exist (Figure 2.41). These fall into rather distinct regimes. In the first, the star and both planets participate in tadpole-like librations around the vertices of an equilateral triangle. The dynamics resembles that of Jupiter’s Trojan asteroids, and the configuration is stable for mass ratios $2M_p/(2M_p + M_\star) < 0.03812$. In the second regime, for larger perturbations, the planets execute symmetrical horseshoe-type orbits similar to those of Janus and Epimetheus. Stability analysis indicates that a pair of Saturn-mass planets could survive in this resonance over long periods (Figure 2.41a).

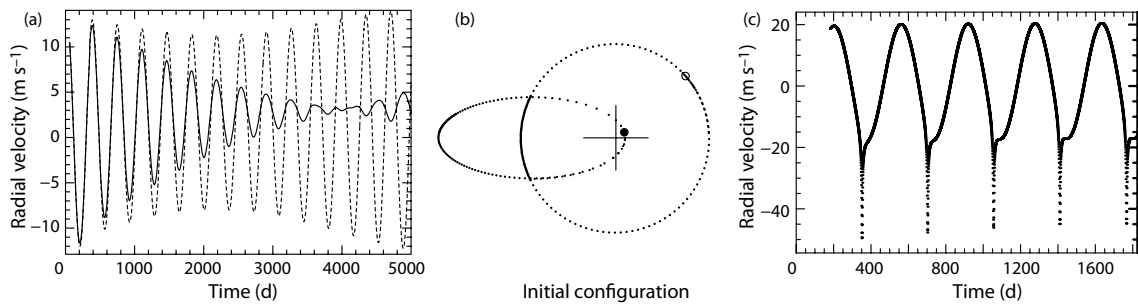


Figure 2.41: Theoretical 1:1 resonance orbits for equal mass planets: (a) synthetic radial velocity variations predicted for the tadpole-type (dotted) and horseshoe-type (solid) orbits; (b) initial configuration for the 1:1 eccentric resonance; solid lines indicate parts of the respective orbits swept out over 15% of the orbital period; (c) associated radial velocity variations for the case of eccentric resonance. From Laughlin & Chambers (2002, Figures 3, 4, 6), reproduced by permission of the AAS.

In practice, recognising such systems from the radial velocity data might be problematic, since the periodogram signal can be indistinguishable from that of a single planet in an eccentric orbit (Laughlin & Chambers, 2002), or from that of a system in 2:1 mean motion resonance (Goździewski & Konacki, 2006).

Their third configuration, described as *eccentric resonance*, is qualitatively distinct from Lagrange's solution (Figure 2.41b). One orbit starts as highly eccentric, the other is more nearly circular, the pericentres are aligned, and conjunctions occur near pericentre; the pair of planets then exchange angular momentum and eccentricity (in the case studied, with a periodicity of ~ 800 yr). With appropriate initial conditions, they can avoid close encounters indefinitely. In this regime, the radial velocity variations are distinctive, perhaps recognisable by a single deviating measurement superposed on an otherwise sinusoidal pattern (Figure 2.41c).

A similar configuration was found independently by Nauenberg (2002b). These eccentric resonances might arise from the same type of planet–planet interactions responsible for non-resonant high eccentricity systems.

The theoretical existence and properties of the various 1:1 resonance configurations (Hadjidemetriou et al., 2009) has led some investigations to consider them as alternative models for certain specific systems (e.g. Érdi et al., 2007), in particular in cases where long-term orbit integration has identified dominant instabilities.

Goździewski & Konacki (2006) examined the long-term stability of HD 128311 ($\mathcal{S}2.6.5$), searching for alternative 2:1 resonances using a genetic algorithm, and examining their long-term stability using MEGNO. They also searched for 1:1 resonances consistent with the observed radial velocity variations of the host star, and identified extended zones of stability in which such a system would survive (Figure 2.42). They used the term ‘Trojan planets’ for all 1:1 mean motion resonances, including the eccentric resonance configurations identified by Laughlin & Chambers (2002), i.e. not only for the

Lagrange-type solutions, but also for those with similar semi-major axes, but possibly with large relative inclinations and time-varying eccentricities.

In the multiple planet system μ Ara studied by Pepe et al. (2007) one of their solutions, located using a genetic algorithm search, gave an inner planet at 9.64 d, an outer planet at 2741 d, and with an intermediate pair of Trojan planets in a 1:1 mean motion resonance. The solution was discarded as a result of its high dynamical instability; orbit integration indicating that it would be disrupted in less than 100 years.

Schwarz et al. (2007a,c,b) demonstrated the stability of Trojan terrestrial planets in mean motion resonance with a giant planet in the habitable zone for a number of systems.

The formation of a hypothetical terrestrial-type body at the Lagrange points of a giant exoplanet has been studied using an N-body code by Beaugé et al. (2007). Although such planets may form *in situ*, the accretion process is inefficient, and the mass of the final planet never exceeds $\sim 0.6M_{\oplus}$.

Retrograde resonances All known exoplanets are generally considered to revolve in the same direction around their respective system barycentres, and most fitted radial velocity orbit elements are derived by assuming *prograde* or *regular* orbits, i.e. all orbiting in the same direction. Such behaviour is expected, at least for unperturbed states, according to current theories of planet formation.

The theoretical existence and dynamical stability of *retrograde* mean motion resonances, i.e. with one planet counter-revolving with respect to the other in a two-planet system, or to all other planets in a multiple planet system, has been investigated and demonstrated (Gayon & Bois, 2008a,b; Gayon et al., 2009).

A planet might conceivably be found in a retrograde mean motion resonance either as a result of capture of a pre-existing free-floating planet directly into a retrograde orbit, or through (violent) dynamical evolution as

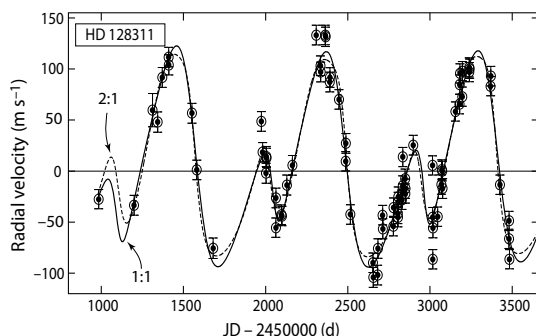


Figure 2.42: The two-planet system HD 128311, showing radial velocity curves for the best-fit solutions corresponding to the 2:1 (dashed) and 1:1 (solid) mean motion resonances. Both curves give an rms of 15 m s^{-1} . Error bars include the stellar jitter of 9 m s^{-1} . From Goździewski & Konacki (2006, Figure 4), reproduced by permission of the AAS.

a result of planet–planet interactions. The slingshot effects seen in the simulations of Nagasawa et al. (2008) show that close-in planets, some in retrograde motion as a result of Kozai resonances, may indeed result.

In the solar system, a small number of asteroids, a number of asteroid-size planetary moons, and Triton (the largest of Neptune’s moons) have retrograde orbits.

Gayon & Bois (2008a) used the genetic algorithm PIKAIA and the fast-chaos indicator MEGNO to search for stable 2:1 retrograde resonant configurations for HD 73526, consistent with the published initial conditions. They found prominent stability islands, which occur when the apsidal longitudes of the two orbits precess at the same average rate, albeit with complex structure, and even though they do not precess in the same direction. The resulting radial velocity curve is very similar to that given by the prograde solutions of Tinney et al. (2006) and Sándor et al. (2007a).

Similar counter-revolving solutions have been investigated by Gayon-Markt & Bois (2009) for eight further compact multiplanetary systems. Their results suggest that six of them (HD 69830, HD 73526, HD 108874, HD 128311, HD 155358 and HD 202206) could be dynamically regulated by such a configuration. Systems with retrograde planets can be packed substantially more closely than prograde systems with an equal number of planets (Smith & Lissauer, 2009).

2.6.6 Interacting doubles

A number of systems contain two giant planets in which planet–planet interactions are significant. These include: HD 12661 (Goździewski & Maciejewski, 2003); HD 14810 (Wright et al., 2009b); HD 155385 (Cochran et al., 2007); HD 169830 (Goździewski & Konacki, 2004); and HD 183263 (Wright et al., 2009b).

2.6.7 Non-interacting doubles

A number of systems contain planet pairs with very large ratios of their orbital periods (sometimes referred to as ‘hierarchical’, but see the footnote on page 43), which are therefore unlikely to be interacting.

These include: HD 11964 with $P = 38 \text{ d}$ and 5.3 yr (Wright et al., 2009b); HD 68988 with $P = 6.3 \text{ d}$ and $11\text{--}60 \text{ yr}$ (Wright et al., 2007); HD 168443 with $P = 58 \text{ d}$ and 4.8 yr (Marcy et al., 2001b); HD 187123 with $P = 3 \text{ d}$ and 10.5 yr (Wright et al., 2009b); HD 190360 with $P = 17.1 \text{ d}$ and 8.0 yr (Vogt et al., 2005); HD 217107 with $P = 7.1 \text{ d}$ and 11.7 yr (Wright et al., 2009b); and HD 38529 with $P = 14.3 \text{ d}$ and 5.9 yr (Fischer et al., 2003) with an infrared excess detected by Spitzer (Moro-Martín et al., 2007).

2.6.8 Super-Earth systems

Planetary systems comprising only low-mass planets are starting to be discovered with improved radial velocity accuracies. The first three such systems discovered are:

HD 69830: the triple Neptune system with $P \sim 9$, 32 , and 200 d (Lovis et al., 2006). The outer planet is located near the inner edge of the habitable zone. Beichman et al. (2005a) reported Spitzer photometry and spectroscopy which shows an infrared excess characteristic of a large cloud of fine silicate dust within a few AU of the star, suggestive of a large asteroid belt.

GJ 581: a five- or six-planet system (Bonfils et al., 2005b; Udry et al., 2007; Vogt et al., 2010a), with some lying within, or close to the inner and outer edges, of the star’s habitable zone (§11.9.2).

HD 40307: the third triple super-Earth system, with masses 4.2 , 6.9 , and $9.2 M_{\oplus}$ (Mayor et al., 2009b).

2.6.9 Stability of habitable zone systems

Numerous investigations have been carried out to examine the stability of planets which might exist in the habitable zones of known exoplanet systems (§11.9.1). Using fast indicators of stability or chaos, stability maps can help to establish, for example, whether and where Earth-like planets could exist in systems having one giant planet (Figure 2.43).

Hypothetical orbits of such terrestrial-type planets within the estimated habitable zone have been classified as fully stable if no mean motion resonance exists within that zone, partially stable if only a few resonances exist, marginally stable if many resonances exist, and very unstable if the region is found to be strongly chaotic (Sándor et al., 2007b).

2.7 Planets around binary and multiple stars

Many stars, perhaps most, occur in binary or multiple star systems. Nearby G-K dwarfs, for example, which have been high-priority targets for many Doppler search

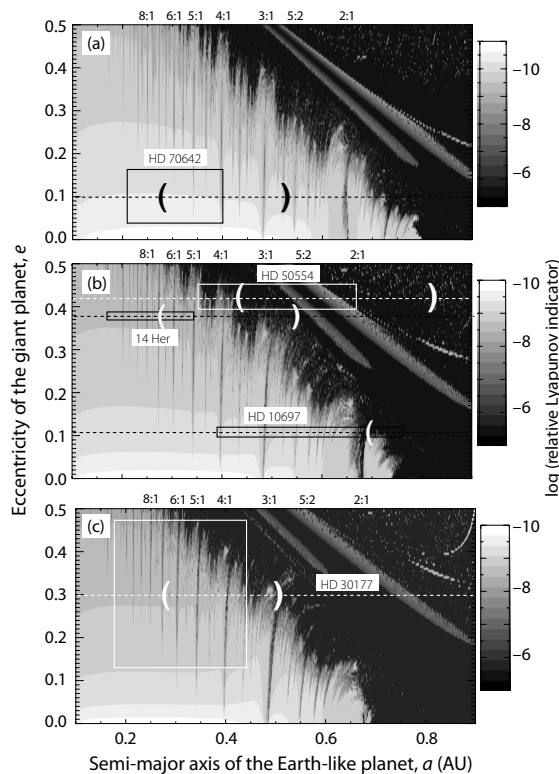


Figure 2.43: Stability maps, computed by the method of relative Lyapunov indicators, for orbits of possible habitable-zone planets interior to the orbit of a known giant planet, for various values of mass ratios $\mu = M_p/(M_\star + M_p)$, where M_p is the mass of the giant planet, and the mass of the hypothetical terrestrial-type planet is negligible: (a) $\mu = 0.002$, (b) $\mu = 0.005$, (c) $\mu = 0.009$. The habitable zones corresponding to the star's zero-age main sequence luminosity are indicated by elongated rectangles, whose heights correspond to the uncertainty on the giant planet's eccentricity, while the current habitable zones are indicated by parentheses. Positions of mean motion resonances are indicated. In (c), for example, the eccentricity of the giant planet in the HD 30177 system is highly uncertain. The present habitable zone is strongly chaotic for large values of e , while its inner part is marginally stable for small e . From Sándor et al. (2007b, Figures 7, 9 and 10), © John Wiley & Sons, Inc.

programmes, are known to occur more often in binary or multiple systems than they do in isolation (Abt, 1979; Duquennoy & Mayor, 1991; Mathieu, 1994; Eggenberger et al., 2004a).

At the same time, many young binaries are known to possess disks—either *circumstellar* (around one of the stars) or *circumbinary* (surrounding both stars)—which, in analogy with single stars, may provide the accretion material necessary for planet formation (e.g. Mathieu, 1994; Akeson et al., 1998; Mathieu et al., 2000; White & Ghez, 2001; Monin et al., 2007).

Understanding the occurrence of planets around binary stars, and how their frequency and properties com-

pare with planets around single stars poses, however, numerous problems: of observability, of quantifying their dynamical stability, and of understanding their detailed origin and formation.

2.7.1 Configurations and stability

The first studies of dynamical configurations for planets in binary systems were carried out well in advance of the first exoplanet detections. Szebehely (1980) classified the possibilities as 'inner orbits' revolving around the primary star, 'satellite orbits' revolving around the secondary, or 'outer orbits' revolving around both.

From studies of the stability of resonant periodic orbits in the circular restricted three-body problem, and as illustrated schematically in Figure 2.44, Dvorak (1982) suggested that a planet may exist in either a circumstellar orbit, i.e. orbiting either the primary or secondary star (also referred to as an *S-type orbit*), or in a circumbinary orbit, i.e. orbiting both stars (*P-type*). Existence in a 1:1 'Trojan' resonance around the L4 or L5 Lagrange points (*L-type*) is also possible dynamically.

Dynamical stability The definition of dynamical stability of a planetary orbit in a binary system has been considered by various authors (e.g. Harrington, 1977; Szebehely, 1980, 1984), following similar stability considerations for multiple planetary systems (§2.6.1).

Harrington (1977), for example, considered a planet orbit as stable if its semi-major axis and orbital eccentricity do not undergo secular changes. More pragmatically, Haghighipour (2008) considered a planet in a binary system stable if, for the duration of study, it does not collide with other bodies, and it does not leave the system's gravitational field.

With the more limited numerical tools then available, the first stability studies were restricted to special cases, or conducted only over a few orbital periods (Graziani & Black, 1981; Black, 1982; Pendleton & Black, 1983). With the development of symplectic integrators (page 44), particular routines have been developed for following the orbits of low-mass bodies in binary systems (Chambers et al., 2002). The long-term stability of planets in various S-type and P-type orbits, at least within certain domains, has been confirmed by such dynamical studies.

Stability of S-type orbits For a planet in an S-type orbit, the gravitational force of the secondary is the principal source of orbital perturbations. Qualitatively, the perturbative effect varies with the companion star mass, and according to the eccentricity and semi-major axis of the binary which together determine the closest approach of the secondary to the planet.

The maximum value of the semi-major axis of a stable planetary orbit is given by (Rabl & Dvorak, 1988; Hol-

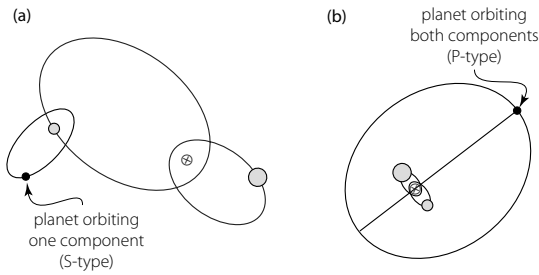


Figure 2.44: Schematic of a planet in a binary star system (⊗ indicates the stellar centre of mass): (a) the planet orbits one component of the stellar binary; (b) the planet is in a circumbinary orbit about both stellar components.

man & Wiegert, 1999; Haghighipour, 2008)

$$a_c/a_b = (0.464 \pm 0.006) + (-0.380 \pm 0.010)\mu \\ + (-0.631 \pm 0.034)e_b + (0.586 \pm 0.061)\mu e_b \\ + (0.150 \pm 0.041)e_b^2 + (-0.198 \pm 0.047)\mu e_b^2, \quad (2.68)$$

where a_c is the critical semi-major axis, a_b and e_b are the semi-major axis and eccentricity of the binary, $\mu = M_1/(M_1 + M_2)$, and M_1 and M_2 are the masses of the primary and secondary stars. The $\pm$ signs together define a transitional region, comprising a mixture of stable and unstable states, where stability depends on orbital parameters, mass ratio, and initial conditions. Slightly different limits have been found in later studies (Moriwaki & Nakagawa, 2004; Fatuzzo et al., 2006).

Such results are applicable to planets of Jupiter-mass, as well as of terrestrial mass (Quintana et al., 2002; Quintana & Lissauer, 2006; Quintana et al., 2007) and, discounting the effects of radiation pressure, dust particles and debris disks (Trilling et al., 2007).

Numerous other stability studies for S-type systems have been reported, both for general cases (e.g. Benest, 1993; Pilat-Lohinger & Dvorak, 2002; Pilat-Lohinger et al., 2004; Musielak et al., 2005; Mudryk & Wu, 2006; Cuntz et al., 2007; Marzari & Barbieri, 2007b,a; Sepinsky et al., 2007), as well as for specific systems, notably for α Cen (Benest, 1988; Wiegert & Holman, 1997), Sirius (Benest, 1989), η CrB (Benest, 1996), and γ Cep (Dvorak et al., 2003a, 2004; Haghighipour, 2006; Verrier & Evans, 2006), and others (Szenkovits & Makó, 2008).

A dynamical classification, distinguishing dynamically rigid systems in which the orbital planes of planets precess together as if they were embedded in a rigid disk, and weakly coupled systems in which the mutual inclination angle between initially coplanar planets grows to large values on secular timescales, has been proposed by Takeda et al. (2008a).

Kozai resonance A specific dynamical effect observed in numerical simulations of S-type binaries is the phenomenon of *Kozai resonance*. In three-body systems with two massive objects and a smaller body in a highly

inclined orbit, such as in the case of a binary–planet system, angular momentum exchange between the planet and the secondary star results in the planet's inclination and eccentricity oscillating synchronously, i.e. increasing one at the expense of the other, whilst conserving the integral of motion (the Delaunay quantity)

$$H_K = (1 - e^2)^{1/2} \cos i. \quad (2.69)$$

The Kozai instability was originally studied in the context of high inclination comets and asteroids in the solar system, and sets in at inclinations greater than a critical value of $i_c = \arcsin \sqrt{0.4} \approx 39.2^\circ$ (Kozai, 1962; Innanen et al., 1997).

This behaviour, in which e and i oscillate with the same periodicity but 180° out of phase, is seen in general numerical simulations of S-type systems (Takeda & Rasio, 2006; Takeda et al., 2008a), as well as for specific configurations of individual systems such as γ Cep (Haghighipour, 2004, 2006; Verrier & Evans, 2006, Figure 2.45).

In multiple planetary systems with highly inclined orbits, the Kozai resonance mechanism can result in highly eccentric outer planets. If their orbits cross those of the inner planets, strong planet–planet interactions can result in the ejection of one or more of the planets (Malmberg et al., 2007a; Parker & Goodwin, 2009; Verrier & Evans, 2009).

Stability of P-type orbits Similar numerical simulations have been carried out for P-type circumbinary orbits (Ziglin, 1975; Szebehely & McKenzie, 1981; Dvorak, 1984, 1986; Dvorak et al., 1989; Kubala et al., 1993; Holman & Wiegert, 1999; Broucke, 2001; Pilat-Lohinger et al., 2003; Musielak et al., 2005; Pierens & Nelson, 2008b), and for more general hierarchical triple systems (Verrier & Evans, 2007). Again, stability criteria are dictated by the perturbative effects of the various components.

The P-type planetary orbit is found to be stable for semi-major axes exceeding a critical value given by (Dvorak et al., 1989; Holman & Wiegert, 1999; Haghighipour, 2008)

$$a_c/a_b = (1.60 \pm 0.04) + (4.12 \pm 0.09)\mu + (5.10 \pm 0.05)e_b \\ + (-4.27 \pm 0.17)\mu e_b + (-2.22 \pm 0.11)e_b^2 \\ + (-5.09 \pm 0.11)\mu^2 + (4.61 \pm 0.36)e_b^2\mu^2, \quad (2.70)$$

with parameters defined as for Equation 2.68.

Outside of these specific regions, numerical simulations show islands of (in)stability that appear in specific configurations (Hénon & Guyot, 1970; Dvorak, 1984; Rabl & Dvorak, 1988; Dvorak et al., 1989), which are now understood to arise from the existence of $n : 1$ mean motion resonances (Holman & Wiegert, 1999; Verrier & Evans, 2006).

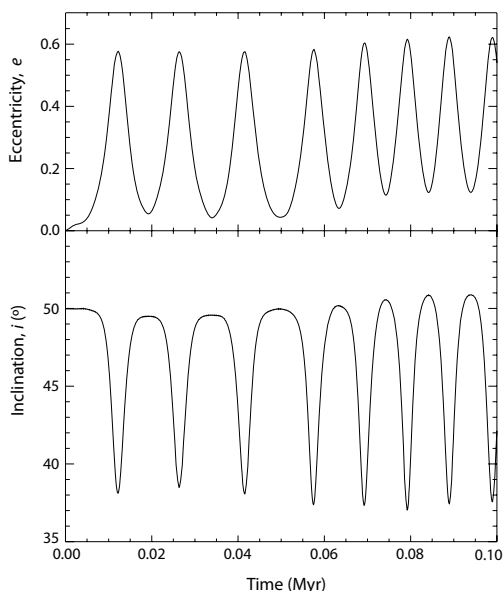


Figure 2.45: Variation of eccentricity and inclination with time for a test particle undergoing Kozai resonance. The configuration is for the system γ Cep, with starting elements $a = 0.5$ AU and $i = 50^\circ$; e and i oscillate strongly, 180° out of phase. From Verrier & Evans (2006, Figure 3), © John Wiley & Sons, Inc.

Planetary formation in binaries In the basic picture of planet formation around single stars, there are generally held to be four key stages of growth: coagulation of dust particles and their growth to cm-size objects; collisional growth to reach km-size planetesimals; the coalescence of planetesimals to form protoplanets or planetary embryos; and the collisional growth of planetary embryos to form terrestrial-size objects.

In a binary star system with moderate to small separation, the secondary star may have a significant effect on each of these processes. Various studies and simulations of the complex processes of planet formation around binary stars have been made (Marzari & Scholl, 2000; Barbieri et al., 2002; Quintana et al., 2002; Lissauer et al., 2004; Pfahl & Muterspaugh, 2006; Haghighipour & Raymond, 2007; Tsukamoto & Makino, 2007; Paardekooper et al., 2008; Pascucci et al., 2008; Xie & Zhou, 2008; Marzari et al., 2009; Payne et al., 2009).

For α Cen A, for example, Thébault et al. (2008) concluded that the planetesimal accretion phase could not have occurred *in situ*, and that outward migration after formation is most likely. A small binary inclination may, however, facilitate accretion (Xie & Zhou, 2009).

Modifying processes Various processes have been identified which are likely to modify planet formation around binary stars. The existence of additional stellar companions may inhibit the formation of circumstellar giant planets with particular orbital separations. Thus Nelson (2000) found that planet formation is unlikely in

equal-mass binary systems with $a \sim 50$ AU; Mayer et al. (2005) considered the effects of modified gravitational instabilities; and Thébault et al. (2006) considered regions in $a \lesssim 50$ AU systems in which runaway accretion is strongly affected.

A stellar component on an eccentric orbit can truncate the circumprimary disk to smaller radii, and remove material that would otherwise be available for the formation of terrestrial planets (Artymowicz & Lubow, 1994; Pichardo et al., 2005). More distant but highly inclined companions can modify their orbits over long time scales (Innanen et al., 1997; Takeda & Rasio, 2005). During the protoplanet formation stage, the perturbation of the secondary may increase the planetesimals' relative velocities, resulting in greater fragmentation, or creating regions where the orbits are unstable and therefore emptied of accretion material (Heppenheimer, 1978; Whitmire et al., 1998; Thébault et al., 2004).

Numerical simulations nevertheless show that giant or terrestrial planet formation should still be possible in binary systems. A binary companion can modify the structure of a planet-forming nebula, and create regions where gas and dust densities are locally enhanced. In turn, this may augment planet formation around the primary by gravitational instability (Mayer et al., 2005; Boss, 2006a), or by core accretion (Thébault et al., 2004).

Observational results from adaptive optic searches also support the idea that a stellar companion within 100 AU tends to inhibit, but not entirely suppress, planet formation (Eggenberger et al., 2007a).

Radial velocity surveys of binary stars Early radial velocity surveys tended to exclude spectroscopic and visual binary stars with separations below 2–6 arcsec because of problems of light contamination from the other component at the spectrograph entrance (Udry et al., 2000; Perrier et al., 2003; Marcy et al., 2005b; Jones et al., 2006b). This resulted in limited or biased surveys for binary star separations ≤ 100 –200 AU.

At larger separations, viz. for visual binaries with separations ≥ 100 AU, circumstellar planet searches can treat the targets as distinct single stars, and such surveys have faced limited technical difficulty (Gratton et al., 2003; Desidera & Barbieri, 2007; Toyota et al., 2009).

Observational confirmation that giant planets existed orbiting one component of a spectroscopic binary star came with the discoveries of GJ 86 Ab (Queloz et al., 2000b) with its white dwarf companion, and γ Cep Ab (Hatzes et al., 2003a), a binary with $P = 57$ yr, with the planet orbiting the primary with $P = 2.48$ yr.

The growing interest in exploring broader planetary domains as a way of constraining formation models has led to some more recent surveys preferentially including, rather than excluding, binary systems (Table 2.4).

Amongst specific radial velocity surveys focusing on binaries are a sample of 101 single-lined spectroscopic

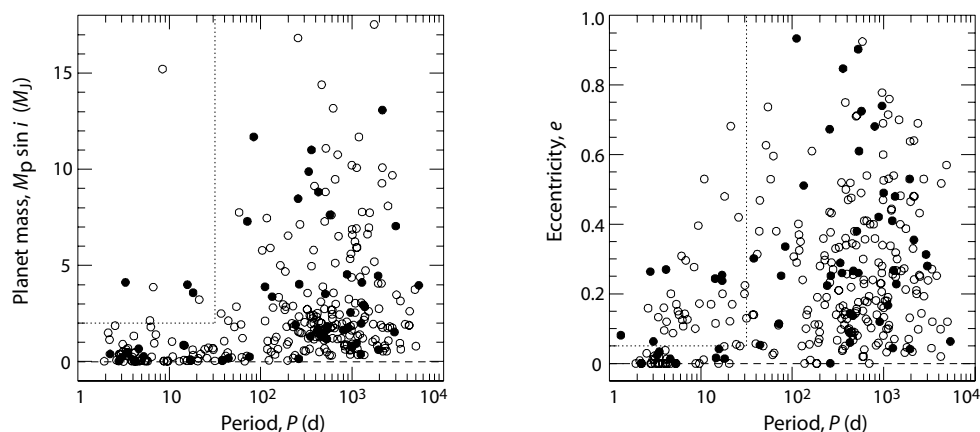


Figure 2.46: Planets detected from radial velocity observations (317 planets from *exoplanets.org*, 2010–11–01), with those orbiting single stars shown as open circles, and those in binary or multiple systems as filled circles. Left: mass ($M_p \sin i$) versus orbital period. Right: eccentricity versus period. Dashed regions correspond to those delineated by Eggenberger (2010, Figure 1).

binaries (SB1) with $P > 1.5$ yr, selected on the basis of the earlier CORAVEL surveys (Duquennoy & Mayor, 1991; Halbwachs et al., 2003). A programme to supplement the original CORAVEL data with 10–15 additional high-precision measurements of each system, using ELODIE in the north and CORALIE in the south, was initiated in 2001 (Eggenberger, 2010). Analysis employs the spectral deconvolution programme TODCOR. No promising planetary candidates have been reported to date, and present constraints indicate that less than 20% of SB1 systems host a short-period, $P < 40$ d circumpriary giant ($M_p \sin i \geq 0.5 M_J$) planet.

No extensive surveys have yet been undertaken for double-lined spectroscopic binaries (SB2) (Eggenberger, 2010). The principal reason for this is related to the problem of wavelength calibration: the standard technique using iodine cells (§2.2.4) is only applicable to single stars, and the time-varying spectra of SB2s cannot provide the template spectrum required by Equation 2.42. A modified procedure applicable to SB2s, using interleaved observations with and without the iodine cell, has been described by Konacki (2005b).

Other surveys for binary identification In parallel, the fraction of planets known to reside in binary systems has risen as a result of searches for common proper motion companions to known planet-hosting stars.

Other search techniques for planets around binary systems include photometric transit studies around eclipsing binaries (§6.2.8) which may allow the detection of transiting circumstellar or circumbinary planets (Deeg et al., 1998; Doyle et al., 2000; Ofir, 2008), or Trojan planets (Caton et al., 2000), and of non-transiting giant planets in circumbinary orbits through eclipse timing, which has resulted in the discovery of a few circumbinary systems (§4.3).

2.7.2 Present inventory

By late 2010, more than 50 planets have been found to be associated with binary or multiple stars (Eggenberger et al., 2004b; Eggenberger, 2010). Although most are gas giants orbiting the primary component, with projected separations in the range 20–12 000 AU, one possibly circumbinary planet, HD 202206, has been reported (Correia et al., 2005).

At small binary separations, $\lesssim 100$ AU, only a small number of planets have been found, and giant planets may be rarer in these systems than around wider separation binaries or single stars (e.g. Bonavita & Desidera, 2007). This may be consistent with studies which suggest that giant planet formation is inhibited under the influence of a nearby orbiting stellar companion (Nelson, 2000; Mayer et al., 2005; Thébault et al., 2006).

At the smallest binary separations, $\lesssim 20$ AU, the absence of circumstellar planets is consistent with the idea that giant planet formation in these systems is suppressed (Nelson, 2000; Thébault et al., 2004; Mayer et al., 2005; Thébault et al., 2006; Boss, 2006a), although Doppler surveys to confirm this remain incomplete.

Properties Zucker & Mazeh (2002) suggested that planets in binaries follow a different mass–period relation than those orbiting single stars, with the most massive short-period planets being found in binary or multiple systems.

More recent studies have decreased the significance of the earliest results (Eggenberger et al., 2004b; Desidera & Barbieri, 2007; Mugrauer et al., 2007a; Eggenberger, 2010), and the most recent distributions (Figure 2.46a) reveal mixed evidence for such an effect.

Meanwhile, short period ($P < 50$ d) planets around binary stars do appear to have smaller eccentricities, $e \lesssim 0.05$, than around single stars (Figure 2.46b).

Still based on somewhat limited statistics, the few planets with the highest eccentricities, $e > 0.8$, tend to be accompanied by a stellar or brown dwarf companion (Tamuz et al., 2008), a correlation that may extend to lower masses (Ribas & Miralda-Escudé, 2007). This has been taken to support the idea of eccentricities being excited by the Kozai mechanism, in which hierarchical triple systems with high relative inclinations cause large-amplitude periodic oscillations of the eccentricity of the inner pair (Holman et al., 1997; Innanen et al., 1997; Mazeh et al., 1997a; Ford et al., 2000b; Wu & Murray, 2003; Takeda & Rasio, 2005; Moutou et al., 2009).

The coupling of Kozai oscillation with tidal friction, also referred to as Kozai migration, might also lead to short-period planets preferentially following nearly circular orbits (Figure 2.46b). In multiple systems this mechanism may be more effective than type II migration in bringing massive planets close to their host stars. This may also explain why the most massive short-period planets are found in binary or multiple systems (Takeda & Rasio, 2006; Fabrycky & Tremaine, 2007).

2.7.3 Specific examples

γ Cep The binary star system γ Cep harbours a stable giant planet orbiting the primary star at a distance some one tenth of the stellar separation (Hatzes et al., 2003a). The primary and secondary stars, of masses 1.6 and $0.4M_{\odot}$, are separated by 18.5 AU ($P \sim 57$ yr, $e = 0.36$), with the planet of mass $1.7M_J$ orbiting the primary with $a = 2.13$ AU, $P = 2.5$ yr, $e = 0.12$.

Numerical integrations as a function of a_b , e_b , and i_b (the inclination of the planet with respect to the stellar binary) by Haghighipour (2006), show that the planet orbit is stable for values of the binary eccentricity $0.2 \leq e_b \leq 0.45$, with the system becoming unstable within a few thousand years for $e_b > 0.5$. Within the stable range of orbital eccentricity, the planet remains stable for $i_b < 40^\circ$, becoming unstable within a few thousand years for $i_b > 40^\circ$. For large values of the inclination, the system may be locked in a Kozai resonance (Haghighipour, 2004). The stability map for test particles interior to the planet orbit show complex stability patterns influenced by mean motion resonances (Figure 2.47).

16 Cyg B The planet around 16 Cyg B has a highly eccentric orbit, $e \sim 0.7$. 16 Cyg B is one component of a widely separated binary system whose eccentricity is itself very large, $e > 0.54$. The orbital period of the binary star is long and difficult to measure accurately, but from astrometric observations made over some 170 years is estimated to be $>18\,000$ yr (Hauser & Marcy, 1999).

The system is an interesting case for planetary formation theories which try to explain the large eccentricities in binary systems as a result of gravitational perturbations by the other stellar component, here 16 Cyg A.

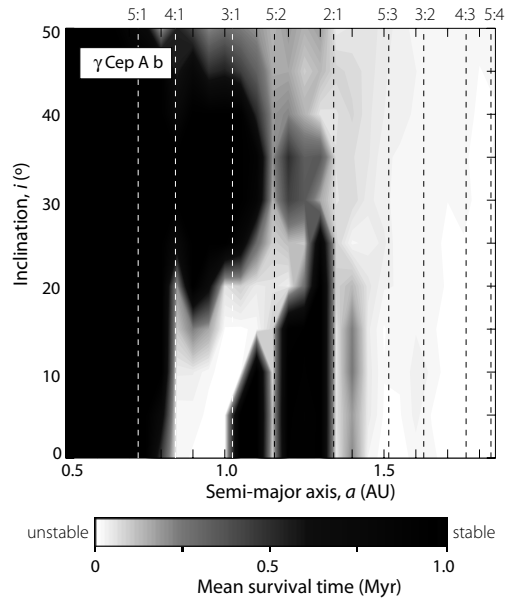


Figure 2.47: Stability map for test particles interior to the orbit of the planet γ Cep Ab, for the coplanar case $i_b = 0$. Stability is indicated by the test particle's survival time averaged over longitudes and normalised to 1 Myr. Darker regions indicate higher stability. Dashed vertical lines are the nominal locations of the mean motion resonances with the planet γ Cep Ab. From Verrier & Evans (2006, Figure 2), © John Wiley & Sons, Inc.

If the planet was originally formed in a circular orbit, with an orbital plane inclined to that of the stellar binary by $>45^\circ$, then the planet orbit is predicted to oscillate chaotically between high- and low-eccentricity states, on time scales of $10^7 - 10^{10}$ yr, (Holman et al., 1997; Mazeh et al., 1997a; Hauser & Marcy, 1999). The discovery of further planets with $M_p \sim M_J$ within 30 AU would require this conclusion to be revisited.

HD 188753 Some controversy surrounds the triple system HD 188753, a 155 d spectroscopy binary (HD 188753B) in a 25.7-yr orbit ($a = 12.3$ AU) about the primary (HD 188753A). Konacki (2005a) reported a 3.35 d $1.1M_J$ planet around the primary component based on 10 m s^{-1} accuracy Keck-HIRES data, unconfirmed by Eggenberger et al. (2007b).

According to theoretical studies, the pericentre distance of the AB pair is too small to have allowed giant planet formation. Modeling by Jang-Condell (2007) has shown that the circumstellar disk around the primary would have been truncated at around 1.5–2.7 AU, leaving insufficient material, and in any case too hot, for a Jovian planet to have formed *in situ*. Alternative formation mechanisms were forthcoming, in particular proposing that the primary could have acquired the planet through interactions within an open star cluster that has now dissolved (Pfahl, 2005; Portegies Zwart & McMillan, 2005).